Dynamics & Simulations HW

Monsoon 2026 | Evolving Document

Updated for Week 11 Homework

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1 Homework 1

1.1 Q1

Problem Statement

Set up (define system, draw FBD, write ODEs) a particle problem. Just one particle. 2D or 3D, your choice. Use a force, or forces that you like (gravity, spring, air friction). Any example of interest. Find a numerical solution. Graph it. Animate it. Try to make an interesting observation.

Interpretation

The system I took conists of a mass 'm' connected to 2 points 'A' and 'B'. It is connected to point 'A' with the help of a spring of spring constant $'k'_a$ with a damper of damping constant $'c'_a$ connected in parallel. The same mass is connected to point 'B' with the help of a another spring of spring constant $'k'_b$

By setting some parameters and solving it with the help of fsolve- I was able to obtain 3 equilibrium points. Each of these points could be either an unstable equilibrium point or a stable equilibrium point. That could be found with the help of the *Jacobian* Matrix.

The jacobian can be calculated at each of these points. If all eigen values of the Jacobian at each point has negative real parts, then that point is a stable equilibrium point, otherwise it's an unstable equilibrium point. Due to time constraints, I wasn't able to attempt this- but will update this solution in the next submission.

Solution

```
Equilibrium position:
x = 4.0000 m
y = 6.0000 m

All equilibrium points found:
   -0.4157    4.6091    5.3480
    6.5349    6.4336    1.0868

Number of equilibria: 3
>>
```

Figure 1: Equilibrium Points. Setup mentioned in matlab code.

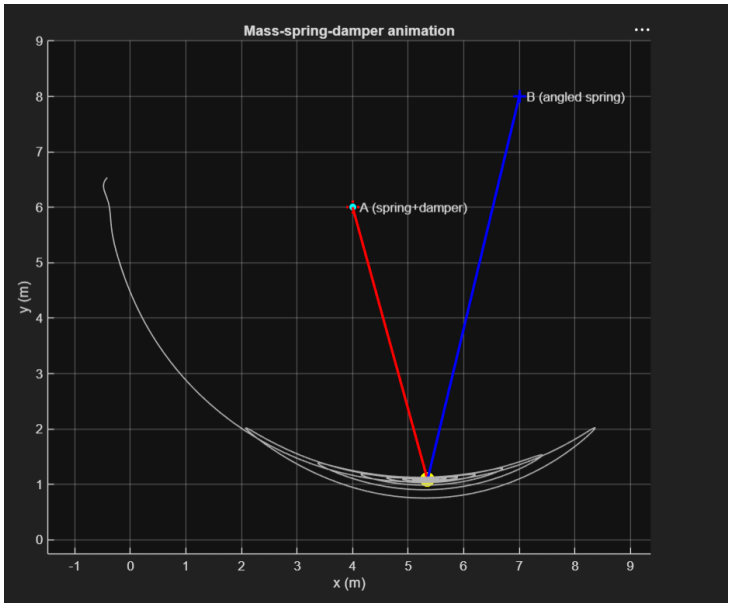


Figure 2: This is when the initial condition is kept at the first equilibrium point- (-0.4157,6.5349). It is unstable- since giving a velocity of (-1,-1)m/s made it shift to the stable point

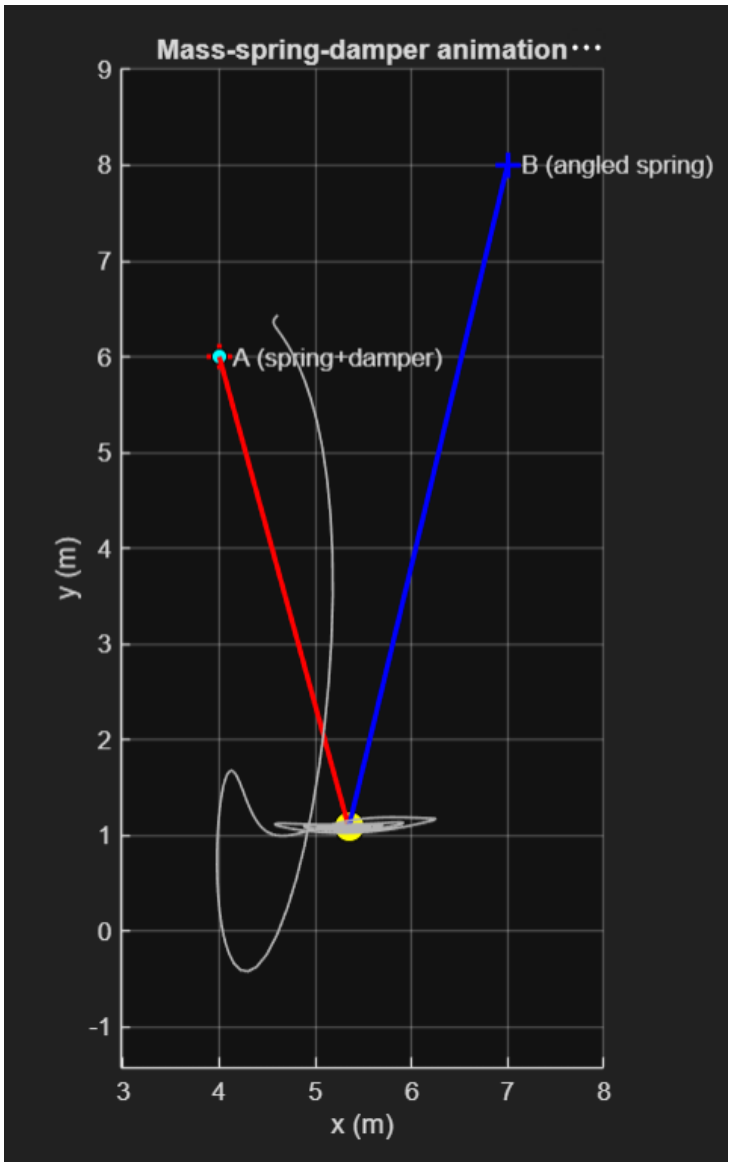


Figure 3: Initial condition- another unstable equilibrium point: (4.6091, 6.4336)

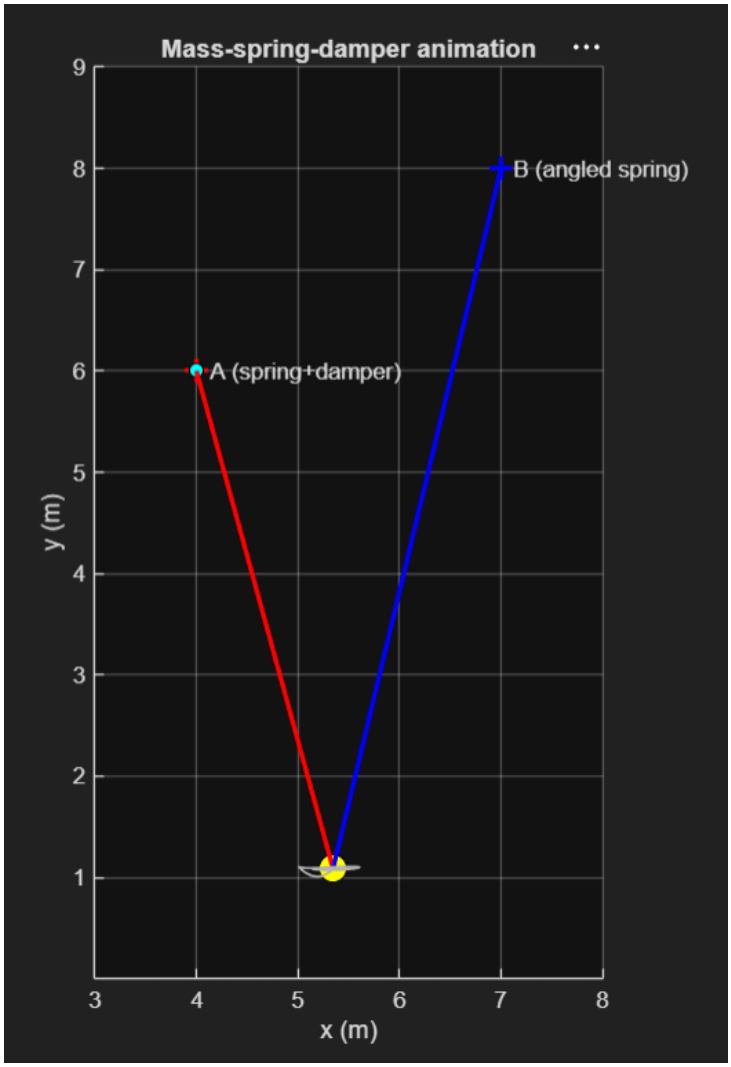


Figure 4: Initial Condition- (5.3480,1.0868). A little nudge resulted in it being brought back to the same point- stable equilibrium.

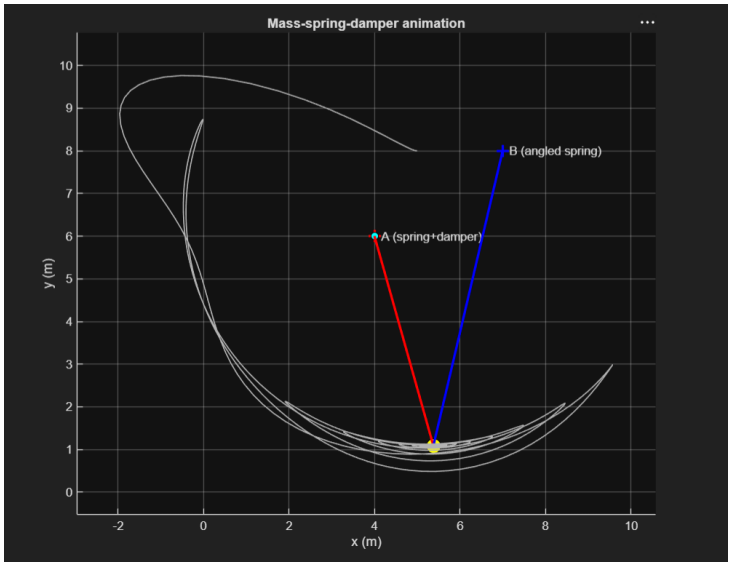


Figure 5: A random point- (3,2) was taken as Initial Condition. It settles at the stable equilibrium point- (5.3480,1.0868).

1.2 Q2

Problem Statement

3D. Get good at vectors. Assume that the positions relative to an origin of four random points, which are randomly located in space, are given as $\vec{r}_A$, $\vec{r}_B$, $\vec{r}_C$ and $\vec{r}_D$. Assume force $\vec{F}$ is given. For each problem below write a single vector formula (one for each problem) that answers the question.

(a) The points A and B define an infinite line. So do the points C and D. Find the distance between these two lines. 'The' distance means 'the minimum distance', that is the length of the shortest line segment connecting the two lines. Either write a formula (or sequence of formulas), or write computer code that gives the answer, or both.

(b) Same problem as above, but also find the end points of the shortest line segment.

(c) Find the volume of the tetrahedron ABCD (you should reason-out and not quote any formulas for the volume of a tetrahedron, that is, see if you can derive the formula: 'volume = one third base times height').

(d) Assume points A, B and C are fixed to a structure. All three are connected, by massless rods, to a ball and socket at each end, to point D. At point D the force $\vec{F}$ is applied. Find the tension in bar AD. Find a formula for the answer, or write computer code to find the answer, or both. The goal is to find a formula for the tension in terms of the positions and the force vector.

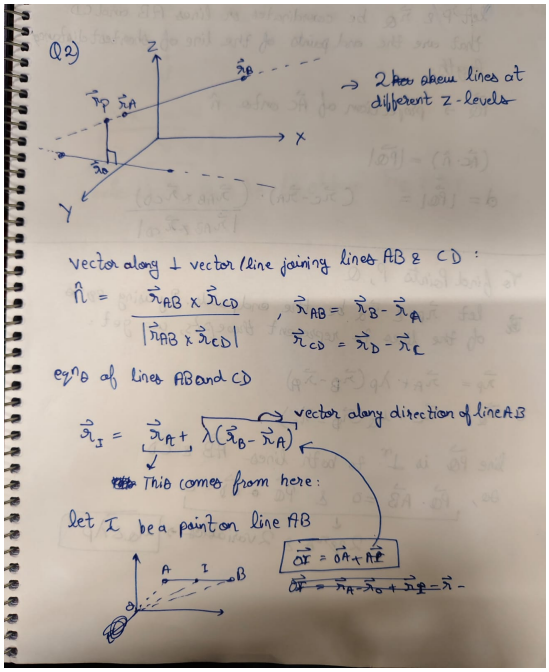
Interpretation

The explanation for the formula for distance between 2 lines and the method to find the end points of the shortest line segent.

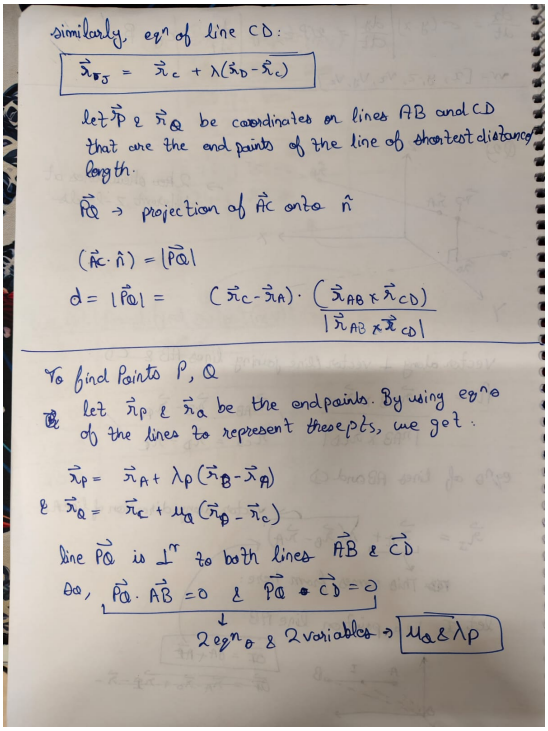
Solution

```
>> Apurv_Q2
The shortest distance between the two lines is: 0.4934
Endpoint P: [1.14, 0.86, -1.58]
Endpoint Q: [1.62, 0.97, -1.54]
Sanity check: Magnitude of PQ is 0.4934
>>
```

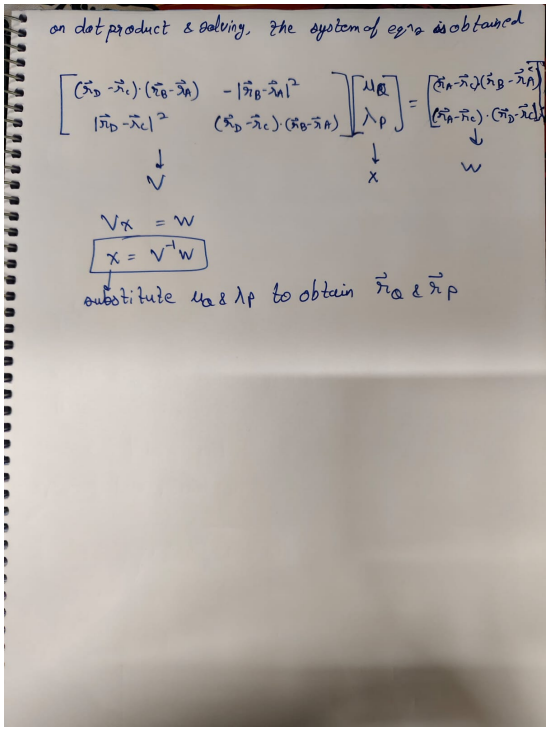
Figure 6: Caption here.



(a)



(b) If the 2 lines intersect, the distance comes out to be 0.



(c)

Figure 7

1.3 Q3

Problem Statement

More ODE & animation practice. Take a simple set of ODEs. Use a set you like, e.g., harmonic oscillator, non-linear pendulum, the Lorentz system (look it up on the internet). Solve this set numerically 3 ways (see below), and understand the accuracy. The goal is that, by the time you hand in the homework, you can write and debug the assignment on your own without looking up anything (outside of trivial syntax things). And you always have a good sense of the accuracy of your solution.

(a) Method 1: as simply as possible, without ODE45, and without calling functions or anything like that. A single function or script file with no function calls (ok, plotting calls are ok). Just write a simple loop that implements Euler's method with your ODE.

(b) With your own Euler solver function. Your main program should call your Euler solver. Your Euler solver should call a RHS (Right Hand Side) function.

(c) With ODE45.

(d) Using (b), solve the equations many times with progressively smaller step size, down to the smallest size you have patience for, and up to the largest size that isn't crazy. As sensibly as possible, compare the results and use that comparison to estimate the accuracy of each solution. You should be able to find a method to estimate the accuracy of a numerical solution even without knowing the exact solution.

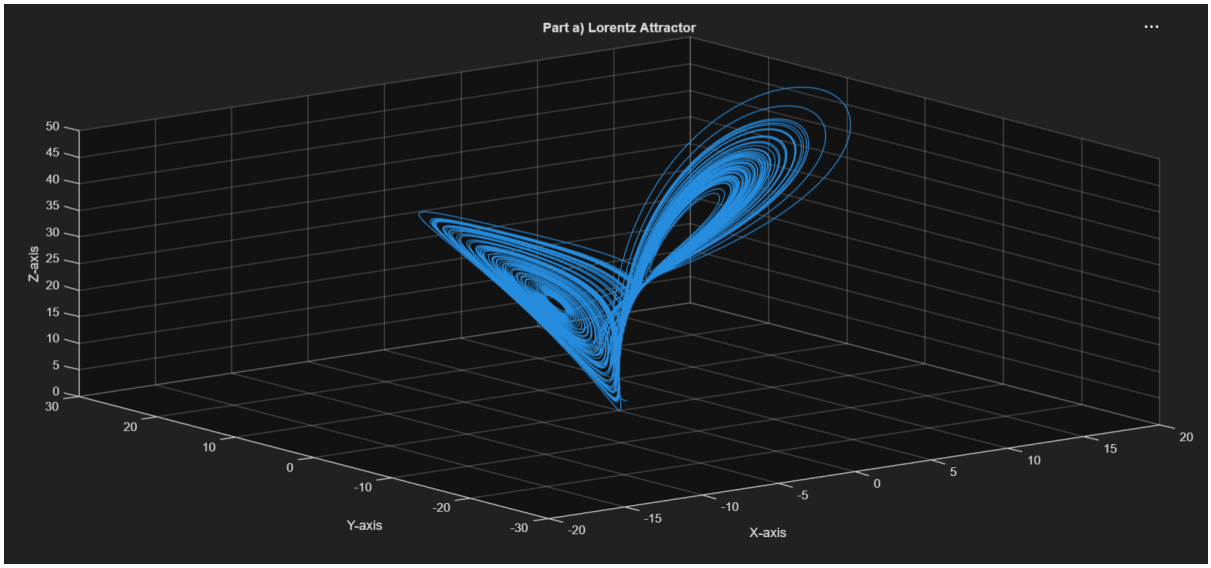
(e) Using ODE45, solve the equations with various accuracies (use 'reltol' and 'abstol', note MATLAB satisfies one or the other, whichever is easiest. So, if you want an accurate solution you need to make both 'reltol' and 'abstol' small). Does MATLAB do a good job of estimating its own accuracy? Use suitable plots to make your point.

Interpretation

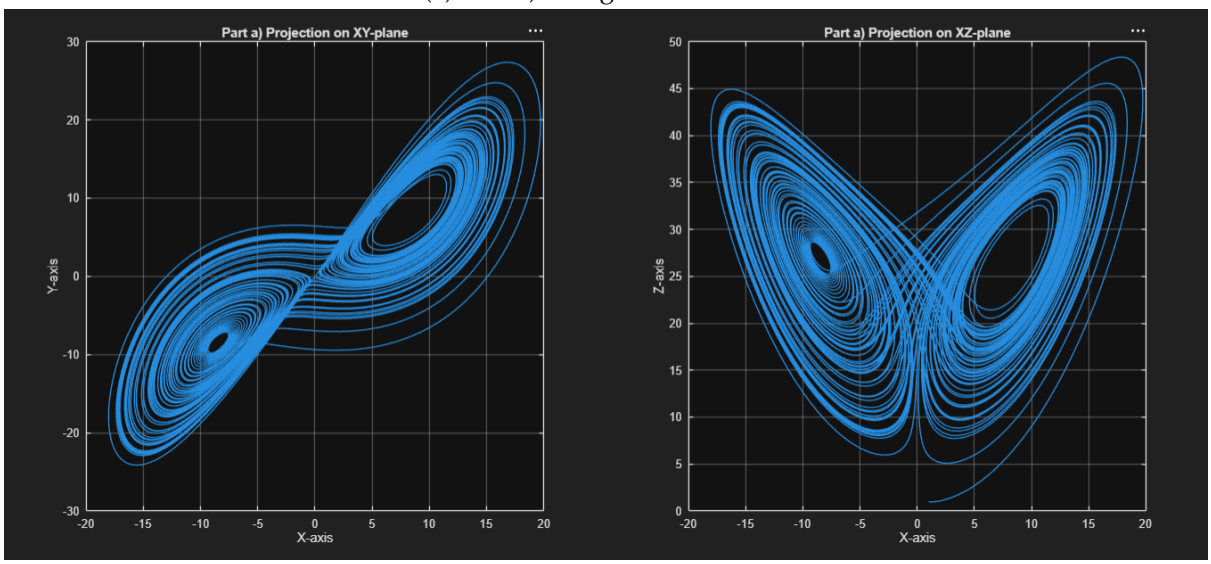
Solution

The System used for this question was the Lorentz System. It is described by the following differential equations:

$$\dot{x} = \sigma(y - x) \tag{1}$$
$$\dot{y} = x(\rho - z) - y \tag{2}$$
$$\dot{z} = xy - \beta z \tag{3}$$

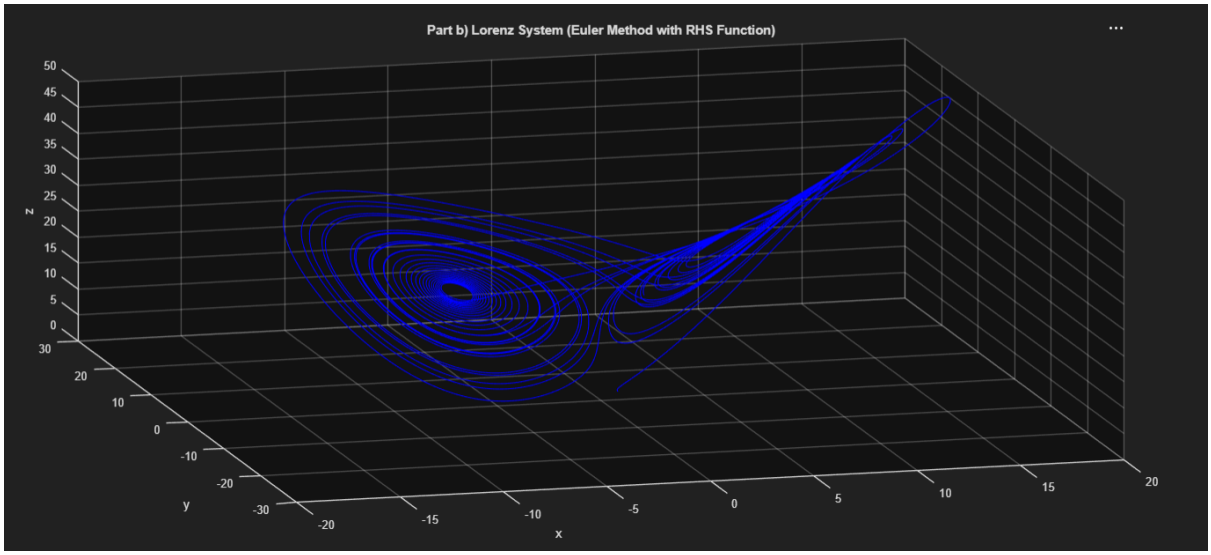


(a) Part a) Using Euler method.

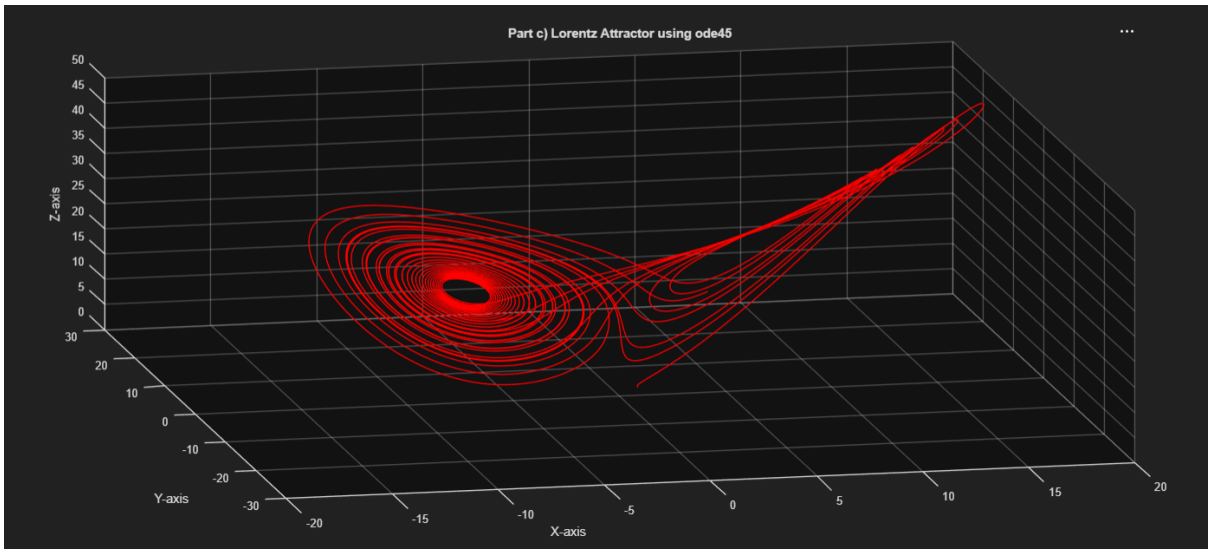


(b) Part a) Continued- Projections on XY and XZ planes.

Figure 8: Caption here.



(a) Part b) Used Euler function and the rhs function



(b) Part c) Using ODE45

Figure 9: Caption here.

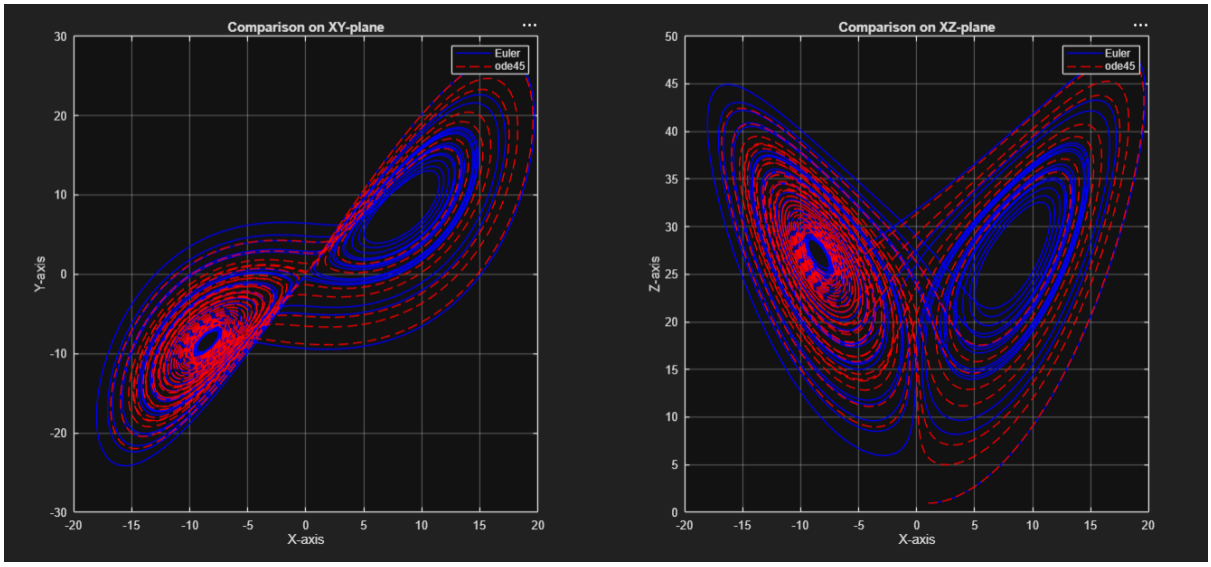


Figure 10: Comparision of Euler and ODE45 methods

2 Homework 2

2.1 Q4

Problem Statement

Cross product: geometry vs components. The geometric definition of cross product is this $\vec{a} \times \vec{b}$ is a vector $\vec{c}$ with magnitude $|\vec{a}||\vec{b}| \sin\theta_{ab}$ that is orthogonal to $\vec{a}$ and $\vec{b}$ in the direction given by the right hand rule. Use this definition to find an alternative geometric definition involving projection (namely: project $\vec{b}$ onto the plane that is orthogonal to $\vec{a}$; then stretch it by $|\vec{a}|$; then rotate it $\pi/2$ around the $\vec{a}$ axis). Use that definition to show the distributive rule $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$.

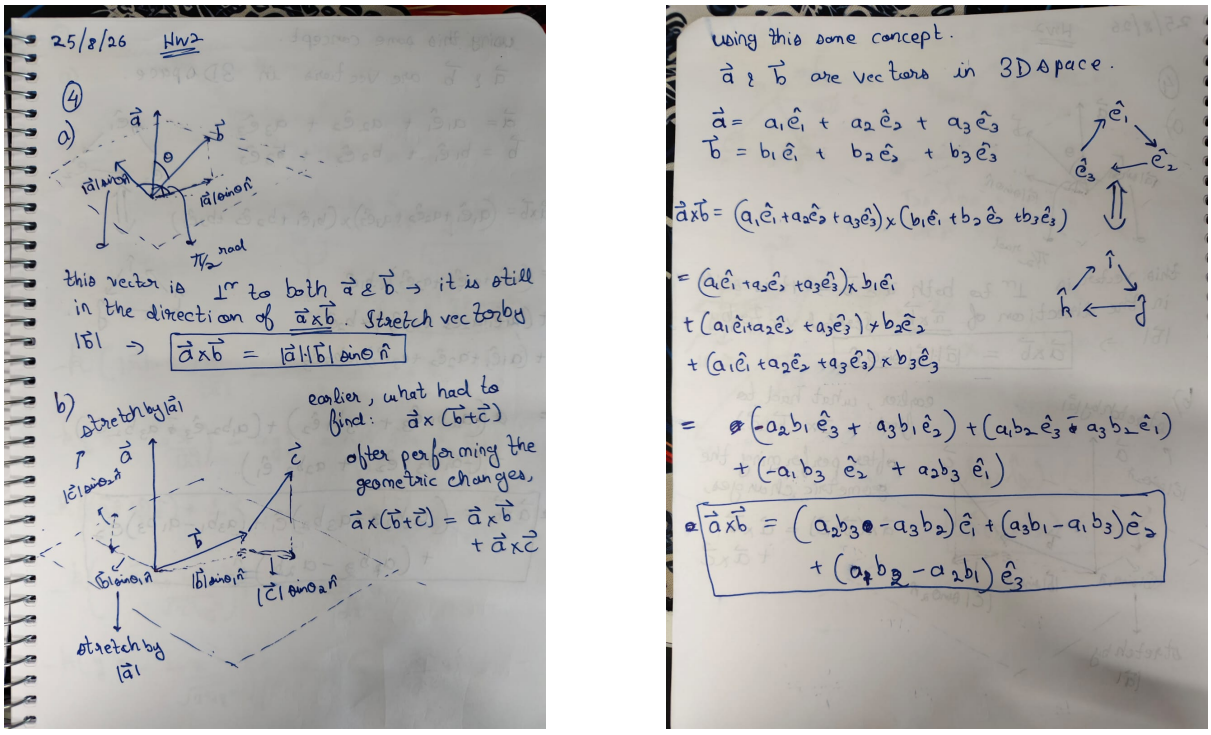
(b) Then use the distributive rule to find the component formula for cross product, namely that

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2)\hat{e}_1 + (a_3 b_1 - a_1 b_3)\hat{e}_2 + (a_1 b_2 - a_2 b_1)\hat{e}_3. \tag{4}$$

Note that this distributive law implies that, for given $\vec{a}$, $\vec{a}\times$ is a linear operator. That is $\vec{a} \times \vec{v}$ is a linear function of $\vec{v}$. Later in the course we will use this to replace the cross product with a tensor product. Hint: You can read about this in, say, the Ruina/Pratap book (box 1.7).

Interpretation

Solution



(a) (b)

Figure 11

2.2 Q6

Problem Statement

Spring and mass (2D). One end of a negligible-mass spring (k, L_0) is pinned to the origin, the other to a mass (m). There is gravity (g). Initial Conditions (ICs): The initial position is $\vec{r}_0 = x_0\hat{\imath} + y_0\hat{\jmath}$, and the initial velocity is $\vec{v}_0 = v_{x0}\hat{\imath} + v_{y0}\hat{\jmath}$. Motion starts at $t = 0$ and ends at t_{end}.

(a) Find the Equations of Motion (EoM);

(b) Assume all parameters and IC's above are given.

   i. Plot the trajectory of the mass.

   ii. Animate the trajectory of the mass.

(c) How many ways can you think of checking the numerical solution, find as many as you can, and do the check. The list is started here:

   i. $k = 0$, all else arbitrary: The motion is parabolic flight (including falling straight down as a special case) [Why? The system is then just ballistics from freshman physics];

   ii. $x_0 = 0$, $v_{x0} = 0$, all else arbitrary: The motion stays on the y axis [Why? There is no force in the x direction if the mass is on the y axis. Because the initial velocity has no x component, the mass never leaves the y axis];

   iii. $g = 0$, $\vec{v}_0 = \vec{0}$, all else is arbitrary: The motion stays on a radial line. And, if the motion does not cross the origin, the motion is that of a harmonic oscillator (sinusoidal oscillations, check by plotting, say x vs t). [Why? Write the EoM in polar coordinates $\Rightarrow m\ddot{r} = -k(r - L_0) \Rightarrow$ the harmonic oscillator equation, $m\ddot{r}^* = -kr^*$, where $r^* \equiv r - L_0$.]

   iv. $L_0 = 0$, all else is arbitrary: ? . [Why? ? .] Hint, this one special case is problem 10, below.

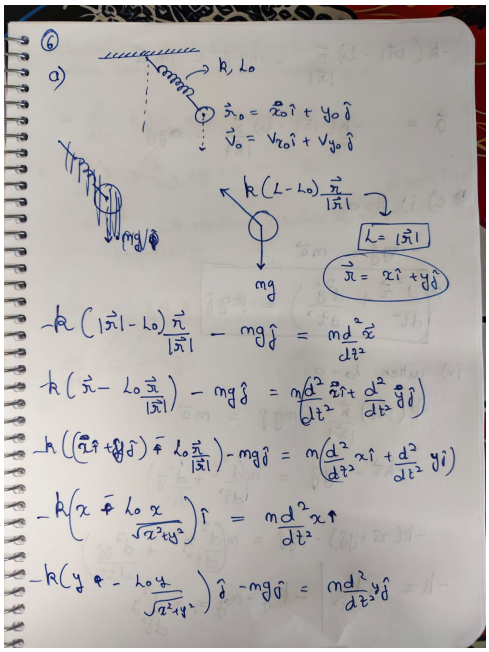
   v. etc.

   vi. etc.

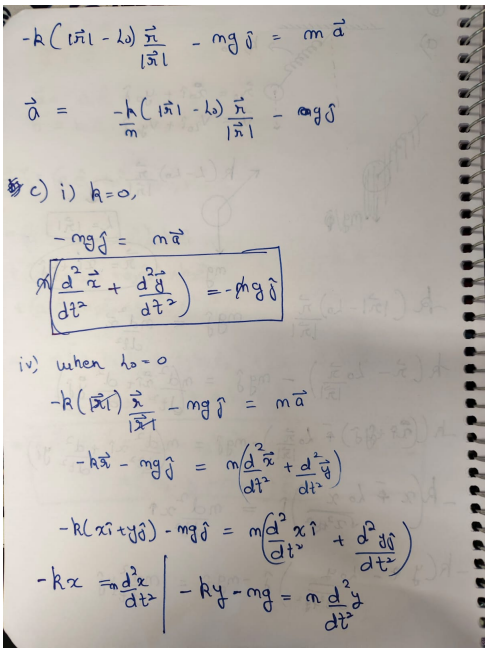
   vii. ...

Interpretation

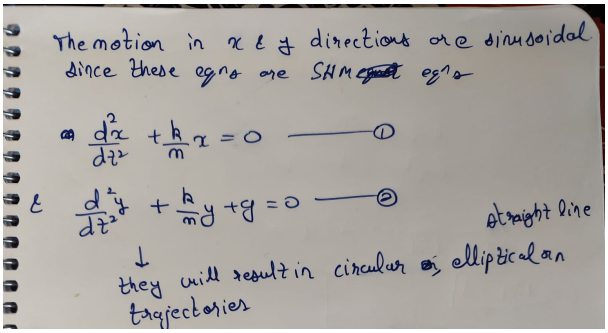
Solution



(a) Equations of Motion in vector form

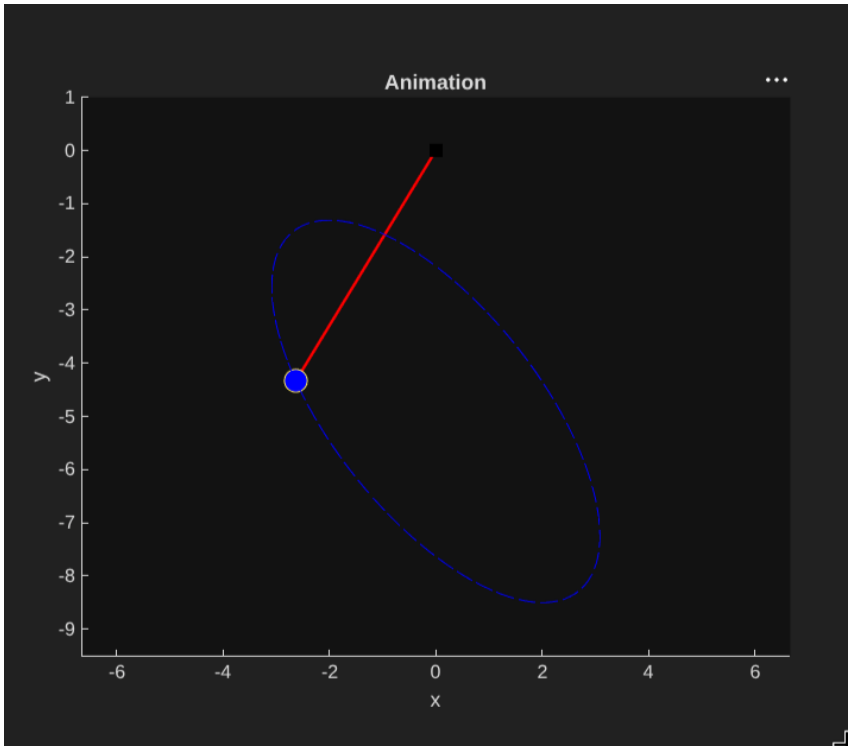


(b) Zero rest length case.

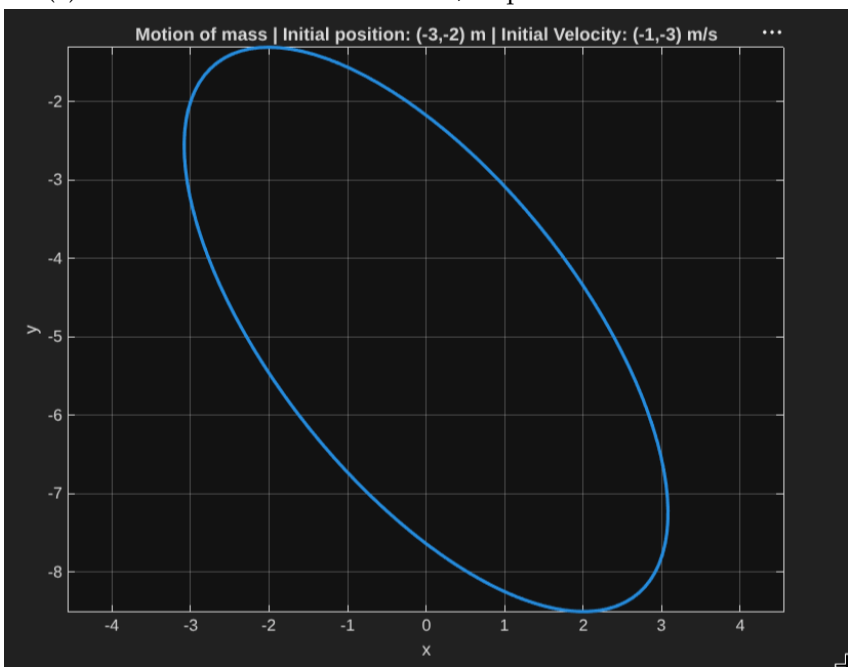


(c) Equations in x and y direction are both SHM equations- meaning x and y will vary sinusoidally w.r.t time. Overall motion will either be circular, elliptical or a straight line.

Figure 12

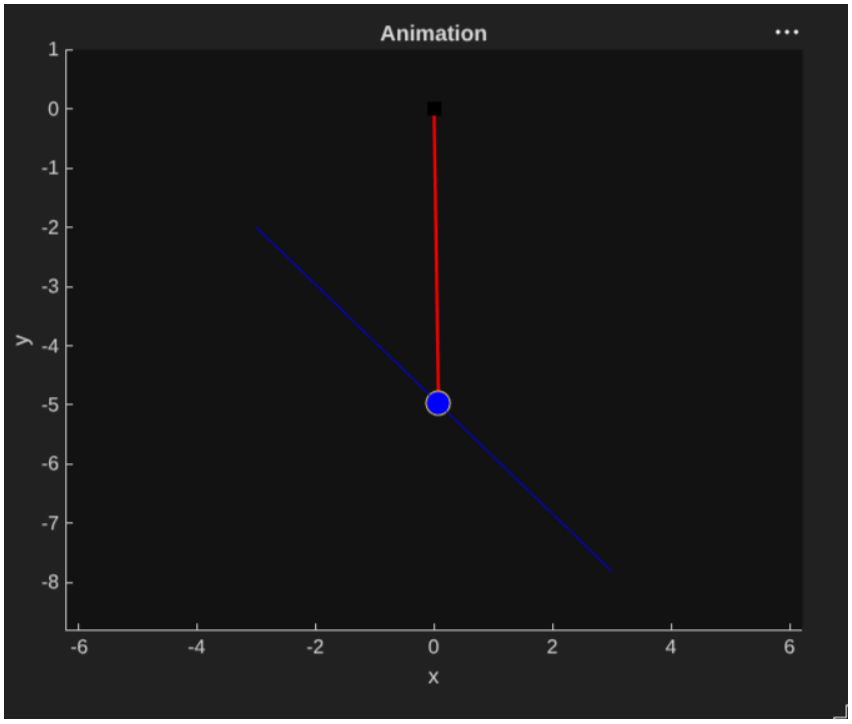


(a) With non-zero initial conditions, elliptical motion is obtained.

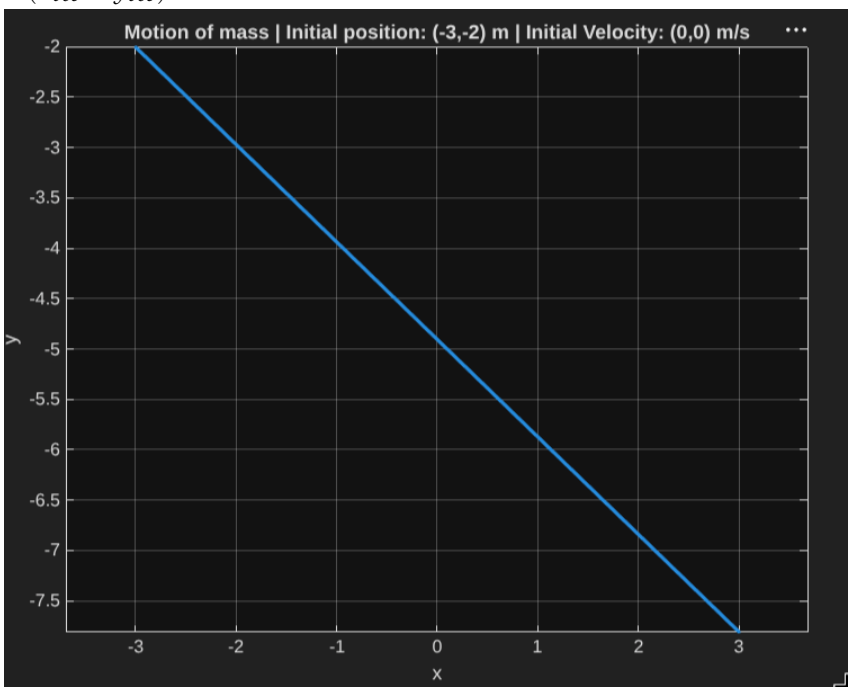


(b)

Figure 13

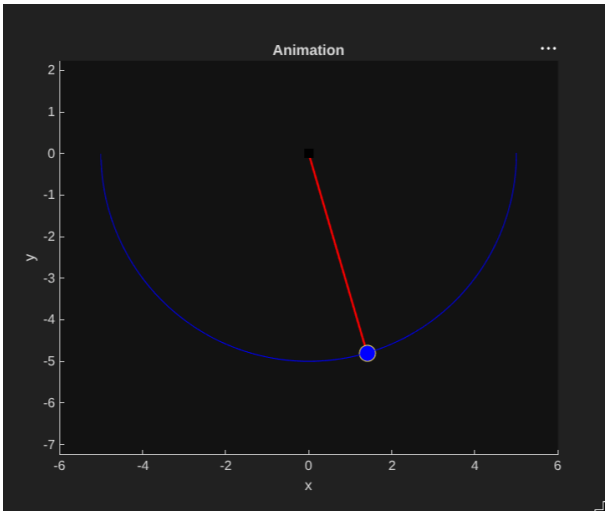


(a) With zero initial velocities, straight line motion is obtained. This is because the spring force is linearly dependent on position vector. F = -$k(x_{vec} + y_{vec})$

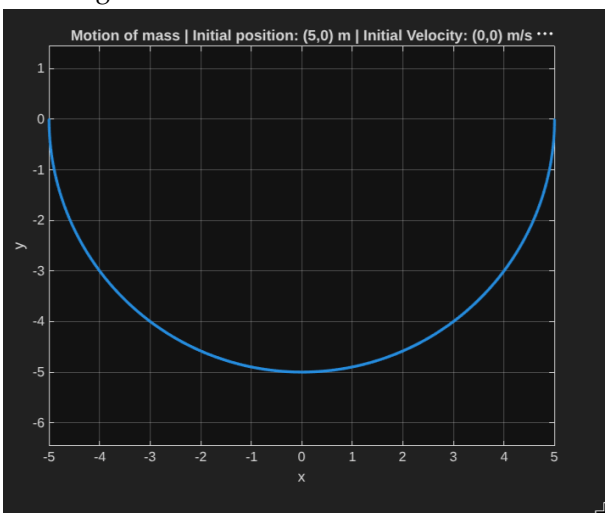


(b)

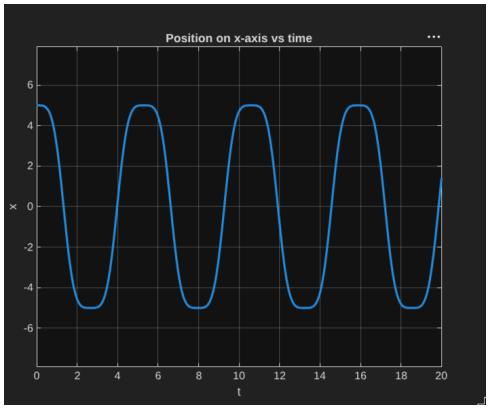
Figure 14



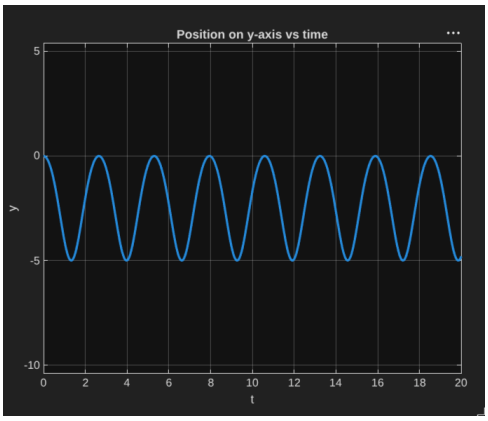
(a) This is the rigid spring case. Over here, the value of the spring constant is set to be a million to represent a rigid rod.



(b)

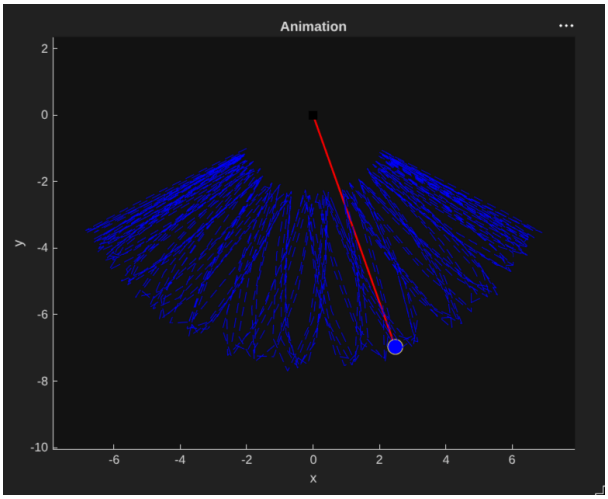


(c)

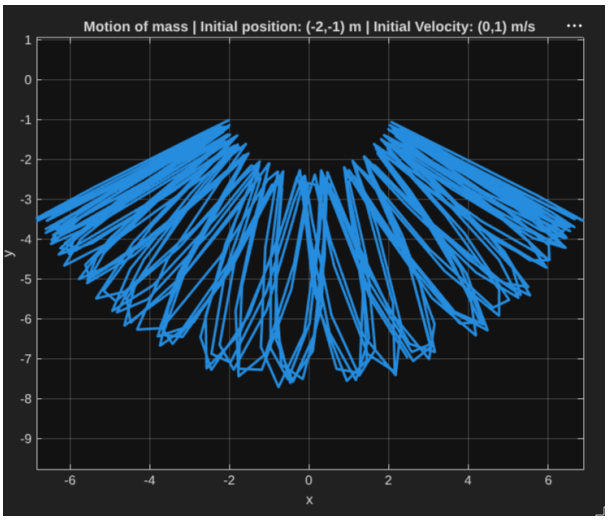


(d)

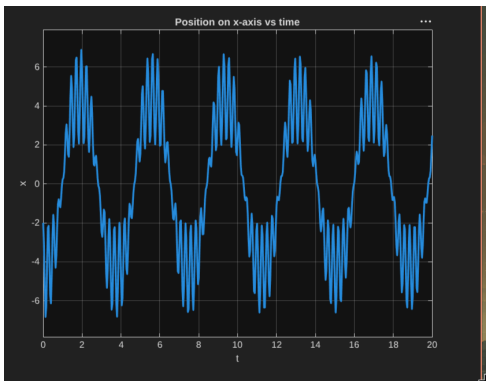
Figure 15: Variation of x and y vs time.



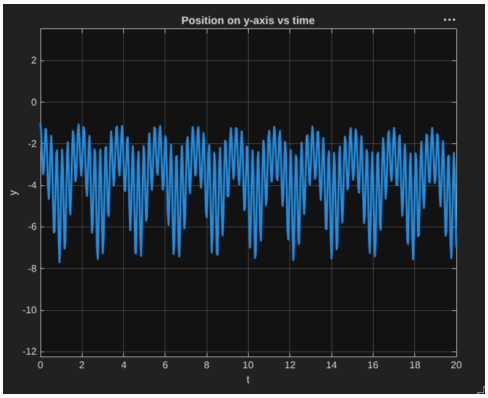
(a)



(b)



(c)



(d)

Figure 16

2.3 Q9

Problem Statement

Canon ball. A cannon ball m is launched at angle θ and speed v_0. It is acted on by gravity g and a viscous drag with magnitude $|cv|$.

(a) Find position vs time analytically.

(b) Find a numerical solution using $\theta = \pi/4$, $v_0 = 1$ m/s, $g = 1$ m/s^2, $m = 1$ kg, $c = 1$ kg/s.

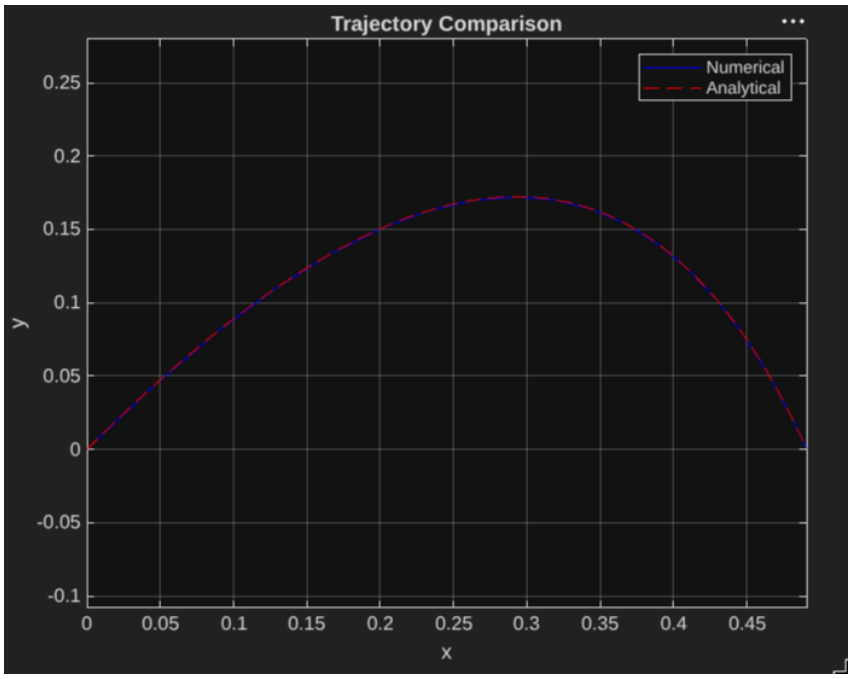
(c) Compare the numeric and analytic solutions. At $t = 2$ how big is the error? How does the error depend on specified tolerances or step sizes?

(d) Use larger and larger values of v_0 and for each trajectory choose a time interval so the cannon at least gets back to the ground. Plot the trajectories (using equal scale for the x and y axis). Plot all curves on one plot. As $v \to \infty$ what is the eventual shape? [Hint: the answer is simple and interesting.]

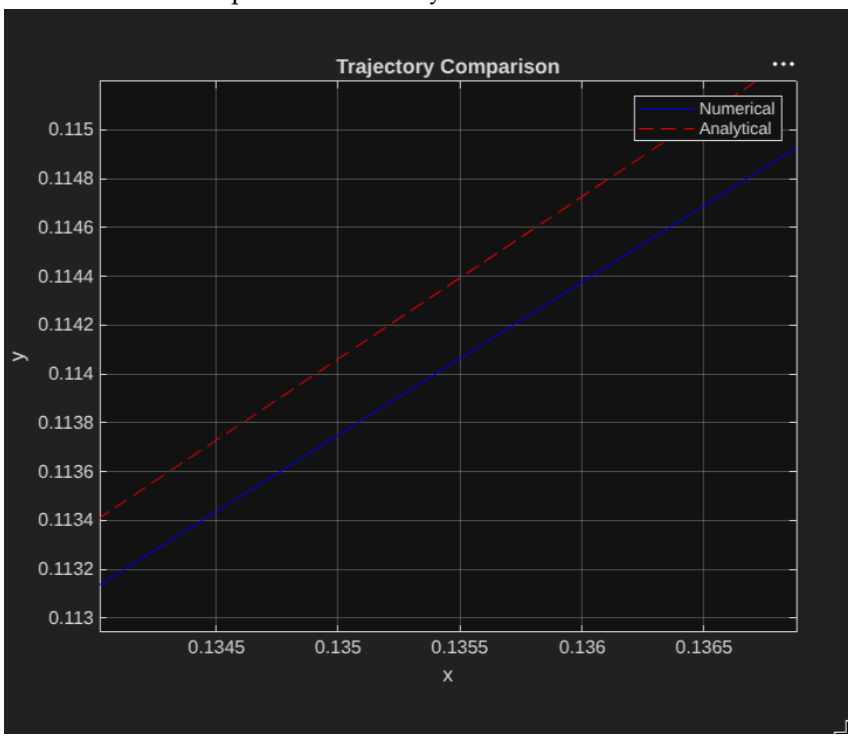
(e) For any given v_0 there is a best launch angle θ^* for maximizing the range. As $v_0 \to \infty$ to what angle does θ^* tend? Justify your answer as best you can with careful numerics, analytical work, or both.

Interpretation

Solution



(a) The numerical setup was solved using ode45 and proved to be very accurate when compared to the analytical solution.



(b) Zoomed in version of the above graph.

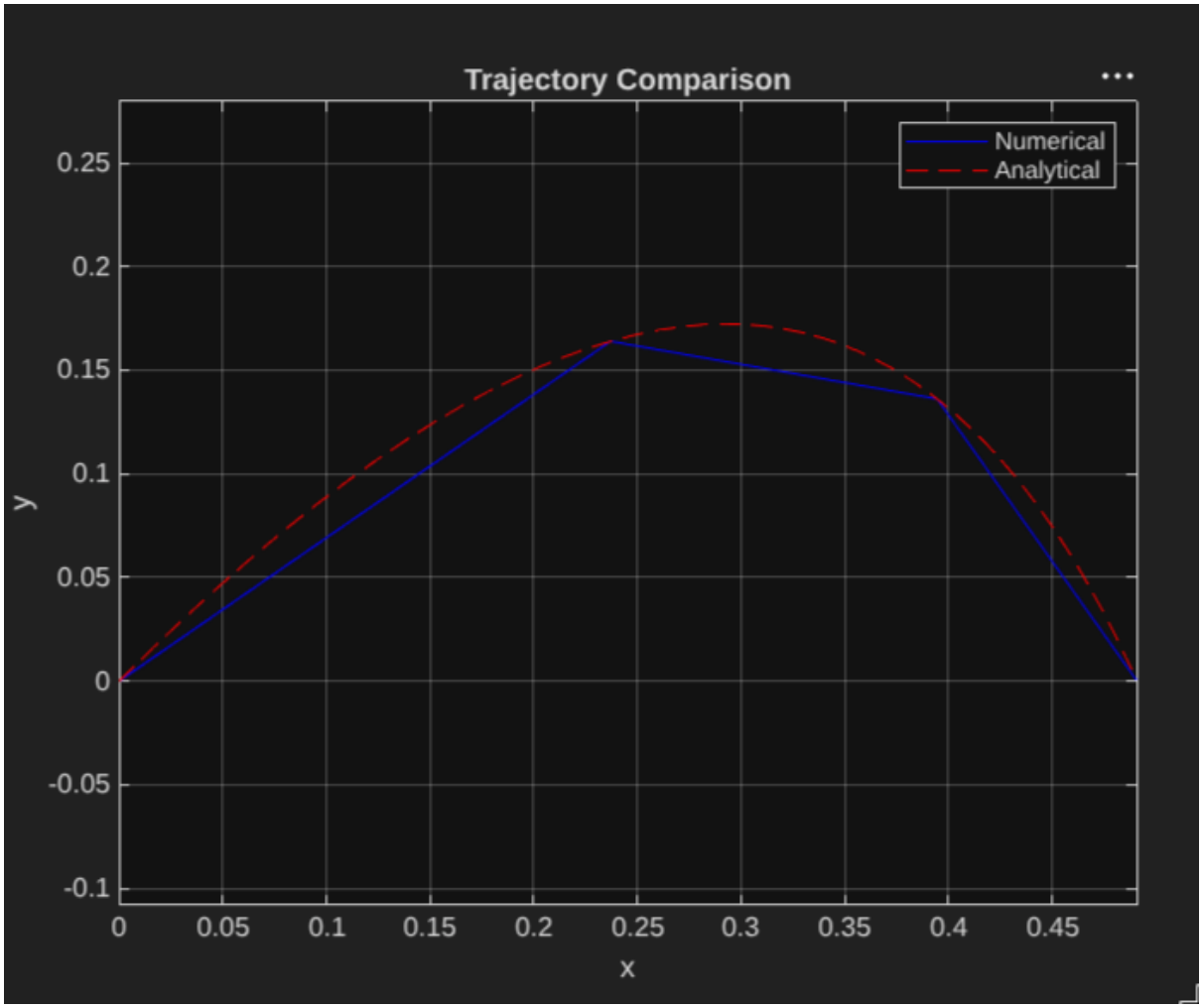


Figure 18: The accuracy of the numberical solution depends on the step sizes- lower the step sizes, more 'coarser' the solution and larger the error.

3 Homework 3

3.1 Q10

Problem Statement

Mass hanging from spring. 3D. Consider a point mass hanging from a zero-rest-length linear spring ($\ell_0 = 0$) in a constant gravitational field.

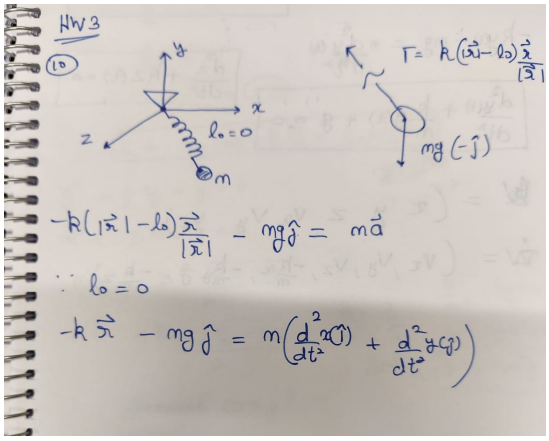
(a) Set up equations. Set up for numerical solution. Plot 2D projection of 3D trajectories.

(b) By playing around with initial conditions, find the most wild motion you can find (wild means most wiggles, or most complicated). Make one or more revealing plots. [Hint: Make sure the features you observe are properties of the system and not due to numerical errors. That is, check that the features do not change when the numerics is refined.]

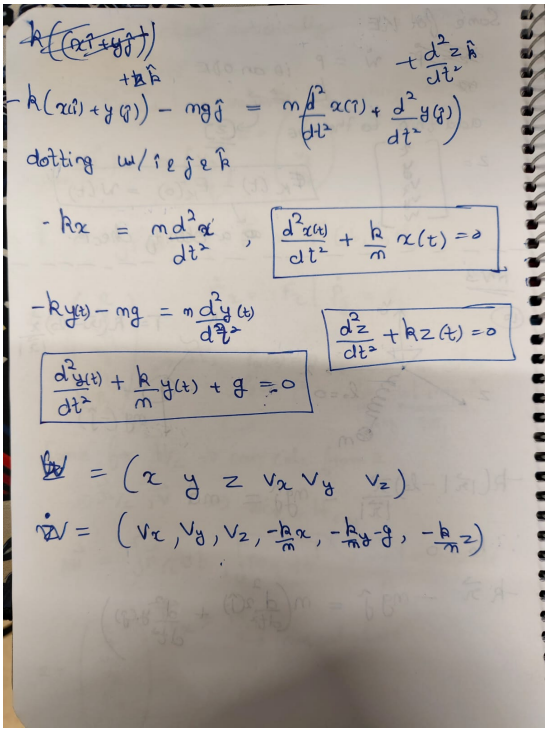
(c) Using analytical methods justify your answer to part (b).

Interpretation

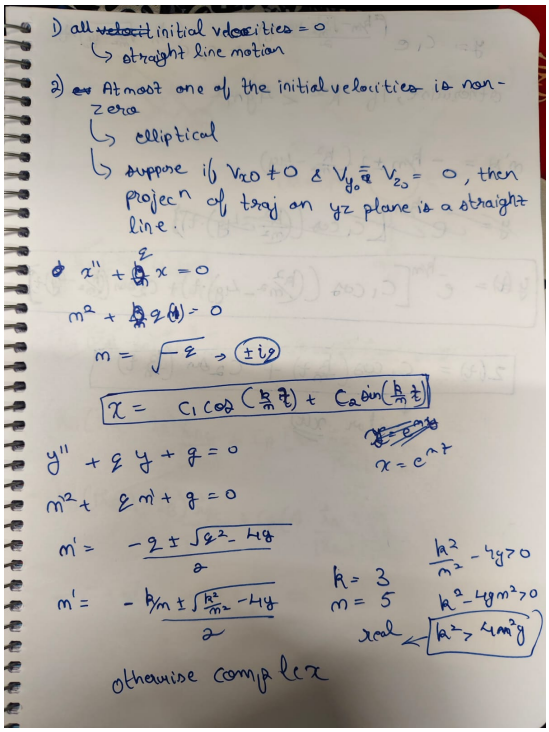
Solution



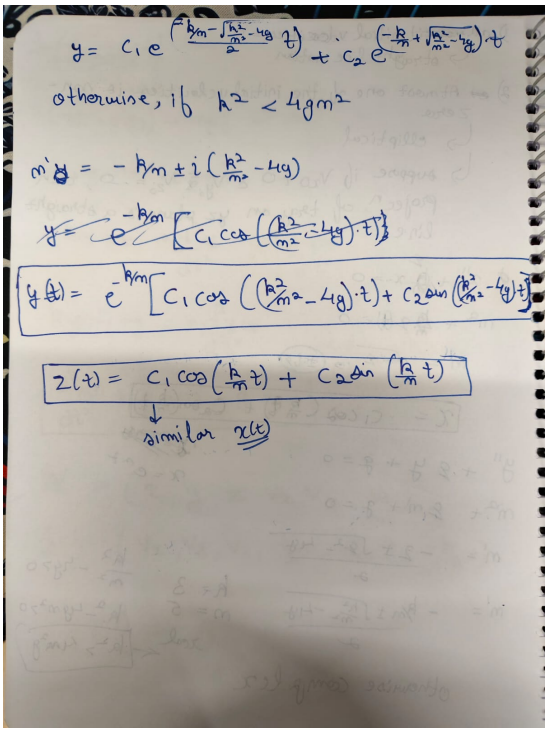
(a) Setting up the free body diagram and the vector equation of motion for the mass on a zero-rest-length spring, then expanding it into component form.



(b) Dotting the vector EoM with $\hat{\imath}, \hat{\jmath}, \hat{k}$ to split it into three scalar ODEs, and writing the resulting state vector $\vec{v} = (x, y, z, v_x, v_y, v_z)$ and its derivative for numerical integration.



(c) Working out the special cases analytically: zero initial velocity gives straight-line motion, and a single non-zero velocity component gives an elliptical/planar trajectory. Solving the $x(t)$ ODE's characteristic equation to confirm it is simple harmonic.



(d) Completing the analytical solution for $y(t)$ and $z(t)$, confirming that all three coordinates are simple harmonic at the same frequency $\sqrt{k/m}$, just offset by gravity in y.

Figure 19: Pen-and-paper derivation of the equations of motion and their analytical solution for the zero-rest-length spring-mass system.

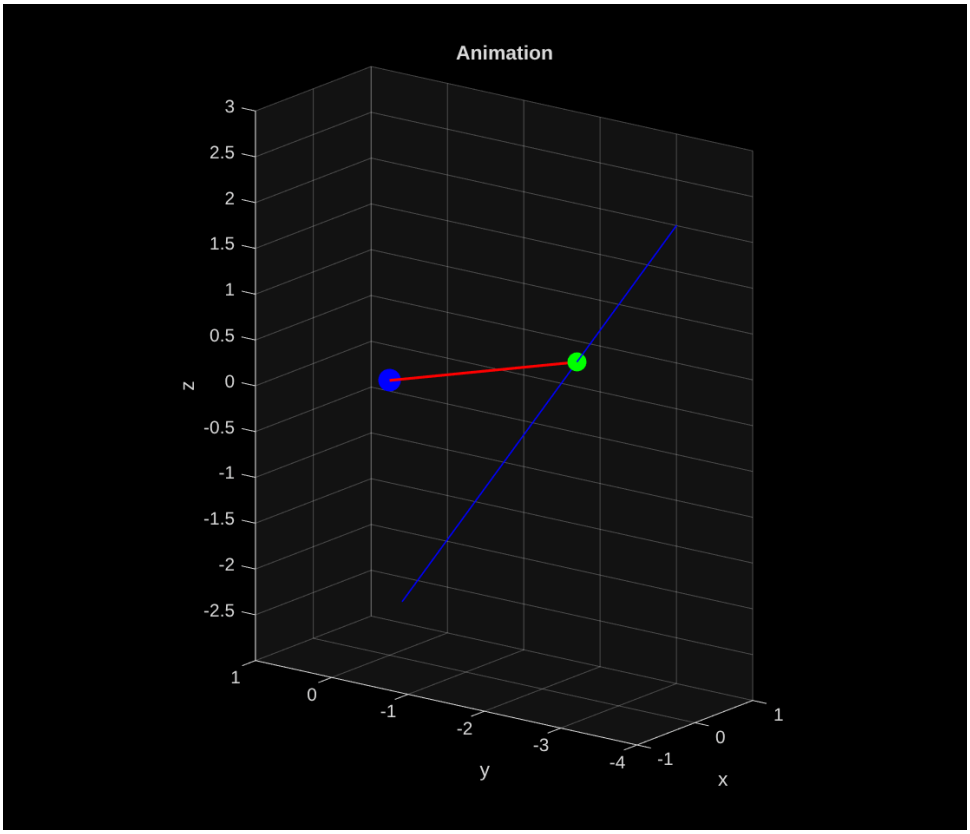


Figure 20: Straight line motion is obtained when all inital velocities are 0.

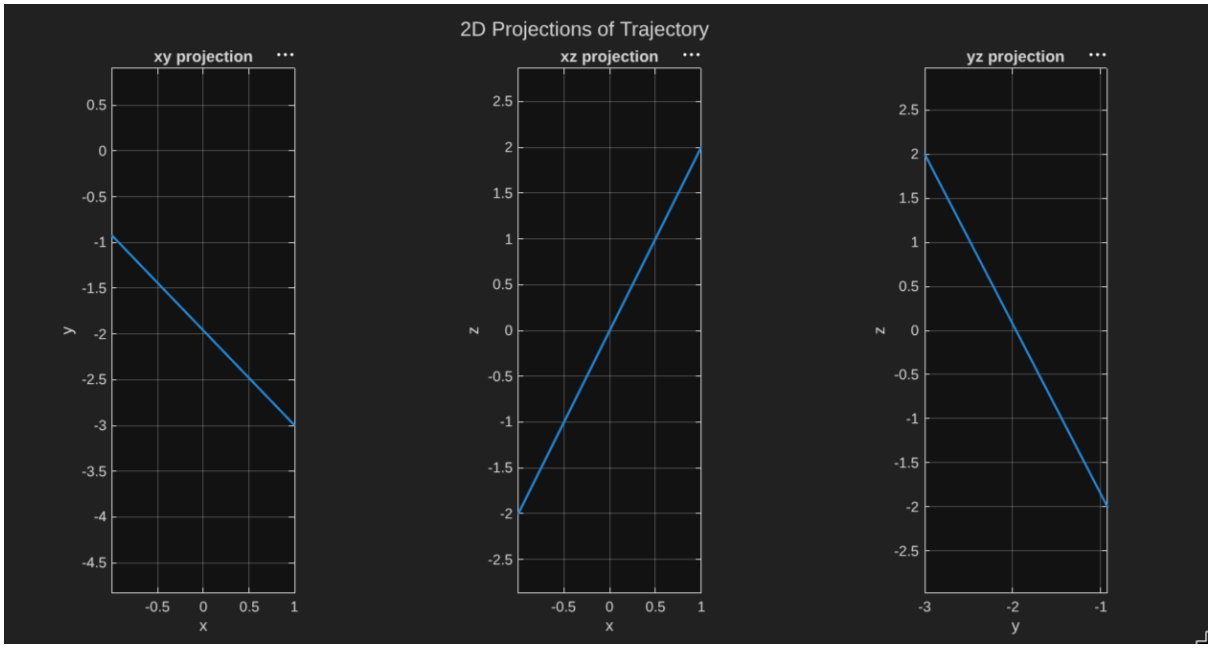


Figure 21: 2D Projectons of the trajectory on each plane

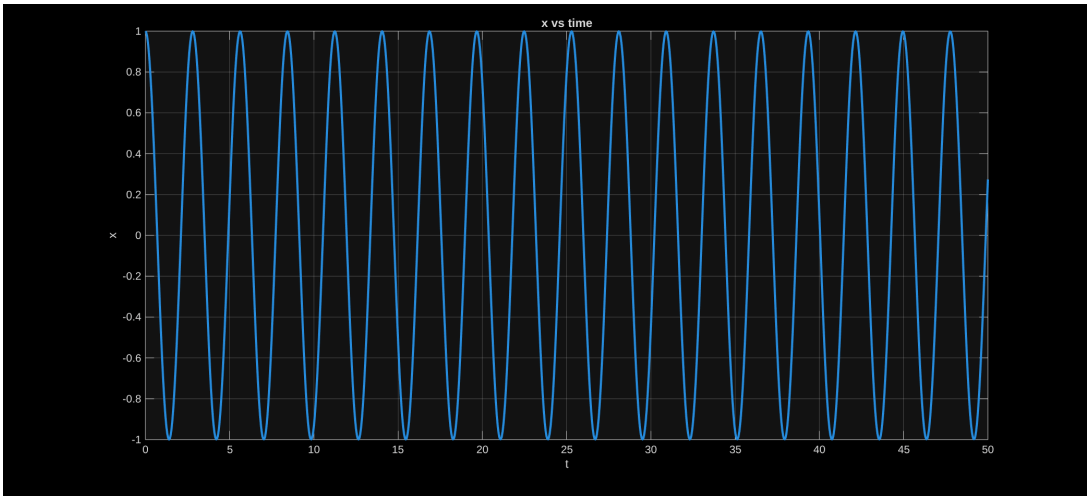


Figure 22: x vs t

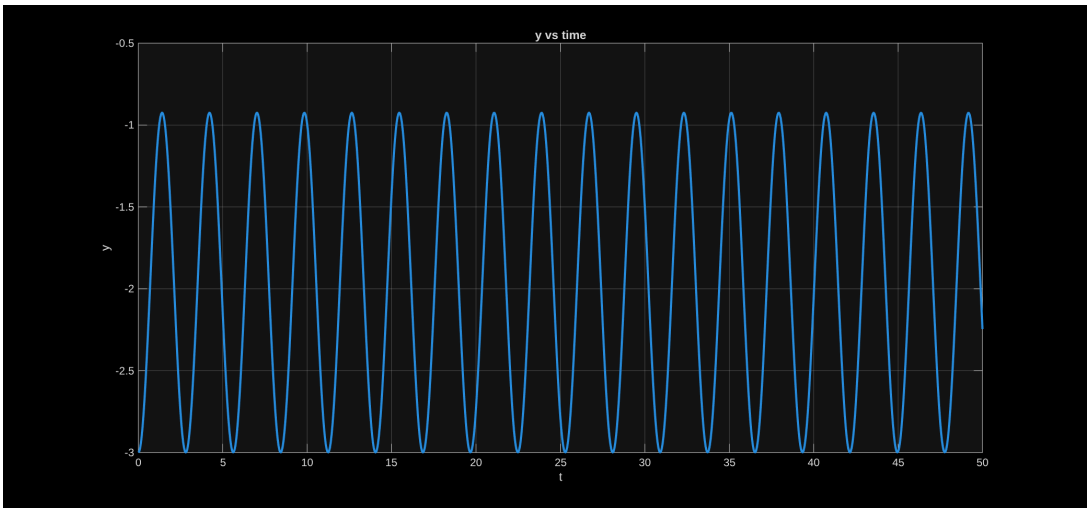


Figure 23: y vs t

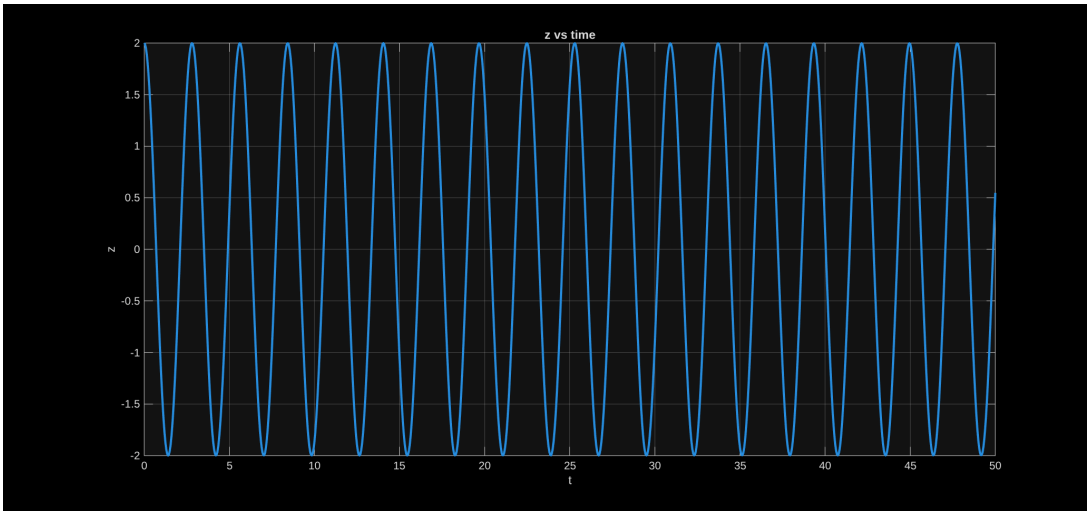


Figure 24: z vs t

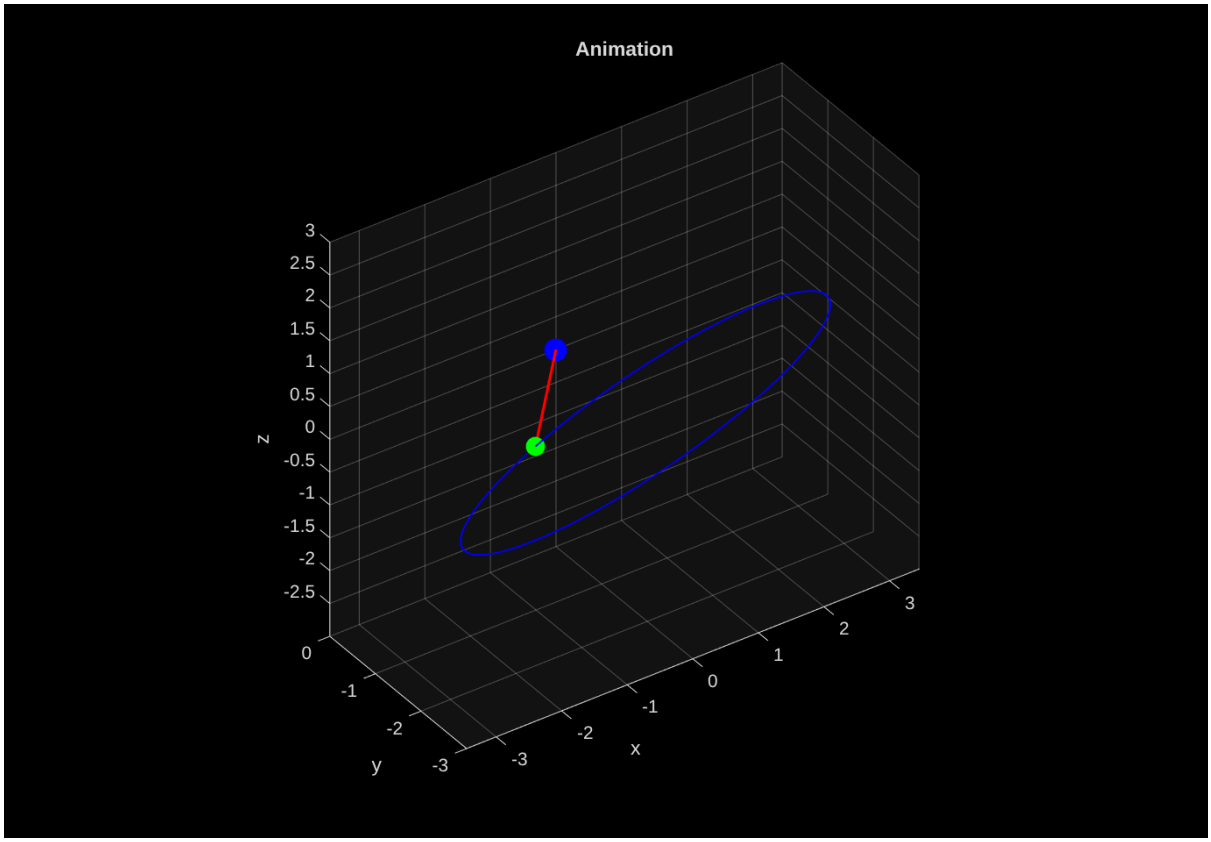


Figure 25: Elliptical path is obtained when initial velocities are non-zero values.

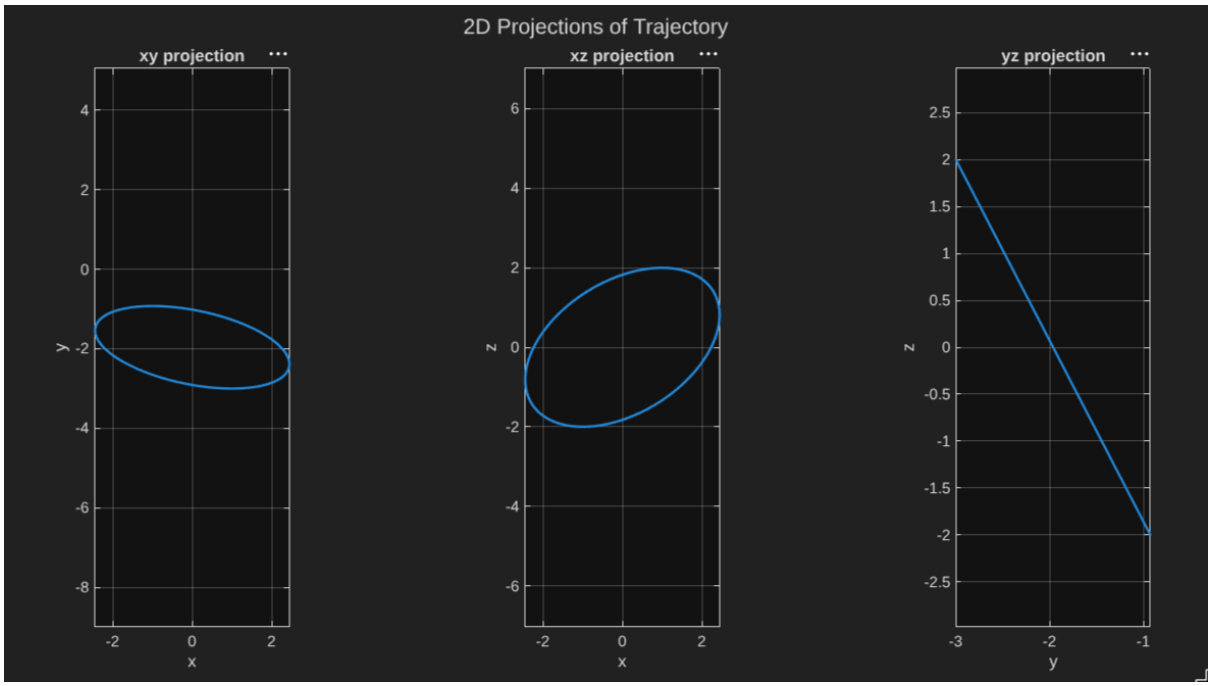


Figure 26: Projections of the trajectory on each plane

3.2 Q13

Problem Statement

Statics vs Dynamics. 2D. Equilibrium and dynamics of a point mass m at $\vec{r}_C$. Gravity g points down in the minus $\hat{j}$ direction. Points A and B are anchored at $\vec{r}_A$ and $\vec{r}_B$. Springs and parallel dashpots connect the mass to A and B. The spring constants are k_A and k_B. Their rest lengths are L_A and L_B. Parallel to the springs are dashpots c_A and c_B. The mass also has a drag force proportional to speed through the air c_d.

The dynamic state of the system is described with $z = [x, y, \dot{x}, \dot{y}]'$ and the static state by $z = [x, y, 0, 0]'$.

(a) **Basic Statics.** Given parameters and an initial guess, find an equilibrium position of this system. If you have an example you like, post it and people can compare solutions.

(b) **Basic Dynamics.** Given parameters and an initial condition, find the motion.

(c) **All equilibria.** Given parameters, attempt to find all equilibria. How many are there? By varying parameters, what is the most and least number of equilibrium points there can be?

(d) **Stability of equilibrium.** Given an equilibrium point, find its stability 2 ways: a) With a dynamic simulation near the equilibrium; and b) looking at the eigenvalues of the matrix describing the linearized equations of motion. Compare.

Make appropriate plots, animations and comparisons.

Interpretation

Solution

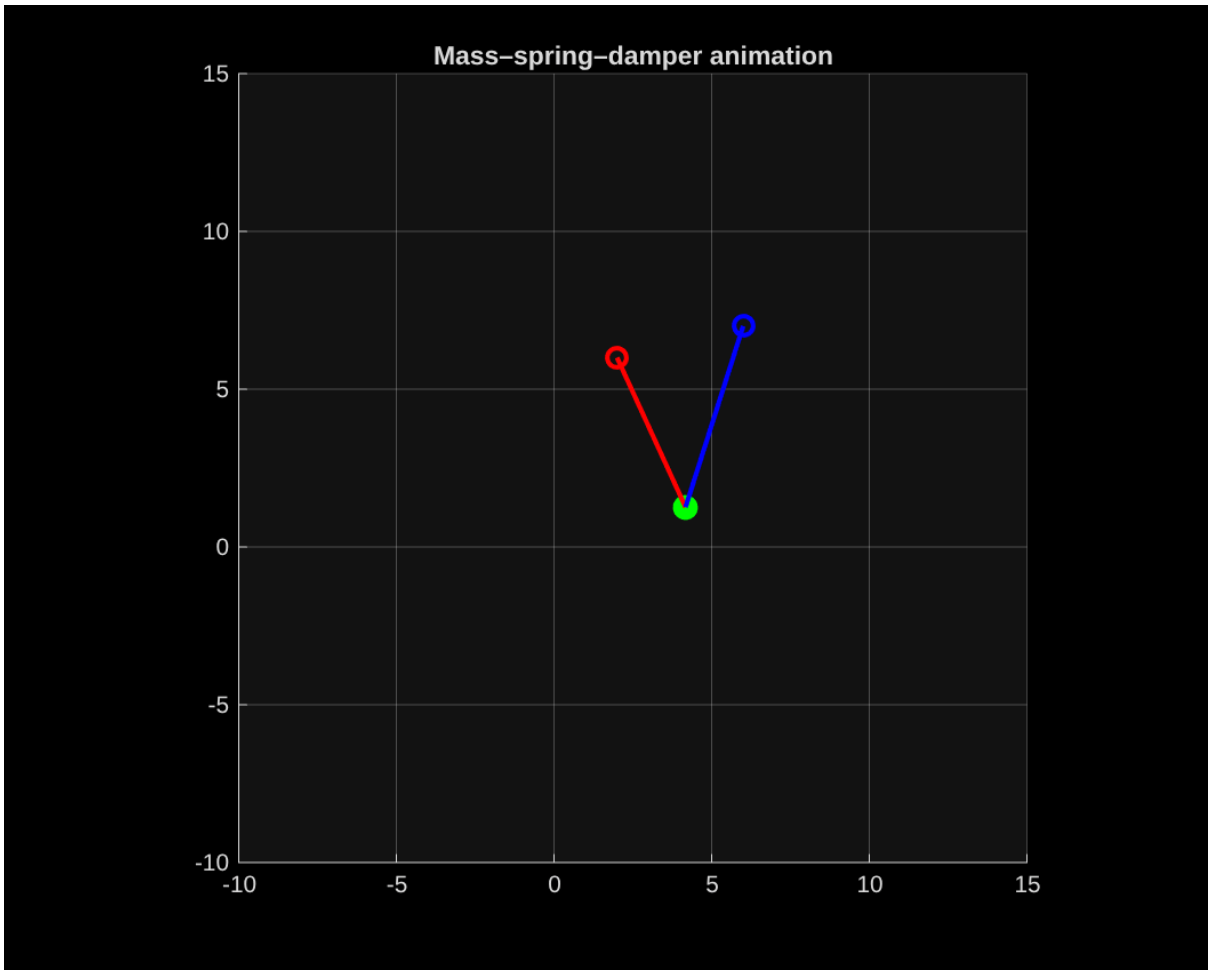


Figure 27: Anchor points are depicted by the red and blue colors. Red Anchor point- (2,6) and Blue anchor point- (6,7). The green ball is the mass 'm'

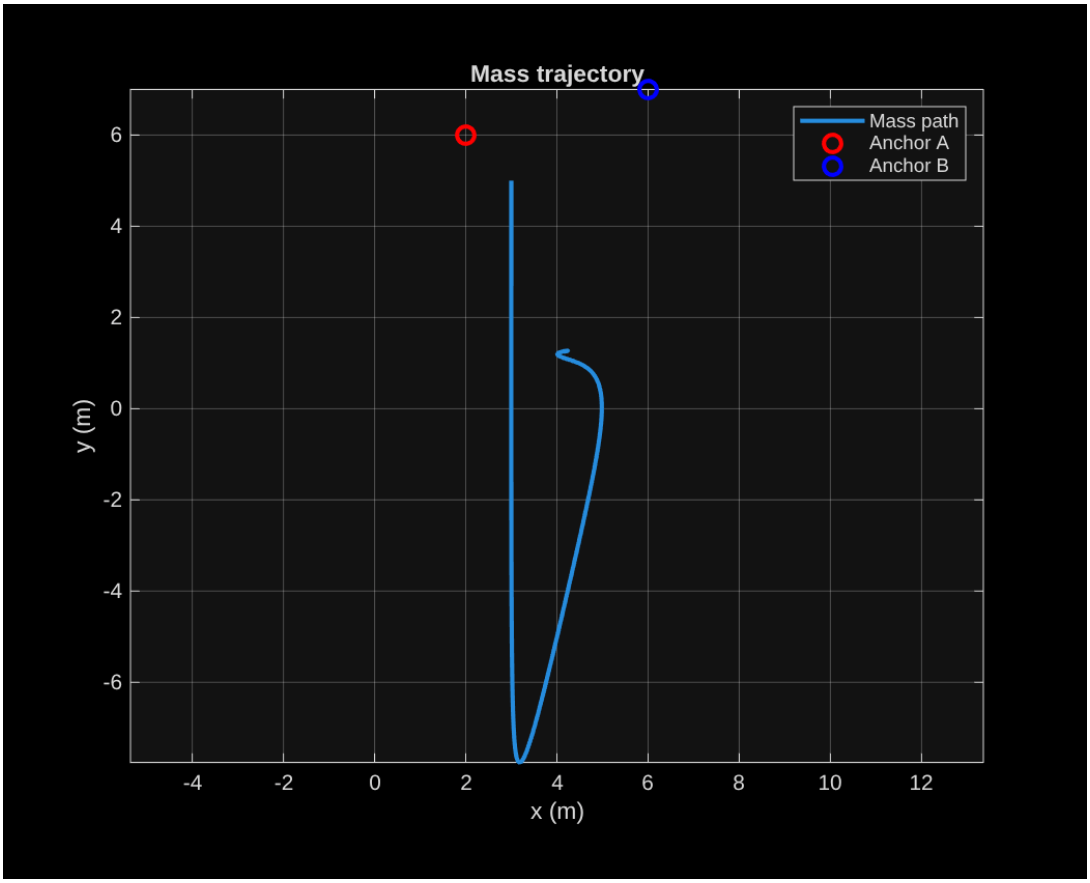
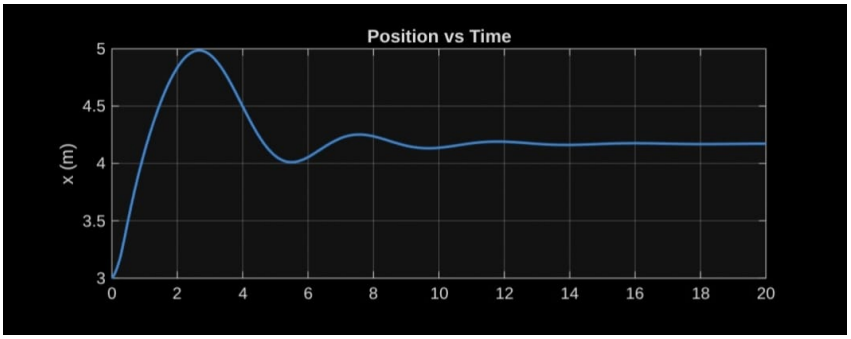
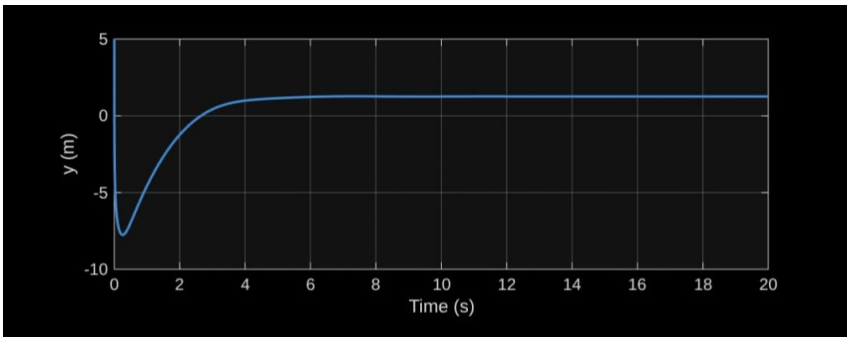


Figure 28: Trajectory of the mass when set to initial conditions $z0 = [x0; y0; vx0; vy0] = [3; 5; -3; -10000]$

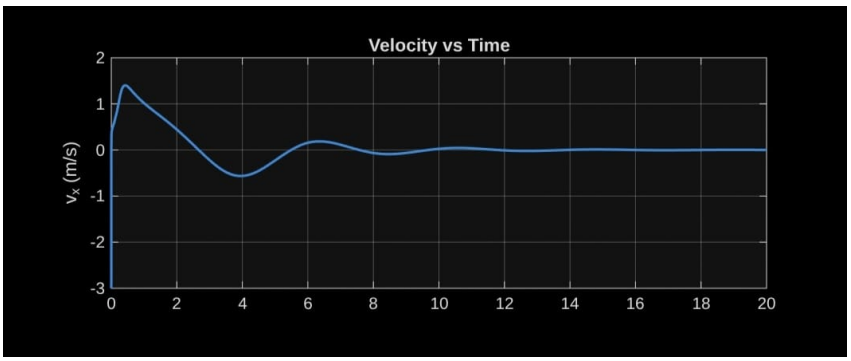


(a)

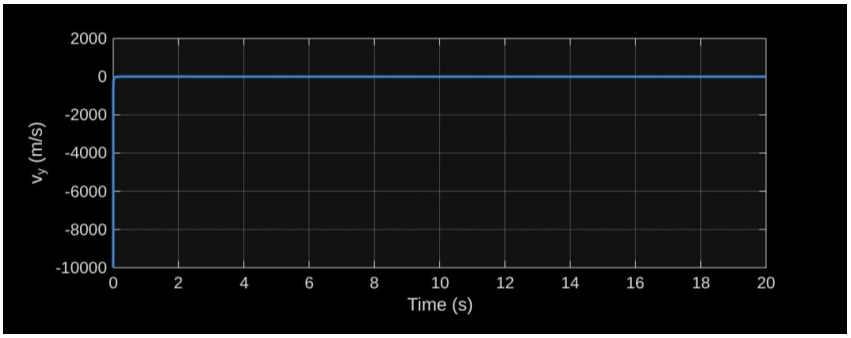


(b)

Figure 29: Position vs time. Top graph shows the variation of the x-coordinate while the bottom graph shows the variation of the y-coordinate with time.



(a)



(b)

Figure 30: Velocity vs time. Top graph shows the variation of the horizontal component of the velocity while the bottom graph shows the variation of the vertical component of the velocity with time.

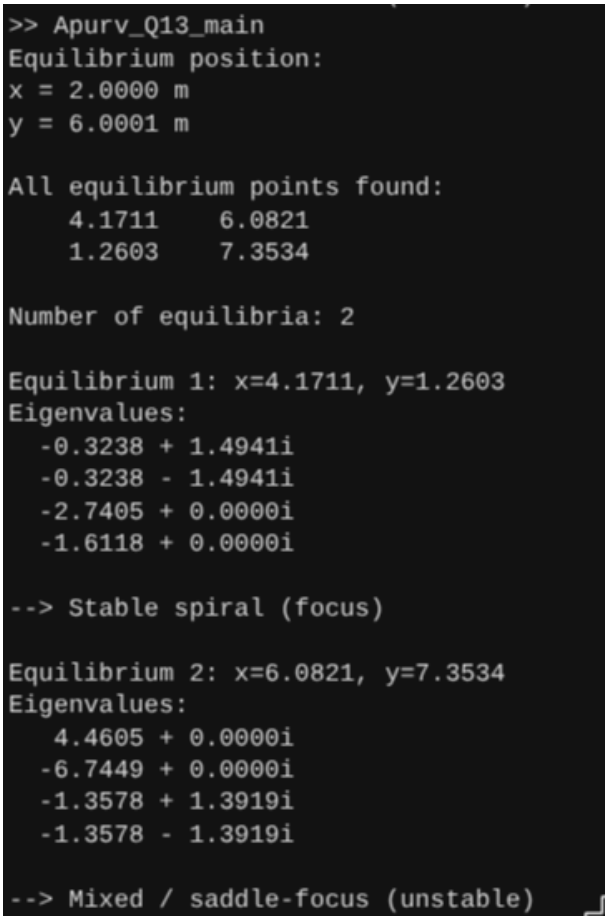
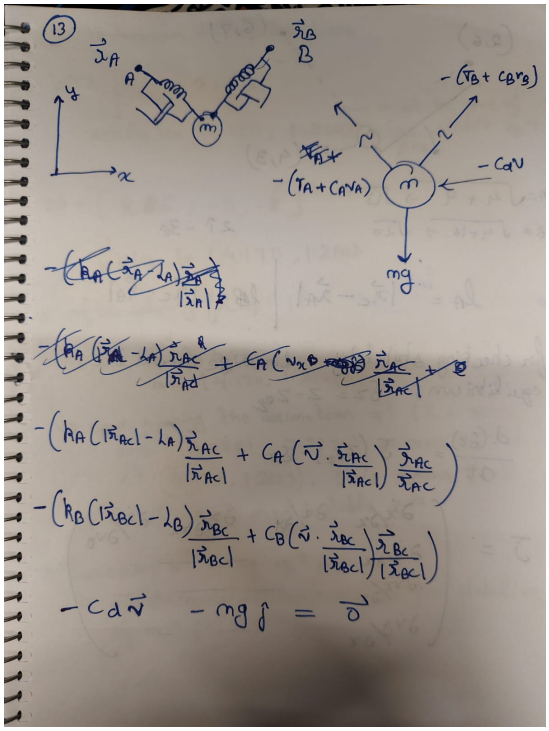


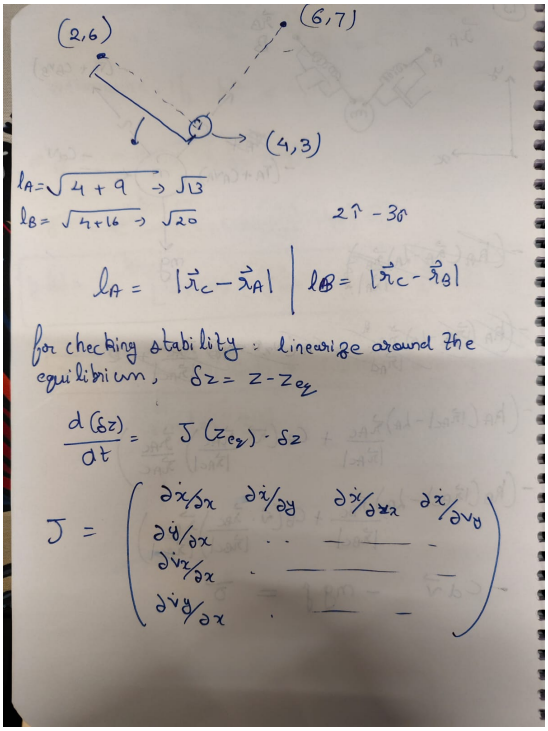
Figure 31: Equilibrium points were calculated using fsolve. Numerical Jacobian was calculated at each point and eigen values were analysed for checking stability of a particular equilibrium point.

Before solving for equilibrium points numerically, multiple runs were done with various initial conditions and every motion resulted in the mass settiling at point (4.1711, 1.2603) meaning that it could be a stable equilibrium point. The remaining 2 points- (2,6.0001) and (6.0821,7.3534) are very close to the anchor points- (2,6) and (6,7).

When tested with those intial conditions, both at 0 initial velocities and very small non-zero intial velocities (little nudge to the mass), the system converge to that stable equilibrium point.



(a) Free body diagram of the mass connected to anchors A and B via springs k_A, k_B (with rest lengths L_A, L_B) and parallel dashpots c_A, c_B, plus air drag c_d and gravity. Assembling the static force-balance equation $\vec{0} = -(k_A(|\vec{r}_{AC}| - L_A)\hat{r}_{AC} + c_A(\vec{v}\cdot\hat{r}_{AC})\hat{r}_{AC}) - (\cdots)_B - c_d\vec{v} - mg\hat{\jmath}$.



(b) Working out ℓ_A and ℓ_B for a chosen anchor geometry ($A = (2,6)$, $B = (6,7)$), then setting up the linearization $\delta\dot{z} = J(z_{eq})\cdot\delta z$ about an equilibrium point to check stability from the Jacobian's eigenvalues.

Figure 32: Pen-and-paper setup of the static equilibrium equation and the Jacobian-linearization approach used to classify equilibrium stability.

4 Homework 5

4.1 Q11

Problem Statement

Periodic orbits under a general central force. Aside from the inverse-square gravitational force and the linear Hooke's-law spring force, both of which are known (Bertrand's theorem) to produce closed elliptical orbits, consider a different central force law $F(r)$. Find initial conditions that produce periodic motion, meaning $r(t)$ oscillates between a fixed minimum and maximum distance forever, even if the trajectory itself does not close into a repeating shape. Use `fsolve` to find the initial condition that gives a desired periapsis (closest approach) distance, and confirm periodicity with suitable plots.

Interpretation

The central force used was $F(r) = -kr^3$ (a "hardening spring," growing stronger with distance rather than weaker). Since Bertrand's theorem rules out closed non-circular orbits for any force law other than $F \propto -1/r^2$ and $F \propto -r$, the resulting orbit is expected to be a precessing rosette: periodic in $r(t)$, but not closed in the (x, y) plane. An initial condition was placed at the (chosen) apoapsis $r_0 = 3$ with a purely tangential velocity, and `fsolve` was used to find the tangential speed v_{y0} that makes the orbit's periapsis land exactly at a target value, $r_{\min} = 1$.

Solution

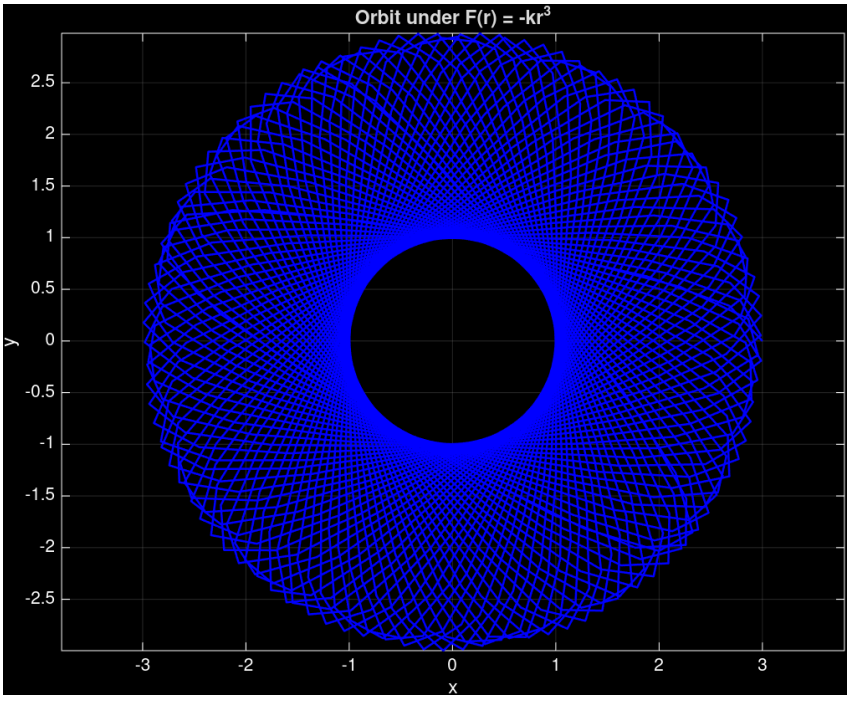


Figure 33: Trajectory under $F(r) = -kr^3$ for the solved initial condition. The orbit forms a precessing rosette rather than a closed curve, bouncing between an inner radius of 1 and an outer radius of 3.

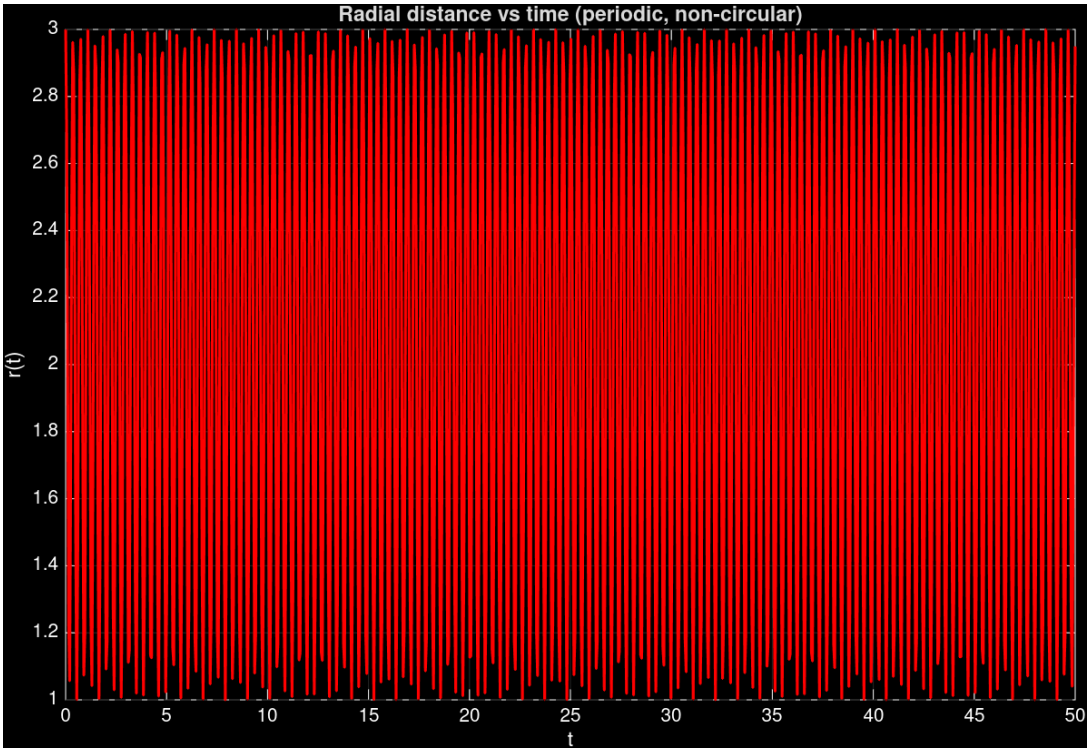


Figure 34: Radial distance $r(t)$ vs. time. Although the orbit above does not close, $r(t)$ is genuinely periodic, oscillating cleanly between the target periapsis ($r = 1$) and the starting apoapsis ($r = 3$).

4.2 Q12

Problem Statement

Cannon ball with quadratic drag. Repeat the cannon ball problem (Q9), but replace the linear viscous drag with a quadratic (speed-squared) drag force, $\vec{F}_d = -c|\vec{v}|\vec{v}$.

(a) Write the equations of motion. Note that, unlike the linear-drag case, the x- and y-equations are now coupled through the speed $|\vec{v}| = \sqrt{v_x^2 + v_y^2}$, so no closed-form analytical solution exists.

(b) Solve numerically, and plot trajectories for a range of increasing launch speeds v_0, using equal-scaled axes.

(c) Numerically verify energy conservation/dissipation: integrate the work done by drag ($\dot{W} = P$) as an additional state variable alongside the equations of motion, and compare $W(t)$ to the independently-computed change in total mechanical energy $\Delta E(t)$. Show, with a suitable plot, that the discrepancy between the two shrinks to zero as the numerical integration is made more accurate.

Interpretation

Since quadratic drag couples v_x and v_y nonlinearly, there is no analytical trajectory to check against (unlike the linear-drag case in Q9). Instead, the check used here is a conservation law: physics guarantees that the energy lost by the system, ΔE, must exactly equal the work done by drag, $W = \int P\,dt$ where $P = \vec{F}_d \cdot \vec{v} = -c|\vec{v}|^3$. These two quantities were computed via two independent numerical paths (one directly from the simulated state, one by integrating

power as its own state variable), so their disagreement is a self-contained, honest measure of numerical error that requires no known exact solution.

Solution

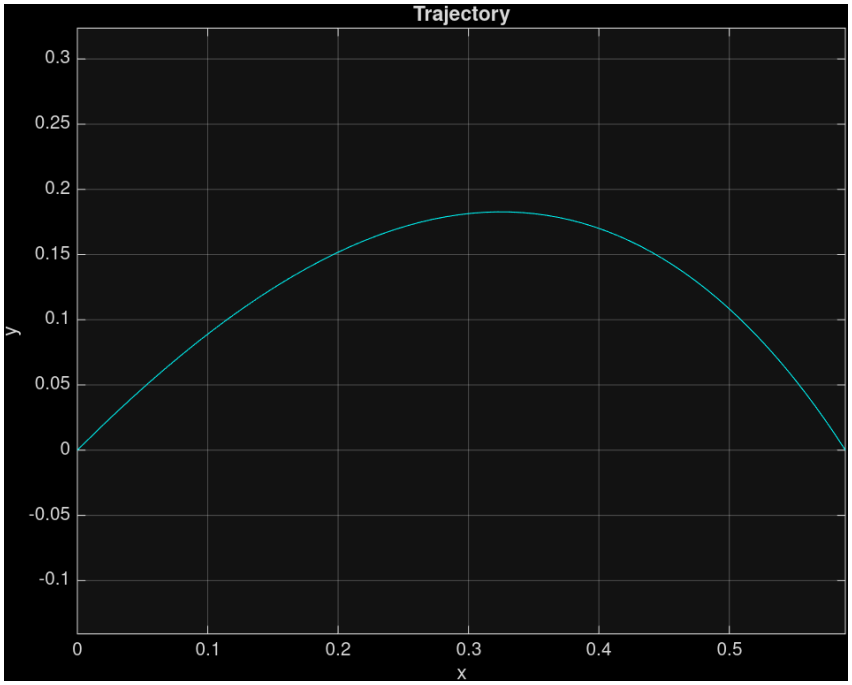


Figure 35: Trajectory under quadratic drag, using the same parameters as Q9 ($\theta = \pi/4$, $v_0 = 1$ m/s, $g = 1$ m/s^2, $m = 1$ kg, $c = 1$ kg/s).

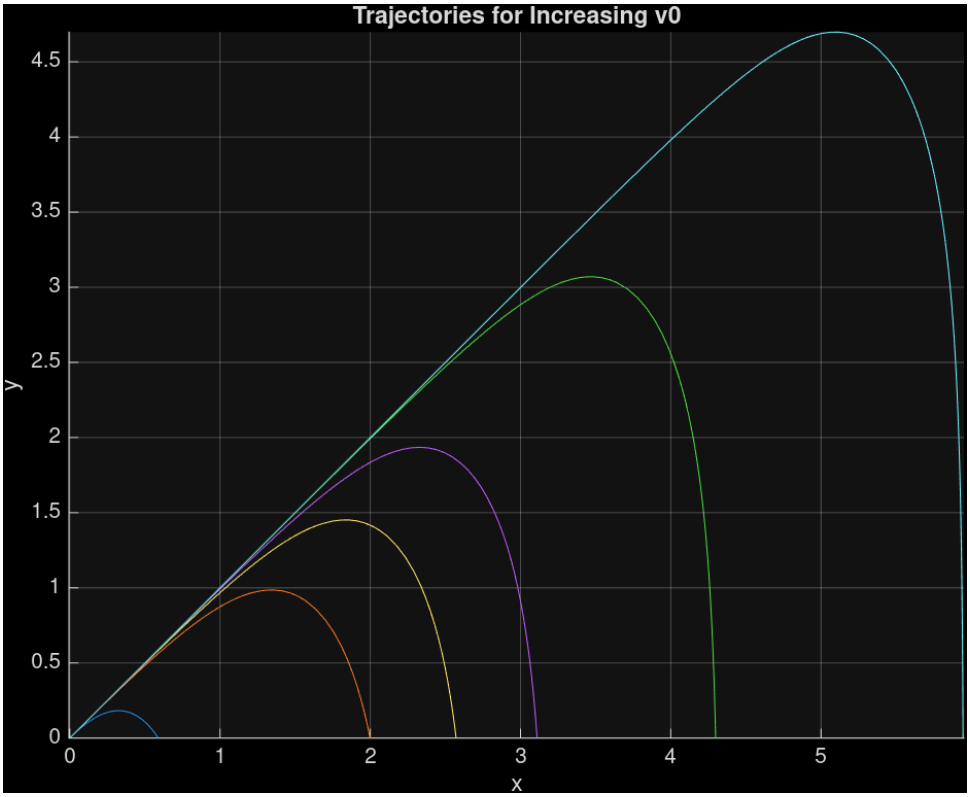


Figure 36: Trajectories for increasing launch speed v_0 (equal-scaled axes). At higher v_0, the early part of the flight approaches a straight line before drag-dominated curvature bends the trajectory back to the ground.

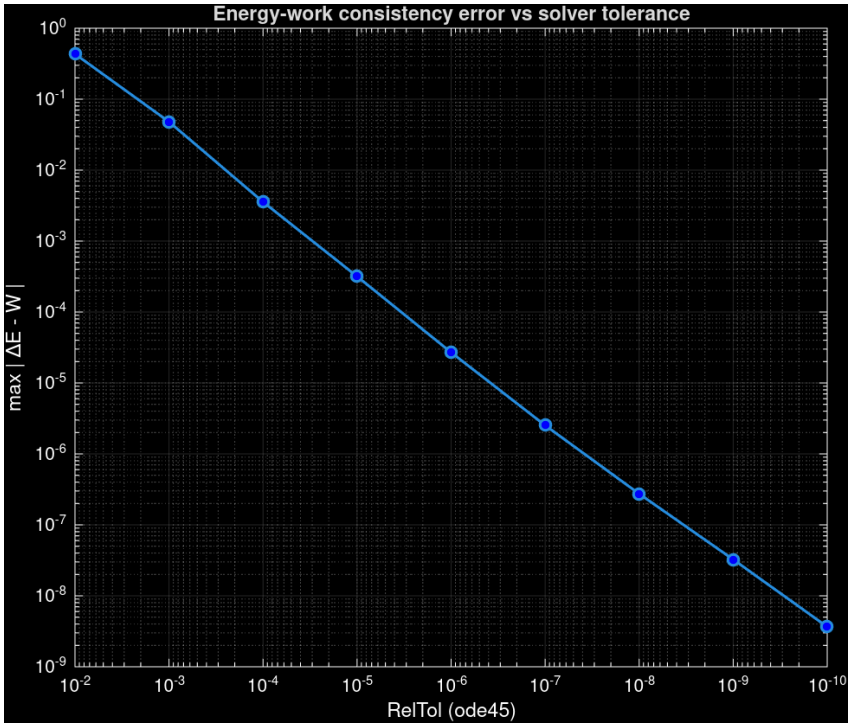


Figure 37: Energy-work consistency check: the maximum discrepancy between the change in total mechanical energy (ΔE) and the independently integrated work done by drag (W) drops by roughly eight orders of magnitude as the `ode45` tolerance is tightened from 10^{-2} to 10^{-10}, confirming both quantities converge to the same true answer.

4.3 Q14

Problem Statement

Angular momentum about a moving point. Let a particle P of mass m move relative to a fixed origin O, and let C be a reference point that may itself be moving. Define the moment of momentum of P about C (using velocity relative to the fixed origin) as $\vec{H}_{/C} \equiv \vec{r}_{P/C} \times m\vec{v}_{P/O}$.

(a) Derive $\dot{\vec{H}}_{/C}$ in general.

(b) Show that $\dot{\vec{H}}_{/C}$ reduces to the moment about C of the force ($\vec{r}_{P/C} \times m\vec{a}_{P/O}$) only under a special condition on the motion of C relative to P; find that condition.

(c) Repeat parts (a)-(b) using the moment of momentum defined with velocity *relative to* C instead, $\vec{N}_{/C} \equiv \vec{r}_{P/C} \times m\vec{v}_{P/C}$, and find the analogous condition under which $\dot{\vec{N}}_{/C}$ reduces to the moment of the force about C.

Interpretation

This is the standard subtlety in rigid-body / particle dynamics of choosing a moving reference point for angular momentum: $\dot{\vec{H}}_{/C}$ picks up an extra term from C's own motion, and that extra term only vanishes under a specific geometric condition, rather than for every choice of C.

Solution

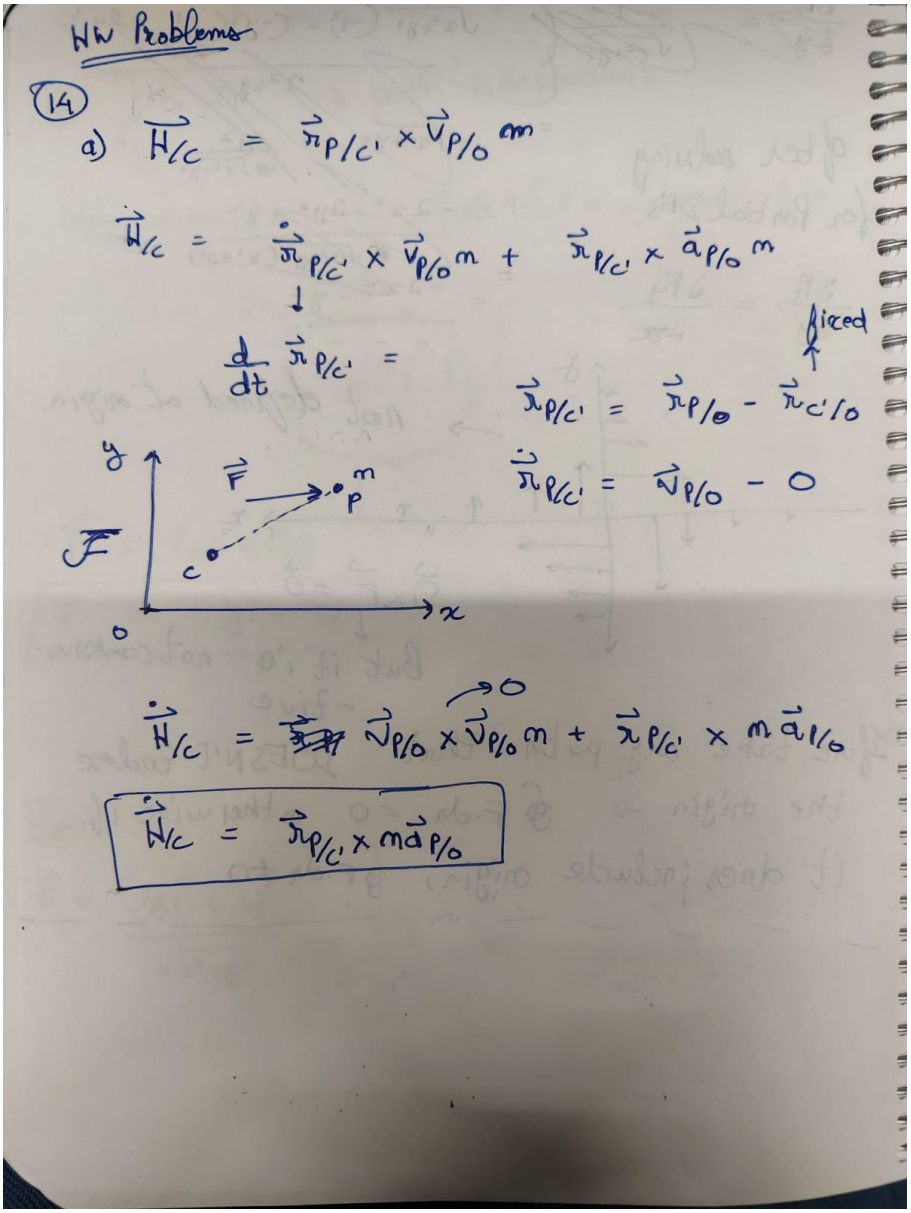


Figure 38: (a) Setting up $\vec{H}_{/C} = \vec{r}_{P/C} \times m\vec{v}_{P/O}$ for a particle P acted on by force $\vec{F}$, with C a fixed point at this stage, and differentiating using $\dot{\vec{r}}_{P/C} = \vec{v}_{P/O} - \vec{v}_{C/O}$.

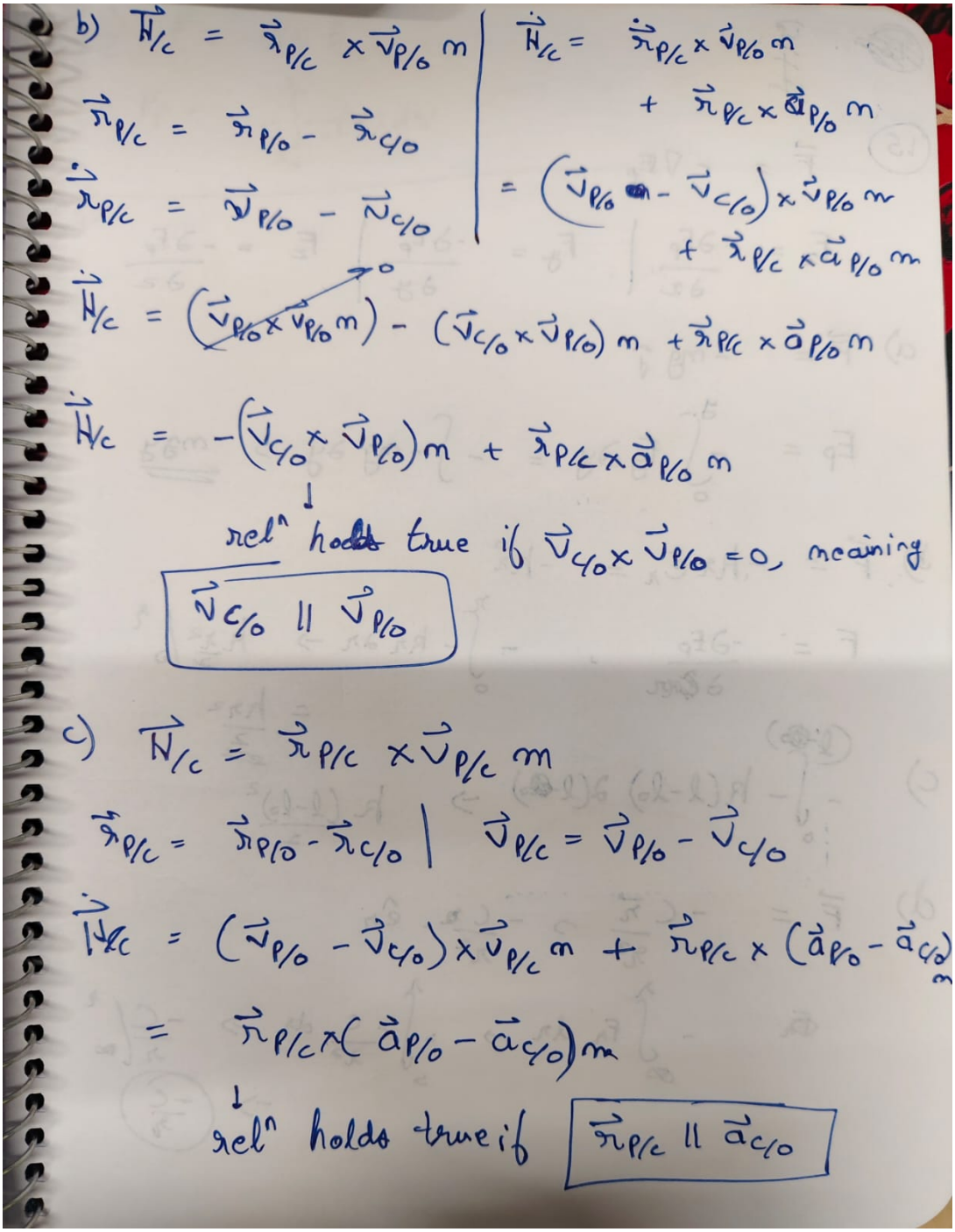


Figure 39: (b) Completing the derivative to get $\dot{\vec{H}}_{/C} = -(\vec{v}_{C/O} \times \vec{v}_{P/O})m + \vec{r}_{P/C} \times m\vec{a}_{P/O}$, showing this reduces to $\vec{r}_{P/C} \times m\vec{a}_{P/O}$ only when $\vec{v}_{C/O} \parallel \vec{v}_{P/O}$. (c) Repeating with $\vec{N}_{/C} = \vec{r}_{P/C} \times m\vec{v}_{P/C}$, showing the analogous simplification $\dot{\vec{N}}_{/C} = \vec{r}_{P/C} \times m(\vec{a}_{P/O} - \vec{a}_{C/O})$ holds when $\vec{r}_{P/C} \parallel \vec{a}_{C/O}$.

4.4 Q15

Problem Statement

Potential energy from a conservative force, $\vec{F} = -\nabla E_p$, so that $F_x = -\partial E_p/\partial x$, $F_y = -\partial E_p/\partial y$, $F_z = -\partial E_p/\partial z$. Find E_p for each of the following forces:

(a) Uniform gravity, $\vec{F} = -mg\,\hat{\jmath}$.

(b) A linear spring through the origin with zero rest length, $\vec{F} = -k\vec{r}$.

(c) A linear spring with rest length ℓ_0, magnitude $F = -k(\ell - \ell_0)$ along the spring's direction.

(d) An inverse-square attractive central force, $\vec{F} = -\dfrac{c}{r^2}\hat{e}_r$.

(e) A general central force, $\vec{F} = -f(r)\,\hat{e}_r$.

Interpretation

Each case follows directly from integrating the defining relation $F = -\partial E_p/\partial(\text{coordinate})$ along the relevant direction; part (e) generalizes all of (a)-(d) into a single formula, from which each earlier result can be recovered as a special case of $f(r)$.

Solution

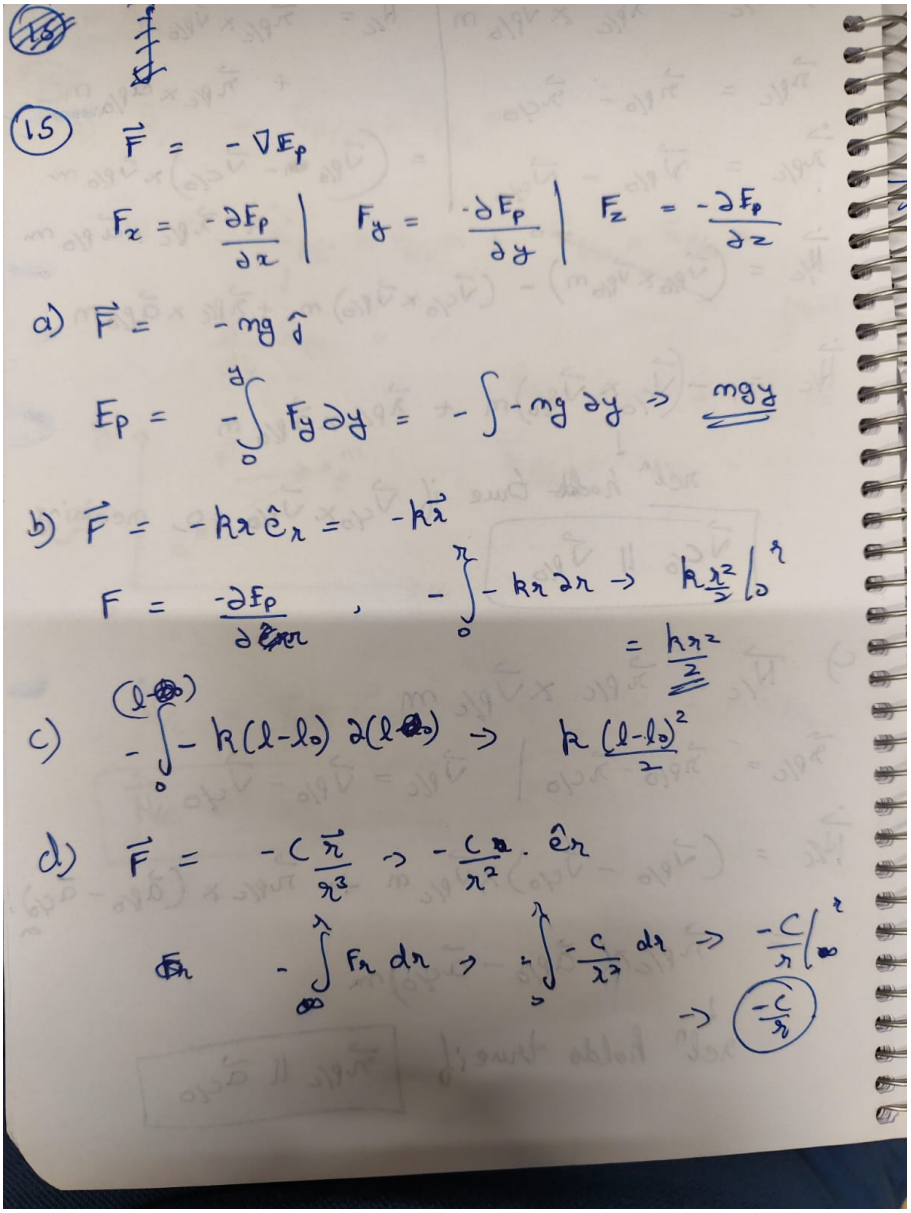


Figure 40: Deriving E_p from $\vec{F} = -\nabla E_p$ for: (a) uniform gravity, giving $E_p = mgy$; (b) a spring through the origin, giving $E_p = \frac{1}{2}kr^2$; (c) a spring with rest length ℓ_0, giving $E_p = \frac{1}{2}k(\ell - \ell_0)^2$; (d) an inverse-square attractive force, giving $E_p = -c/r$.

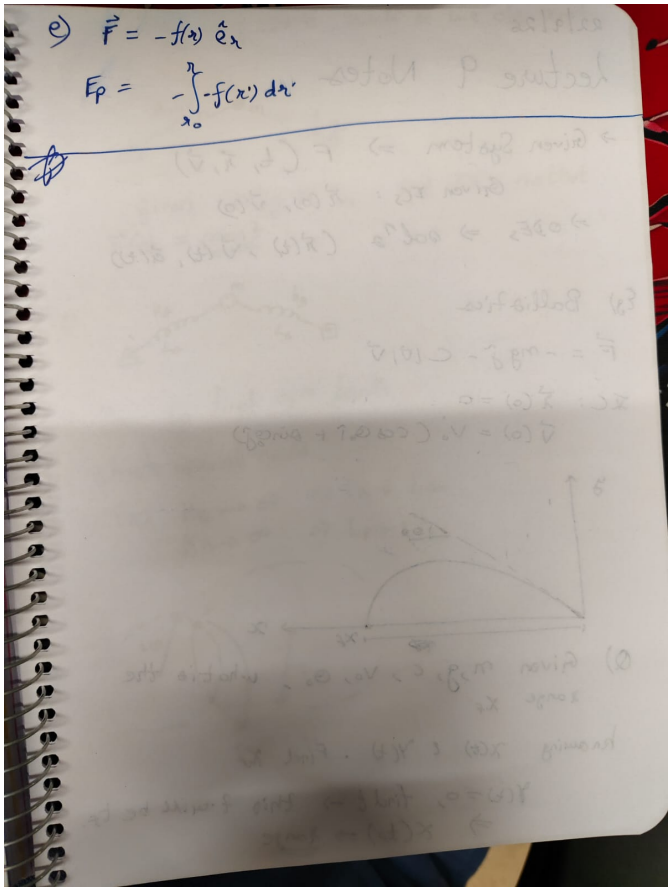


Figure 41: (e) Generalizing to an arbitrary central force $\vec{F} = -f(r)\,\hat{e}_r$, giving $E_p(r) = -\int_{r_0}^{r} f(r')\,dr'$, of which parts (a)-(d) are all special cases.

5 Homework- Week 7

5.1 Q16

Problem Statement

Mechanics of two or more particles.

(a) For two unequal particles with mass m_1 and m_2 what is the period of circular motion if the distance between the particles is d and the only force is the force between them from classical inverse-square Newtonian gravity,

$$F = \frac{Gm_1m_2}{r^2}?$$

(b) Pick numbers for G, m_1, m_2 and r and, using appropriate initial conditions, test your analytical result with a numerical simulation. Make any plots needed to make the agreement of your numerical and analytical results convincing.

(c) For three equal particles, $m_1 = m_2 = m_3 = 1$ and $G = 10$ what is the angular speed for circular motion on a circle with diameter of $d = 3$?

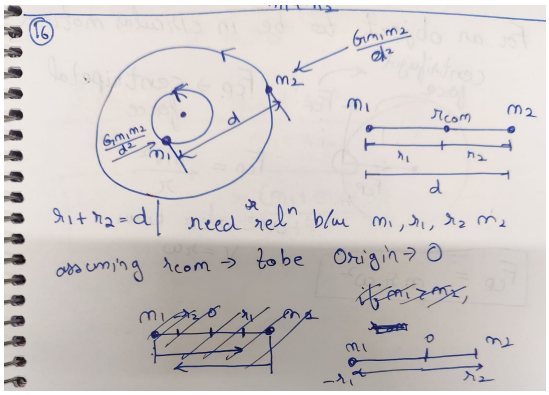
(d) Check your result for three particles with a numerical simulation, as you did for 2 particles.

Interpretation

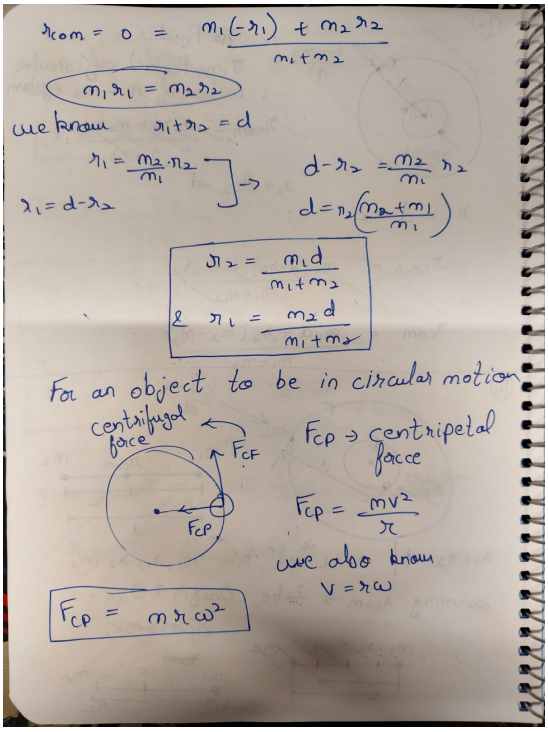
The problem is divided into two parts. First, the circular motion of two unequal masses is analysed by considering their motion about the centre of mass and equating the gravitational force to the required centripetal force. The resulting analytical expression is then checked numerically.

The same approach is then extended to three equal masses arranged symmetrically on a circle. By symmetry, the resultant gravitational force on each mass is directed towards the centre of the circle, allowing the required angular speed to be determined.

Solution

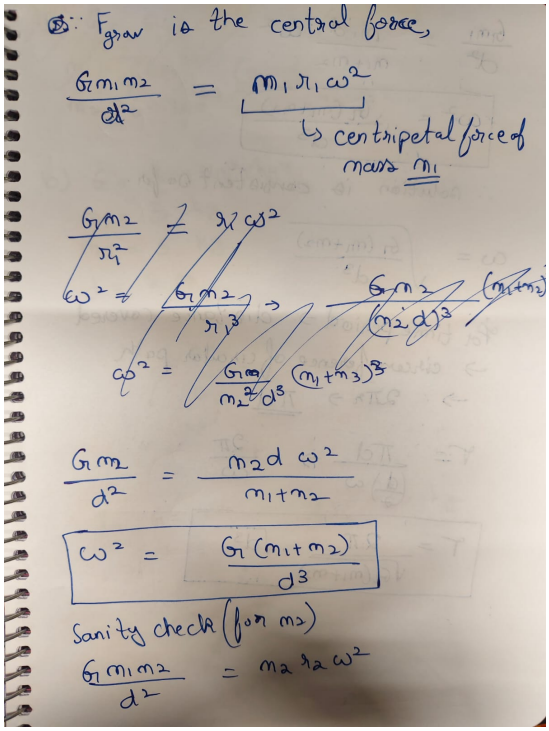


(a) Setting up the two-body system and locating the centre of mass for two unequal masses separated by a distance d.

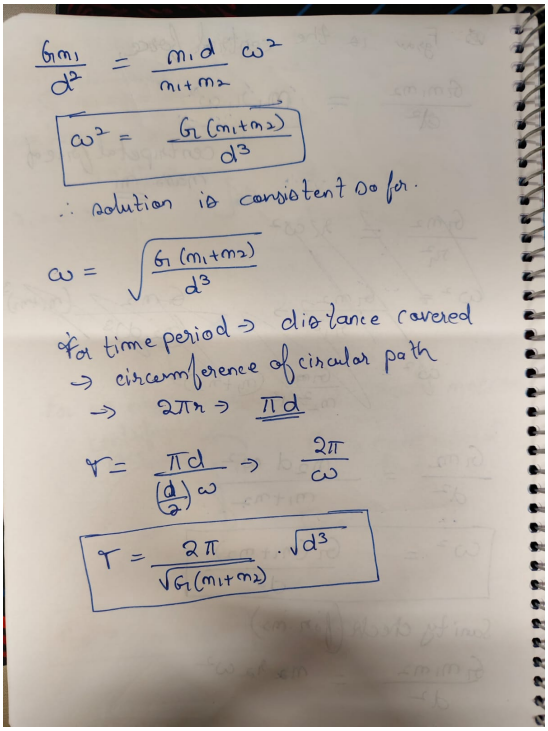


(b) Relating the distances of the two masses from the centre of mass to their respective masses and setting up the centripetal-force condition.

Figure 42: Initial setup for the two-body circular-motion problem.

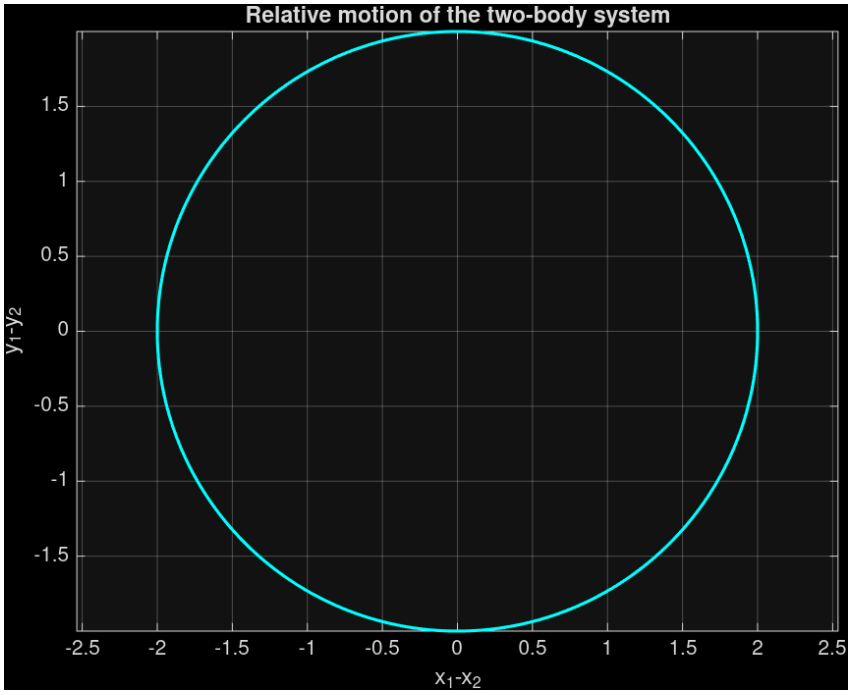


(a) Equating the gravitational force between the two masses to the required centripetal force and solving for the angular speed.

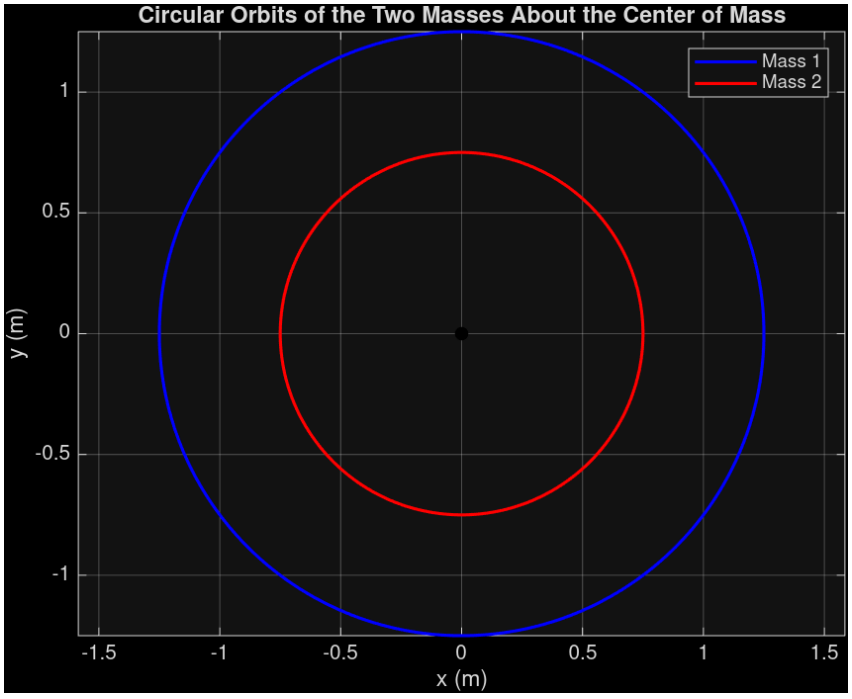


(b) Obtaining the period of circular motion from the derived angular speed.

Figure 43: Analytical derivation of the angular speed and period for the two-body system.

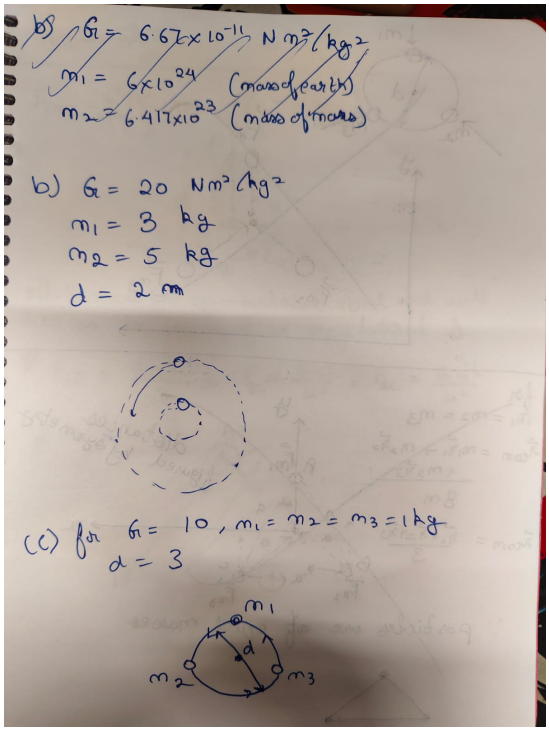


(a) Relative motion of the two-body system. The relative coordinate follows a circular trajectory.

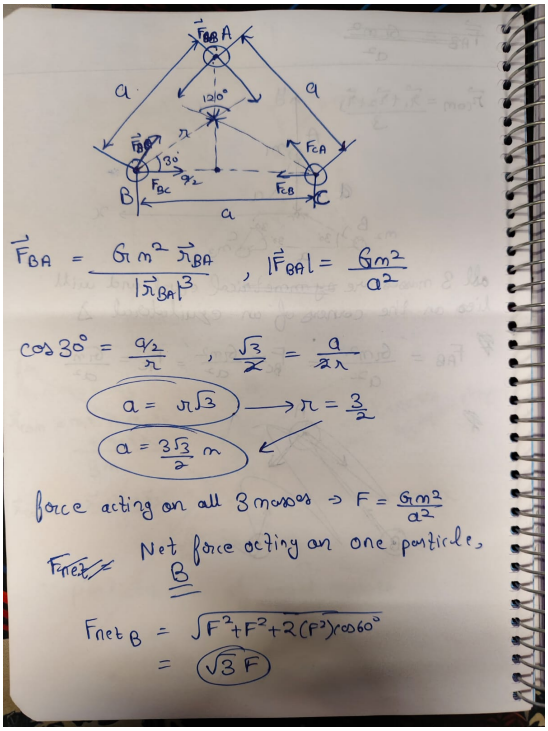


(b) Individual circular trajectories of the two masses about the centre of mass.

Figure 44: Numerical verification of the circular motion of the two-body system.

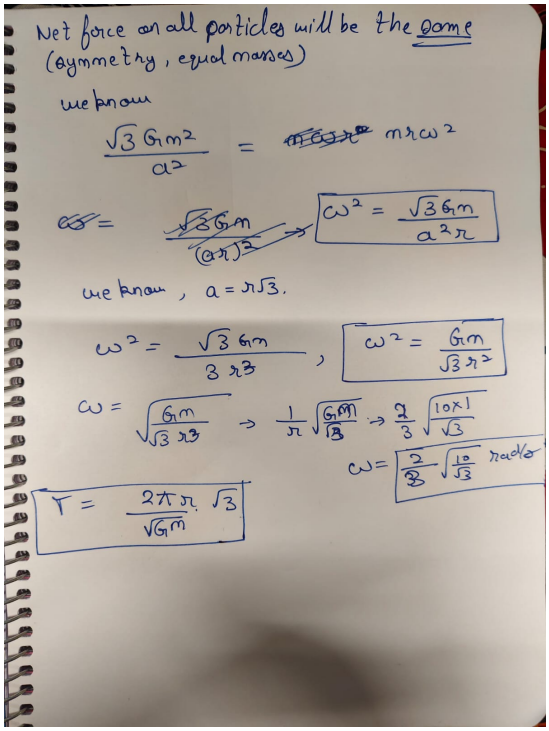


(a) Setting up the three-body problem for three equal masses arranged symmetrically on a circle of diameter $d = 3$.

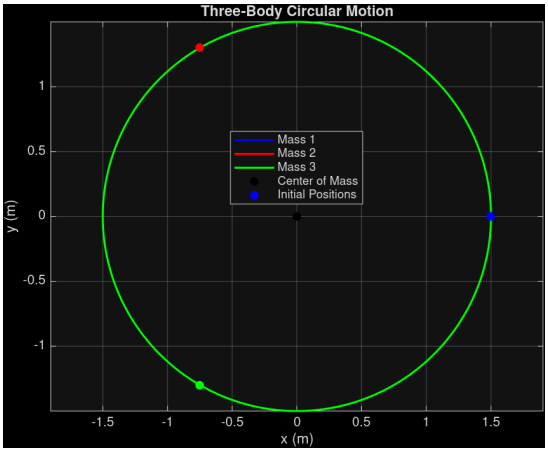


(b) Resolving the gravitational forces acting on one of the particles. By symmetry, the resultant force is directed towards the centre of the circle.

Figure 45: Force analysis for the three equal-mass circular system.



(a) Equating the resultant gravitational force to the centripetal force and obtaining the angular speed for the three-body system.



(b) Numerical simulation of the three-body circular motion. The three masses maintain their relative positions while rotating about the centre of mass.

Figure 46: Analytical and numerical verification of the three-body circular motion.

5.2 Q22

Problem Statement

Simple pendulum. Derive the simple pendulum equation

$$\ddot{\theta} + \frac{g}{\ell} \sin\theta = 0$$

as many ways as you can without looking anything up in books. For example, in all cases using polar coordinates:

(a) Linear momentum and manipulate the equations to eliminate the constraint force.

(b) Linear momentum, dot with $\hat{e}_\theta$.

(c) Linear momentum, cross with $\vec{r}$.

(d) Angular momentum.

(e) Conservation of energy.

(f) Power balance.

(g) Lagrange equations (if you know them already, not if you don't).

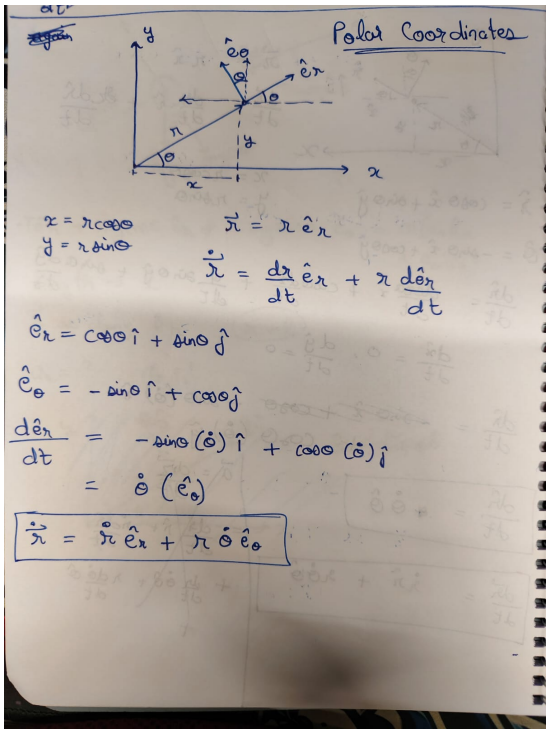
Interpretation

The simple pendulum is analysed using several different formulations of mechanics. Although the starting point is different in each method, all approaches should lead to the same nonlinear equation of motion,

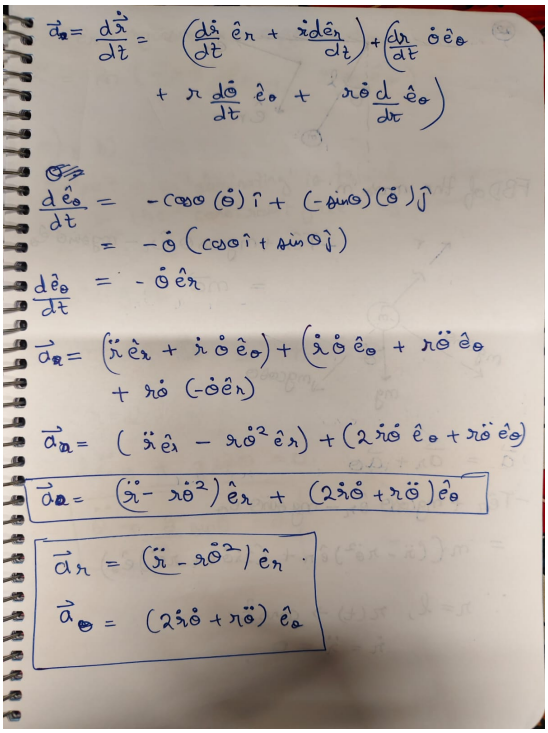
$$\ddot{\theta} + \frac{g}{\ell} \sin\theta = 0.$$

The first three approaches begin with linear momentum, while the remaining approaches use angular momentum, conservation of energy, power balance, and Lagrange equations.

Solution

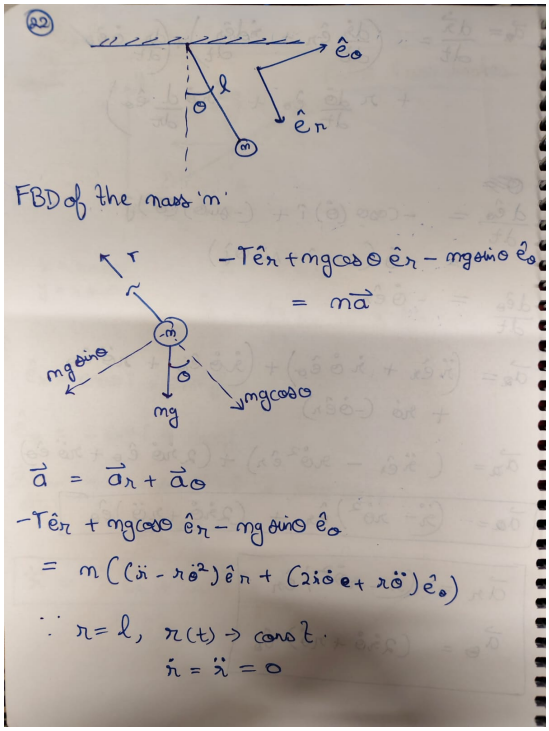


(a) Setting up the simple pendulum using polar coordinates and expressing the position and velocity of the mass in terms of θ.

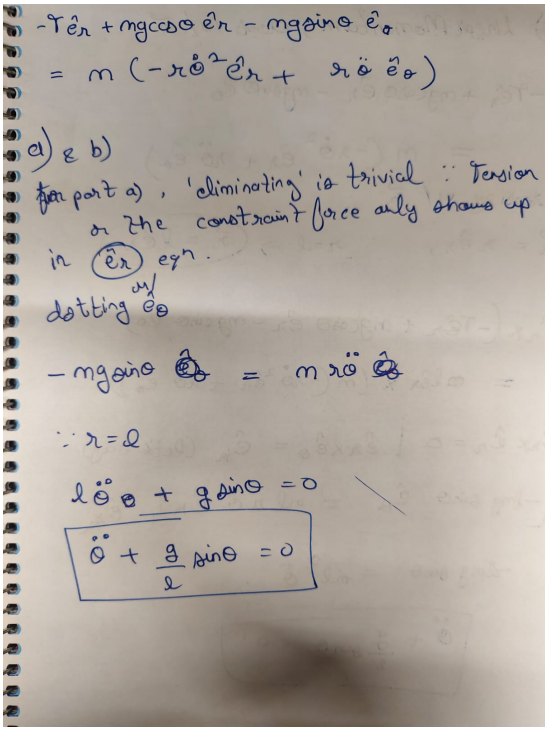


(b) Differentiating the polar-coordinate expressions to obtain the velocity and acceleration of the pendulum mass.

Figure 47: Polar-coordinate setup and kinematic relations for the simple pendulum.

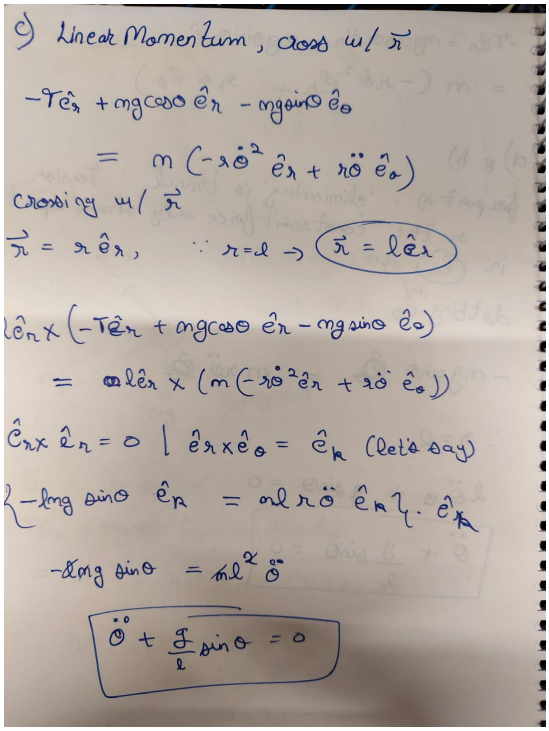


(a) Free-body diagram of the pendulum mass and application of linear momentum balance to obtain the equations in the radial and tangential directions.



(b) Dotting the linear momentum equation with $\hat{e}_\theta$ to eliminate the constraint tension and obtain the simple pendulum equation.

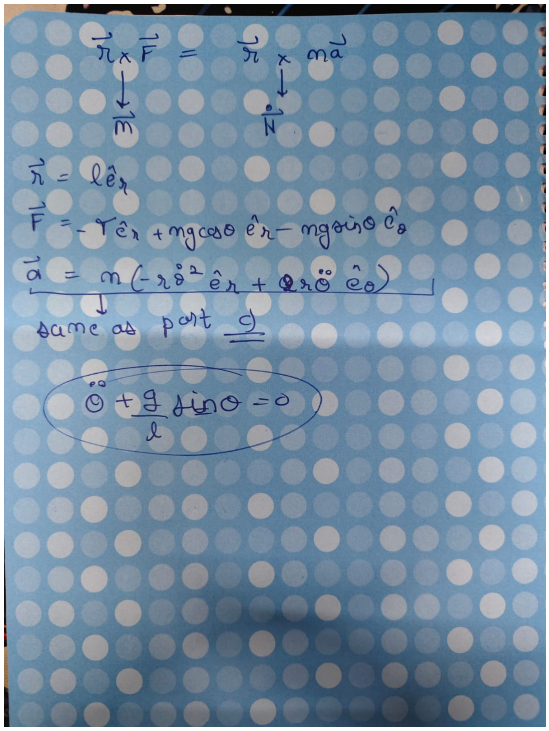
Figure 48: Derivation of the pendulum equation using linear momentum.



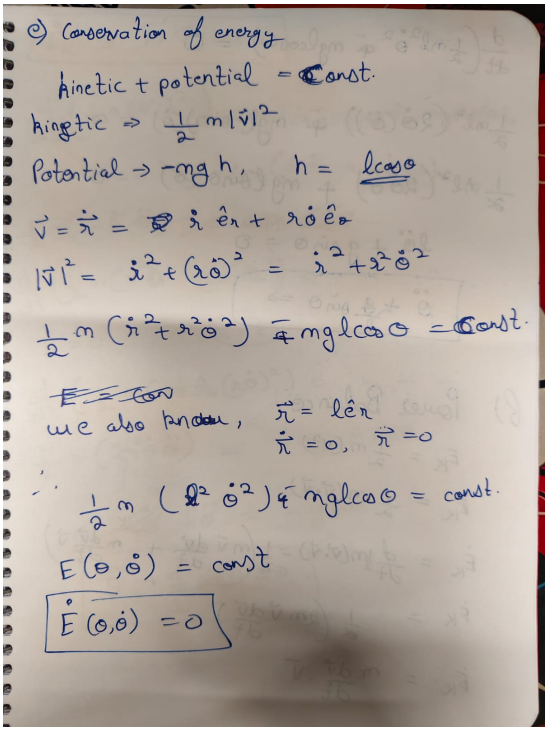
(a) Taking the cross product of the linear momentum equation with $\vec{r}$ to eliminate the radial constraint force.

(b) Continuing the cross-product derivation and simplifying the resulting equation to obtain the angular equation of motion.

Figure 49: Derivation of the simple pendulum equation using linear momentum and the cross product with $\vec{r}$.

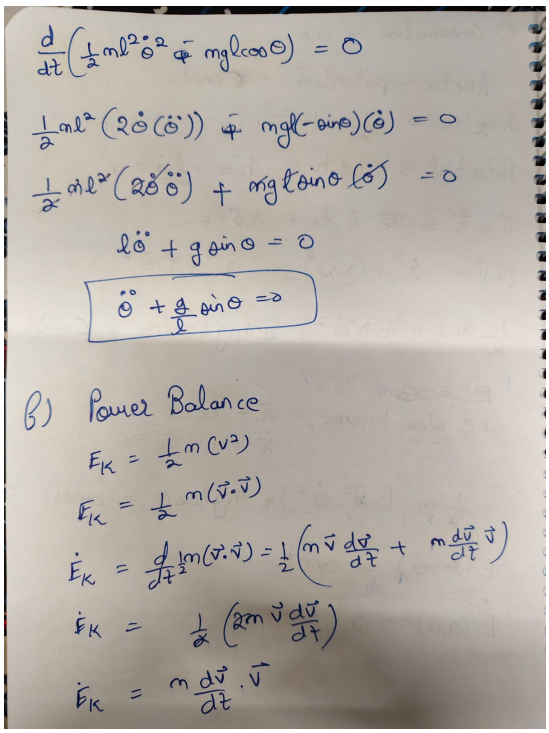


(a) Using conservation of mechanical energy, with kinetic energy and gravitational potential energy expressed in terms of θ.

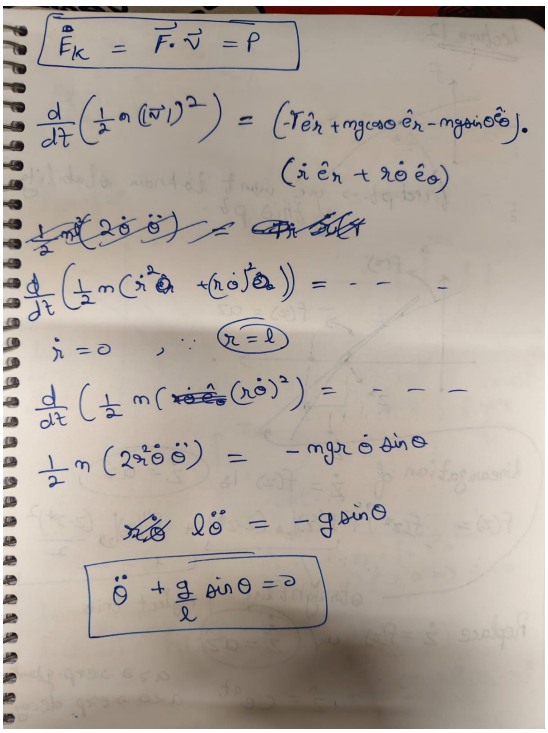


(b) Differentiating the conserved-energy expression with respect to time and simplifying to recover the pendulum equation.

Figure 50: Derivation using conservation of mechanical energy.



(a) Setting up the power balance by taking the dot product of force and velocity.



(b) Simplifying the power-balance equation to obtain the same nonlinear simple pendulum equation.

Figure 51: Derivation of the simple pendulum equation using power balance.

6 Homework- Week 8

6.1 Q17

Problem Statement

Montgomery's eight. (From Ruina/Pratap.) Three equal masses, say $m = 1$, are attracted by an inverse-square gravity law with $G = 1$. That is, each mass is attracted to the other by $F = Gm_1m_2/r^2$ where r is the distance between them. Use these unusual and special initial positions:

$$(x_1, y_1) = (-0.97000436,\ 0.24308753)$$
$$(x_2, y_2) = (-x_1,\ -y_1)$$
$$(x_3, y_3) = (0, 0)$$

and initial velocities

$$(v_{x3}, v_{y3}) = (0.93240737,\ 0.86473146)$$
$$(v_{x1}, v_{y1}) = -(v_{x3}, v_{y3})/2$$
$$(v_{x2}, v_{y2}) = -(v_{x3}, v_{y3})/2.$$

For each of the problems below show accurate computer plots and explain any curiosities.

(a) Use computer integration to find and plot the motions of the particles. Plot each with a different color. Run the program for 2.1 time units.

(b) Same as above, but run for 10 time units.

(c) Same as above, but change the initial conditions slightly.

(d) Same as above, but change the initial conditions more and run for a much longer time.

Interpretation

The three bodies interact only through mutual inverse-square gravity, so the state of the system is the twelve quantities $(x_i, y_i, v_{xi}, v_{yi})$, $i = 1, 2, 3$, and the acceleration of each mass is the sum of the pulls from the other two. The system is integrated numerically with `ode45`. The special initial conditions are chosen so that the three masses chase each other along a single figure-eight curve. The later parts change the initial conditions and the integration time to see how long this motion survives.

Solution



Figure 52: This is the motion for 2.1 seconds



Figure 53: This is the motion for 10 seconds, an '8' figure motion is still observed.

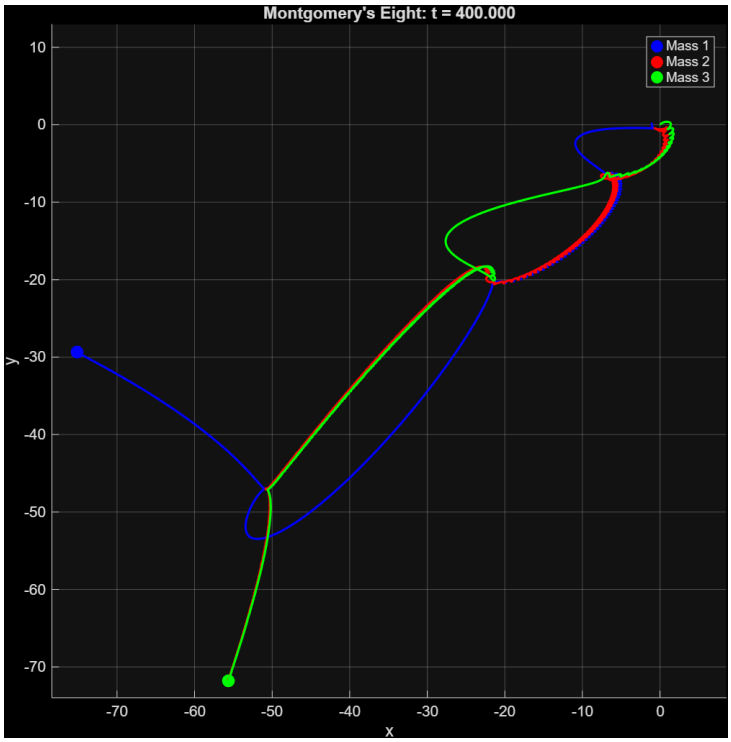


Figure 54: The only change in the initial condition for this case is vx2 and vy2 are set as -vx3 and -vy3 (Double from IC given in the question). 'Funny' fact- this was actually my first simulation for this question, I initially thought this was the motion- thg masses just tend to loop in 8 in a very weird manner, but as pointed out by Suraj- there was actually an error in the inital condition itself!

About the weird motion- the blue mass is the first one to shoot away from its initial state and the green and red masses bounce off of each other. All 3 masses converge roughly around x = -8, where the 'Bouncing Pair' change to red and blue masses and the green mass shoots away.

All 3 masses converge again where red and green masses become the 'Bouncing Pair' again! I showed it to other people who though the red mass would be the one to 'shoot away'.

The third time the masses converge, red and green **still** remain the 'Bouncing Pair' which was really intriguing to see.

6.2 Q18

Problem Statement

Konig's Theorem. The total kinetic energy of a system of particles is

$$E_K = \frac{1}{2}\sum m_i v_i^2.$$

(a) Derive an expression of this form

$$E_K = \frac{1}{2} m_{tot} v_G^2 + \ldots \text{you fill in the rest} \ldots$$

(b) Is it always true that

$$\left(\sum \vec{F}^{ext}\right) \cdot \vec{v}_G = \frac{d}{dt}\left(\frac{1}{2} m_{tot} v_G^2\right)?$$

Defend your answer with unassailable clear reasoning (that is, a proof or a counter-example).

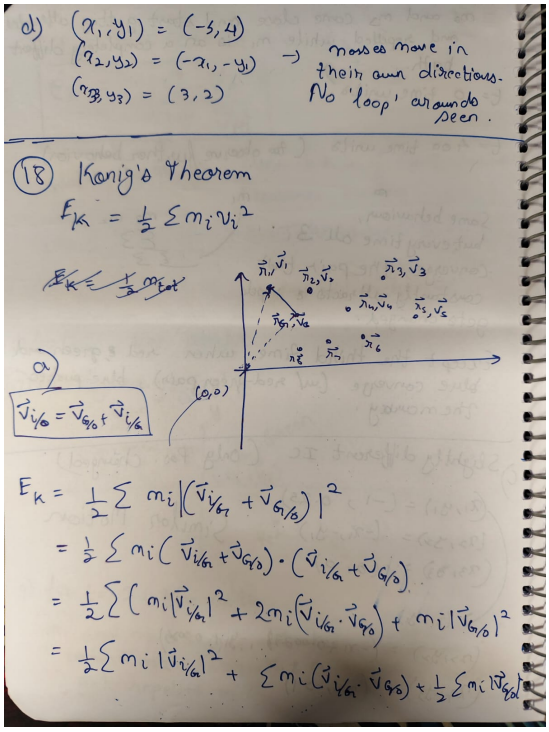
(c) Is it always true that the power of internal forces is equal to the rate of change of the quantity you filled in part (a) above (just the second half of the full expression)? Provide a proof or a counter-example.

Interpretation

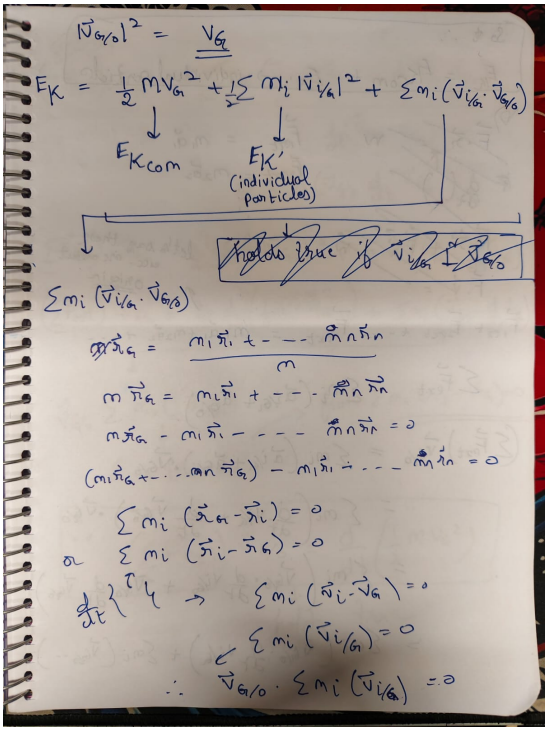
Each particle's velocity is split into the velocity of the centre of mass G plus its velocity relative to G, $\vec{v}_{i/O} = \vec{v}_{G/O} + \vec{v}_{i/G}$. Substituting this into E_K and expanding gives a centre-of-mass term, a term from the motion of the particles relative to G, and a cross term. The cross term vanishes because $\sum m_i \vec{v}_{i/G} = \vec{0}$, which follows from the definition of the centre of mass.

Parts (b) and (c) test whether two work–energy style statements hold in general, by differentiating the corresponding kinetic energies and checking whether the remaining terms vanish.

Solution

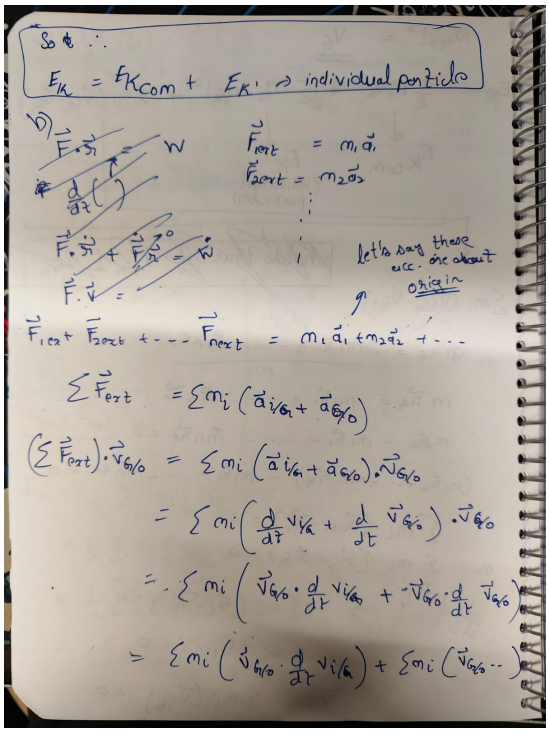


(a)

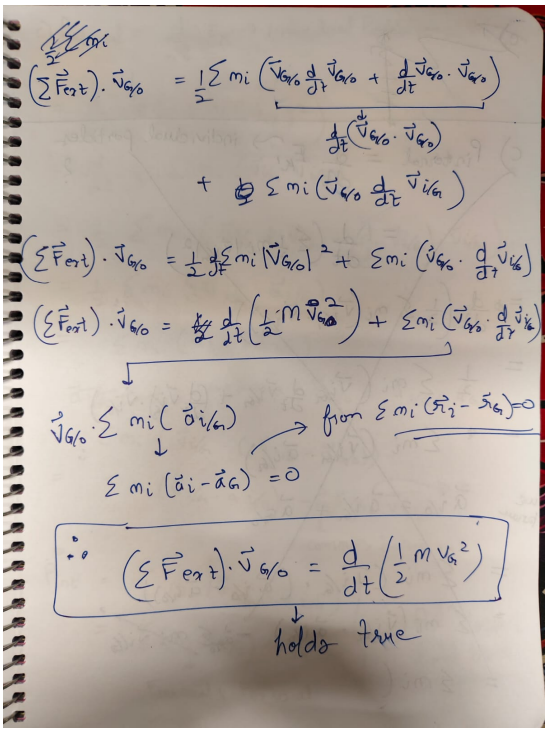


(b)

Figure 55



(a)



(b)

Figure 56

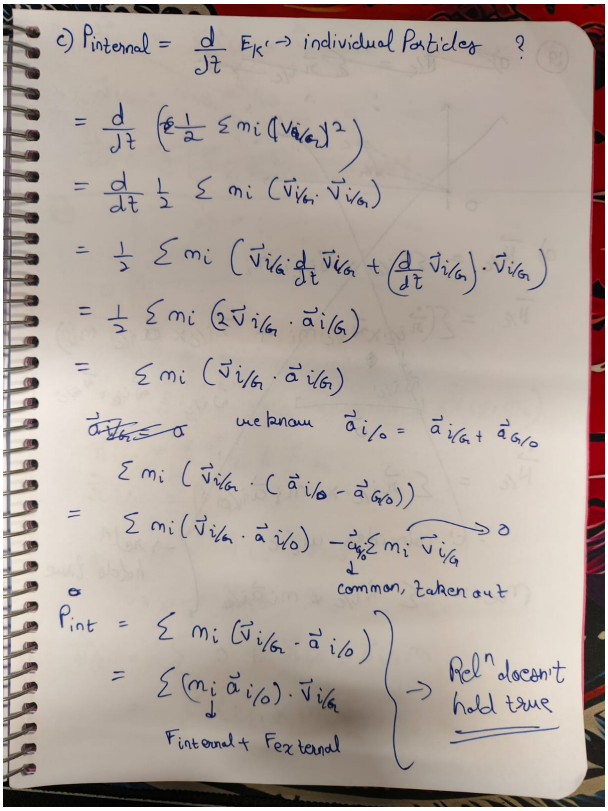


Figure 57

6.3 Q19

Problem Statement

What means "rate of change of angular momentum" for a SYSTEM of particles? You should master problem 14 before looking at this problem. Consider a system of moving particles with moving center of mass at G. Consider also a moving point C (moving relative to a Newtonian frame $\mathcal{F}$ that has an origin 0). For which of these definitions of $\vec{H}_{/C}$ is the following equation of motion true (that is, consistent with $\vec{F} = m\vec{a}$)?

$$\vec{M}_C = \dot{\vec{H}}_{/C}$$

In each case say whether the definition works i) in general, or ii) for some special cases (that you name) concerning the motions of P and C.

(a) $\vec{H}_{/C} = \sum \vec{r}_{i/C'} \times \vec{v}_{i/C'} \, m_i$, where C' is a point fixed in $\mathcal{F}$ that instantaneously coincides with C. (Hint: this definition is a good one, always!)

(b) $\vec{H}_{/C} = \sum \vec{r}_{i/C} \times \vec{v}_{i/0} \, m_i$. (This strange definition is used in the classic, but in this case odd, Dynamics book by Housner and Hudson.)

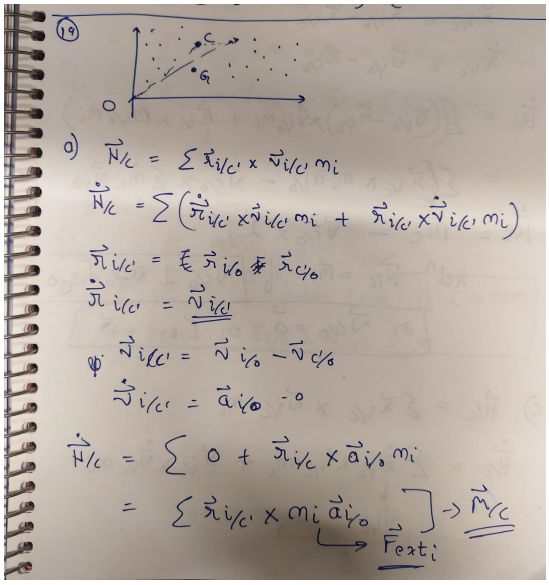
(c) $\vec{H}_{/C} = \sum \vec{r}_{i/C} \times \vec{v}_{i/C} \, m_i$. (Hint: this is the most important candidate definition, but it is only good for special kinds of C, namely: C = COM, C is fixed and…?)

That is, for each possible definition of $\vec{H}_{/C}$ you need to calculate $\dot{\vec{H}}_{/C}$ by differentiation and see if and when you get $\sum \vec{r}_{i/C} \times \vec{a}_{i/0} \, m_i$.

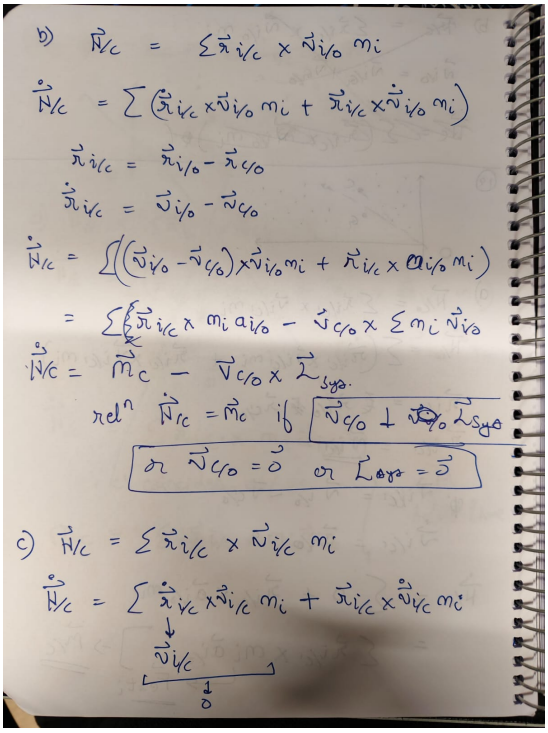
Interpretation

For each candidate definition, $\dot{\vec{H}}_{/C}$ is found by the product rule, using $\vec{r}_{i/C} = \vec{r}_{i/0} - \vec{r}_{C/0}$ and the corresponding relations for velocity and acceleration. The target is $\sum \vec{r}_{i/C} \times m_i \vec{a}_{i/0}$, which is the moment $\vec{M}_C$ of the external forces about C. Any extra terms left over show when the definition fails, and setting those terms to zero gives the special cases (a fixed point, a particular choice such as the centre of mass, or parallel vectors) in which it still works.

Solution



(a)



(b)

Figure 58

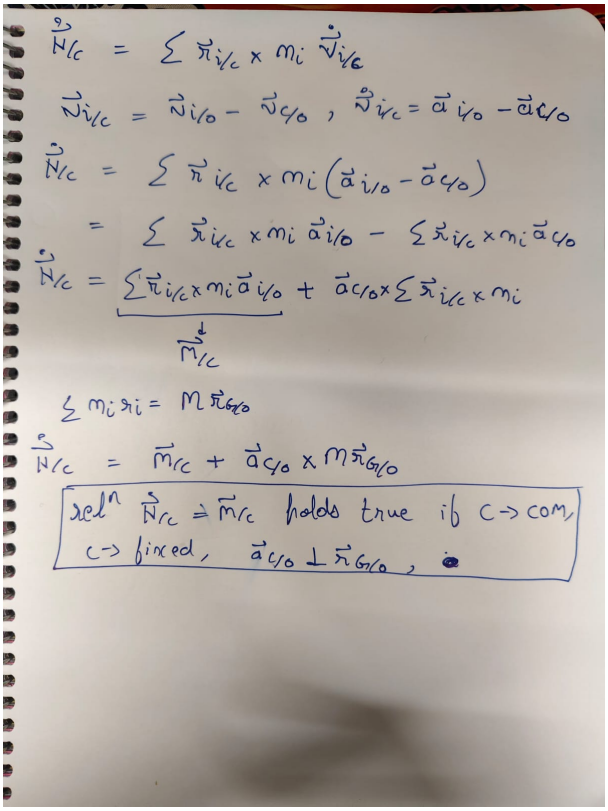

$\dot{\vec{N}}_{/c} = \sum \vec{r}_{i/c} \times m_i \dot{\vec{v}}_{i/c}$
$\vec{v}_{i/c} = \vec{v}_{i/o} - \vec{v}_{c/o}$, $\dot{\vec{v}}_{i/c} = \vec{a}_{i/o} - \vec{a}_{c/o}$
$\dot{\vec{N}}_{/c} = \sum \vec{r}_{i/c} \times m_i (\vec{a}_{i/o} - \vec{a}_{c/o})$
$= \sum \vec{r}_{i/c} \times m_i \vec{a}_{i/o} - \sum \vec{r}_{i/c} \times m_i \vec{a}_{c/o}$
$\dot{\vec{N}}_{/c} = \underbrace{\sum \vec{r}_{i/c} \times m_i \vec{a}_{i/o}}_{\vec{M}_{/c}} + \vec{a}_{c/o} \times \sum \vec{r}_{i/c} \times m_i$
$\sum m_i r_i = M \vec{r}_{G/o}$
$\dot{\vec{N}}_{/c} = \vec{M}_{/c} + \vec{a}_{c/o} \times M \vec{r}_{G/o}$
rel^n $\dot{\vec{N}}_{/c} = \vec{M}_{/c}$ holds true if c -> com, c -> fixed, $\vec{a}_{c/o} \perp \vec{r}_{G/o}$,

Figure 59

7 Homework- Week 9

7.1 Q20

Problem Statement

20. Two masses This problem has 3 independent educational goals:

(a) Motivate the use of kinematic constraints.

(b) Introduce the simplest of a class of vibrations problems which you should master. At this point, it is this aspect: mastery of derivation of the equations of motion. You should check that you can reproduce the lecture example with no sign errors without looking up anything.

(c) Development of critical exploration using analytical and numerical methods.

Two masses m_1 and m_2 are constrained to move frictionlessly on the x axis. Initially they are stationary at positions $x_1(0) = 0$ and $x_2(0) = \ell_0$. They are connected with a linear spring with constant k and rest length ℓ_0. A rightwards force is applied to the second mass. It is a step, or 'Heaviside' function

$$F(t) = F_0 H(t) = \begin{cases} 0 & \text{if } t < 0, \\ F_0 & \text{if } t \geq 0. \end{cases}$$

(a) Write code to calculate, plot and (optionally) animate the motions for arbitrary values of the given constants.

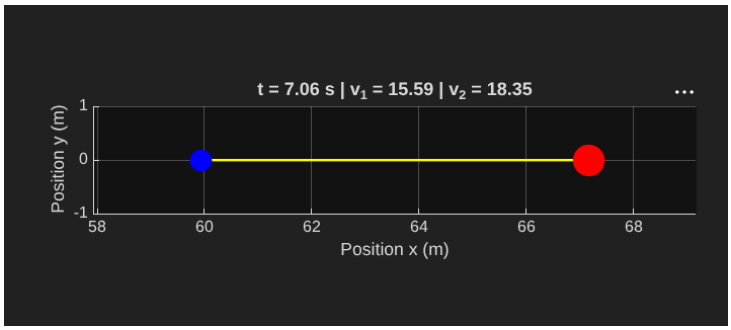
(b) Within numerical precision, should your numerical solution always have the property that $F = (m_1 + m_2)a_G$ where $x_G = (x_1 m_1 + x_2 m_2)/(m_1 + m_2)$? (As always in this course, "yes or no" questions are not multiple choice, but need justification that another student, one who got the opposite answer as you, would find convincing.)

(c) Use your numerics to demonstrate that if k is large the motion (displacement relative to its starting position) of each mass is, for time scales large compared to the oscillations, close to the center of mass motion.

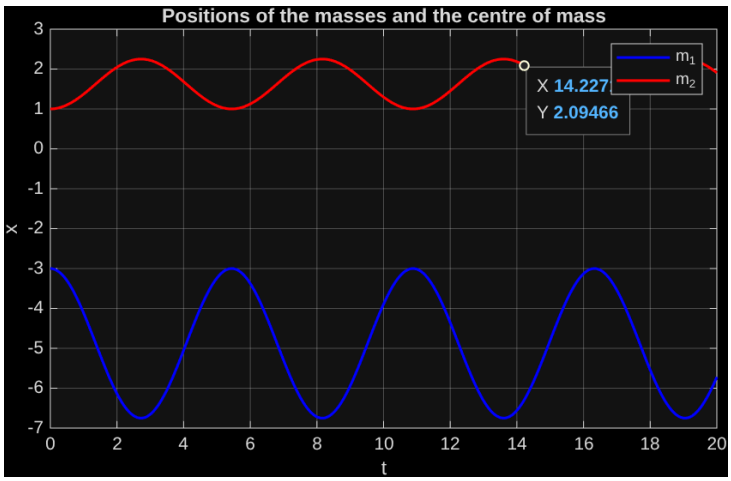
(d) 5730 only: Using analytic arguments, perhaps inspired by and buttressed with numerical examples, make the following statement as precise as possible: For high values of k the system nearly behaves like a single mass. Of course, in detail, the system has 2 degrees of freedom (DOF). So you are looking for a way to measure the extent to which the system acts like it has 1 DOF, and in which conditions (for which extreme values of parameters and times) the system is close to 1 DOF by that measure. There is not a simple single unique answer to this question.

Interpretation

Solution



(a) Screenshot of the animation for the motion. The masses had a to-and-fro motion when a force was being applied on mass 2 (Red mass)



(b) Plot of relative position of both masses with the center of gravity. The to-and-fro motion is sinusoidal.

Figure 60

Upon FBD analysis, both equations show a coupled SHM equation and hence the sinusoidal plot for relative position of both masses.

$$\ddot{x}_1 = \frac{k}{m}(x_2 - x_1 - l_0) \tag{5}$$

$$\ddot{x}_2 = \frac{F_0}{m} - \frac{k}{m}(x_2 - x_1 - l_0) \tag{6}$$

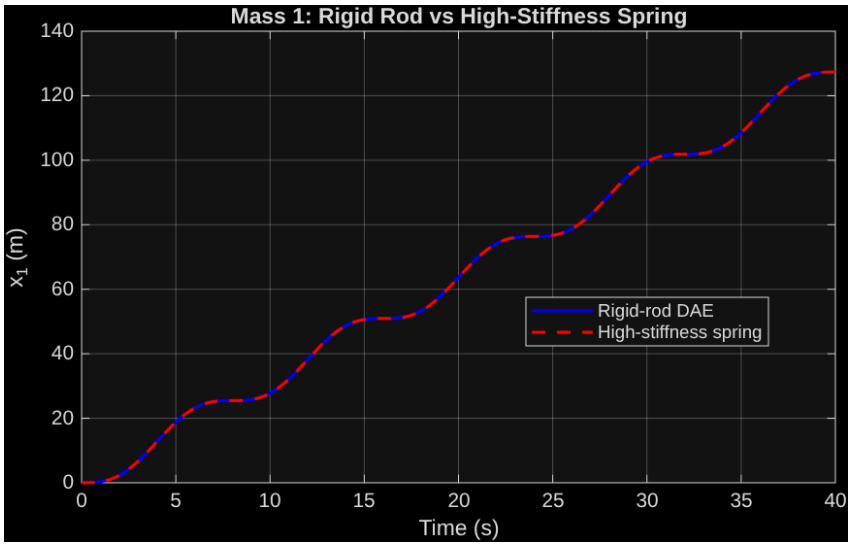
7.2 Q21

Problem Statement

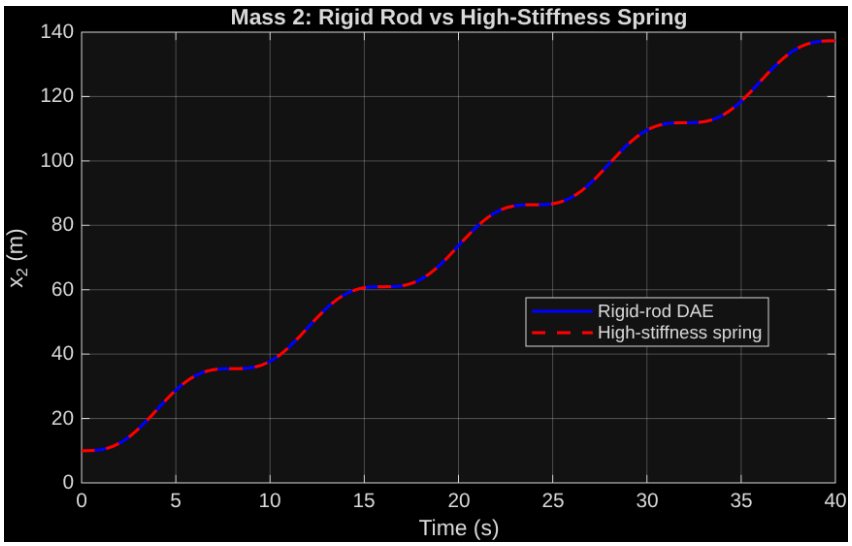
21. Two masses constrained This is an elaboration of the problem above, replacing the spring with a rigid rod. As per lecture, set up the DAEs and solve them using Matlab using numbers of your choice. Use a forcing F that varies sinusoidally in time. Note the increasing error (as time progresses) in the satisfaction of the constraint. Compare this solution with the the method from the problem above (where you use some very large value of k). Which one is is faster to compute on the computer? Which is more accurate in predicting COM motion? (Again, as always, justify your answer so someone who disagrees would be convinced.) Try putting in initial conditions in which the masses have unequal initial velocity; explain as best you can the result you get, and why.

Interpretation

Solution

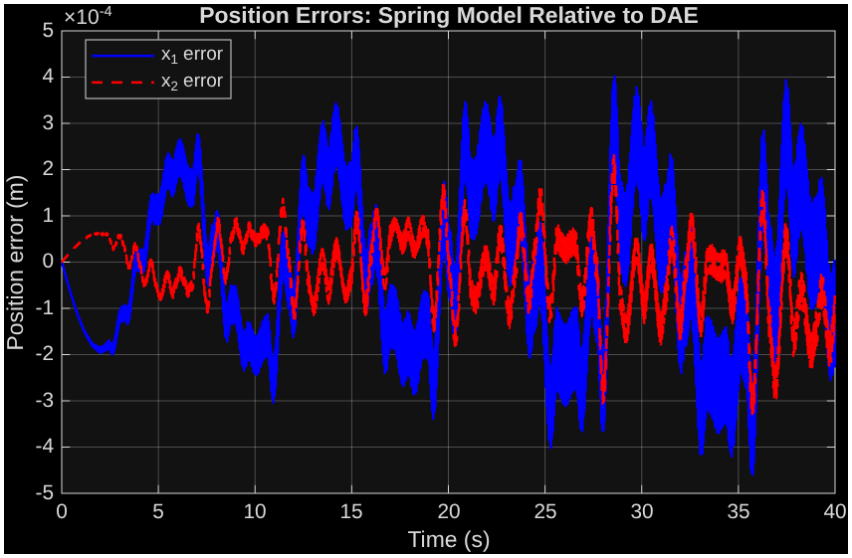


(a) Mass m1

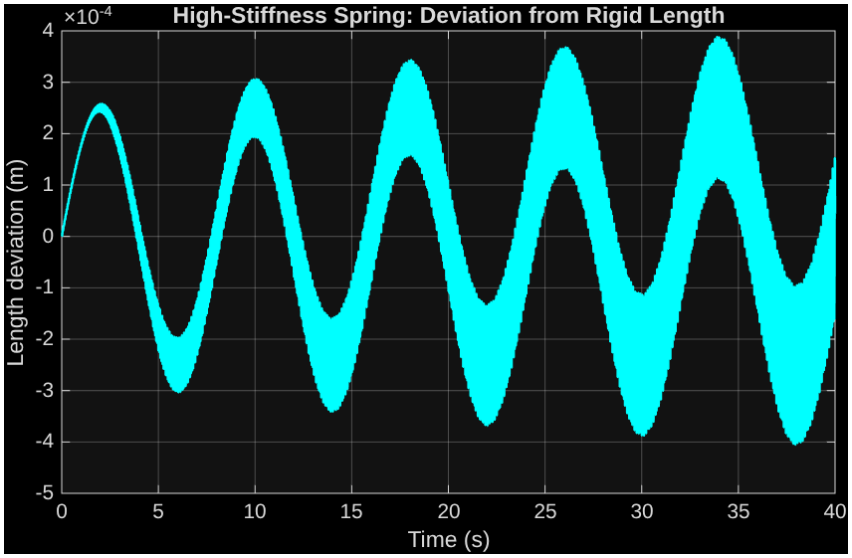


(b) Mass m2

Figure 61: Position Comparison of the Masses of both Rigid rod and High-Stifness spring cases



(a)



(b)

Figure 62: The error in the high stiffness spring case increases as time goes on. Since the system oscillates at a really high frequency

Let q be the relative displacement from the spring's natural length:

$$q = x_2 - x_1 - l_0$$

$$\ddot{q} = \ddot{x}_2 - \ddot{x}_1$$

$$\ddot{q} = \frac{F(t)}{m_2} - \frac{k}{m_2}q - \frac{k}{m_1}q$$

$$\ddot{q} + k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)q = \frac{F(t)}{m_2}$$

Comparing with the standard equation of an oscillator,

$$\ddot{q} + \omega_n^2 q = \frac{F(t)}{m_2}$$

we obtain the natural angular frequency:

$$\omega_n = \sqrt{k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}$$

The corresponding oscillation period is

$$T_n = \frac{2\pi}{\omega_n} = \frac{2\pi}{\sqrt{k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}}$$

Hence, as the spring stiffness increases,

$$\omega_n \propto \sqrt{k}, \qquad T_n \propto \frac{1}{\sqrt{k}}$$

8 Homework- Week 10

8.1 Q23

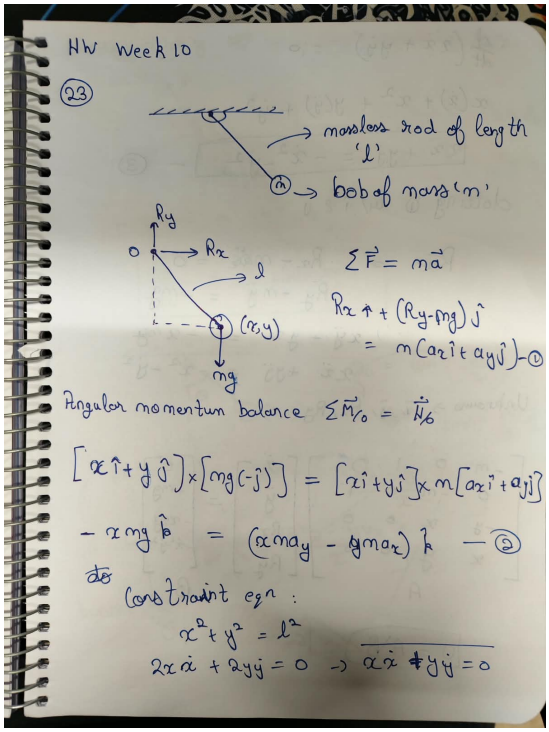
Problem Statement

23. Pendulum numerics. Set up the pendulum in cartesian coordinates. Express the constant length constraint as a set of linear equations restricting the acceleration. Solve these (3 2nd order) DAE equations with numerical integration and initial conditions and parameters of your choosing. No polar coordinates allowed. Quantitatively compare your solution with a solution of the simple pendulum equations (for the comparison you need to either compute x from θ or *vice versa*). Integrate for a long enough time so you can detect drift away from satisfying the kinematic constraint.

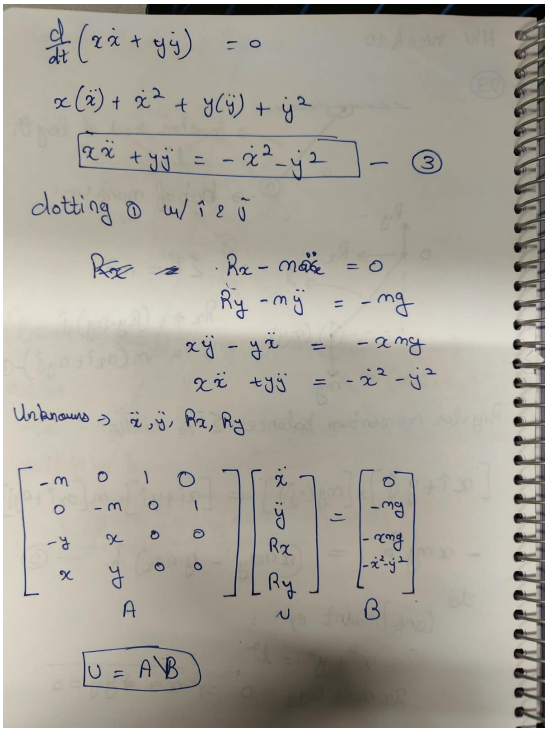
Interpretation

A point mass m hangs from a massless rod of length ℓ hinged at the origin O. Instead of the angle θ, the position (x, y) of the bob is used as the coordinate, and the rod appears as an unknown reaction force (R_x, R_y) at the hinge. The unknowns are $\ddot{x}, \ddot{y}, R_x, R_y$. They are found from four equations that are linear in the unknowns: two from $\sum \vec{F} = m\vec{a}$, one from angular momentum balance about O (which removes the moment of the reaction), and one from the constraint $x^2 + y^2 = \ell^2$ differentiated twice. At every time step the system is solved as $U = A \backslash B$ and passed to `ode45`. The result is compared with the simple pendulum equation $\ddot{\theta} + \frac{g}{\ell} \sin\theta = 0$.

Solution



(a) Free body diagram, linear momentum balance, angular momentum balance about O, and the constraint $x^2 + y^2 = \ell^2$ differentiated once.



(b) Constraint differentiated a second time, the four scalar equations, and the matrix form $AU = B$ with $U = A \backslash B$.

Figure 63: Setting up the pendulum in Cartesian coordinates as a DAE.

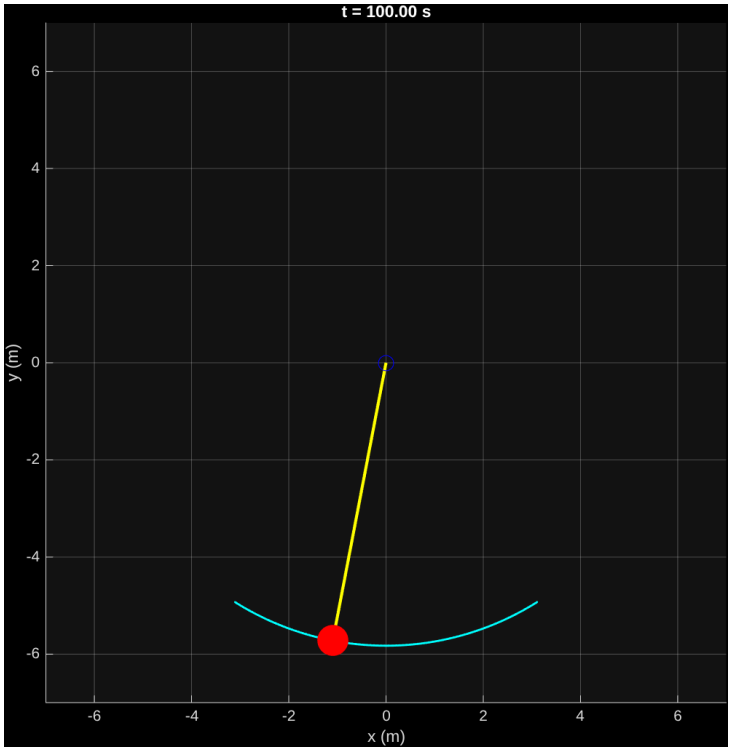


Figure 64: Screenshot of the pendulum animation at $t = 100$ s. The cyan curve is the path of the bob.

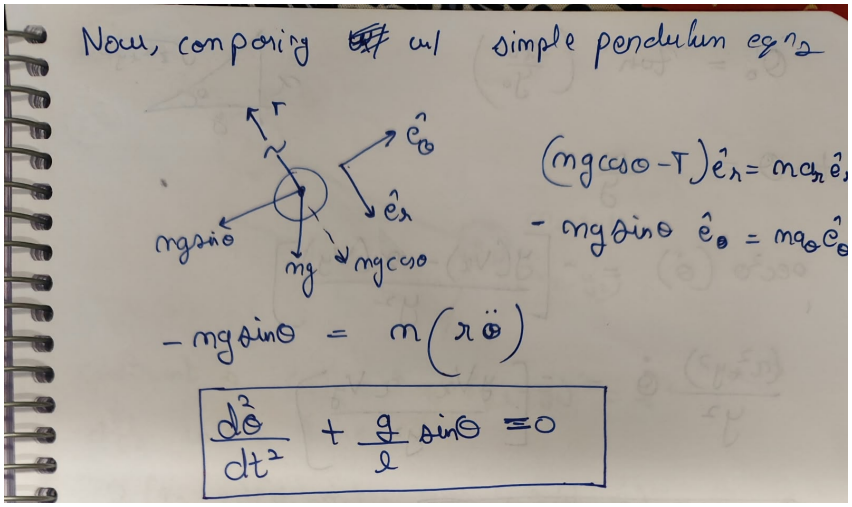


Figure 65: Free body diagram in polar components and the resulting simple pendulum equation $\frac{d^2\theta}{dt^2} + \frac{g}{\ell}\sin\theta = 0$, used as the reference solution.

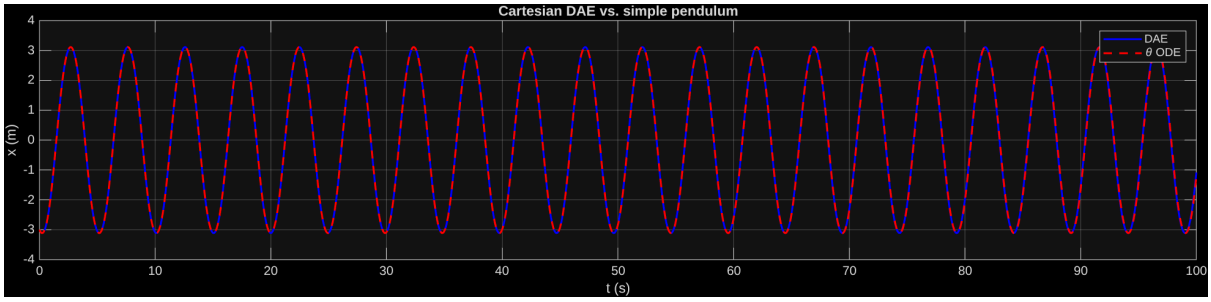


Figure 66: $x(t)$ from the Cartesian DAE method and from the θ equation over 100 s. The two solutions are nearly identical.

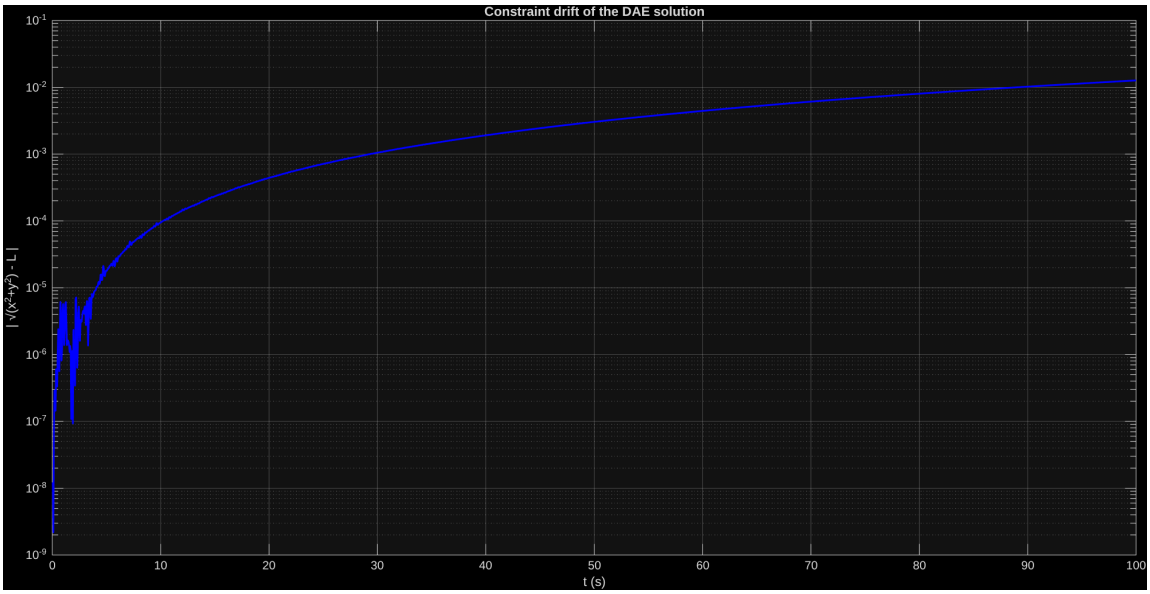


Figure 67: Constraint error $\left|\sqrt{x^2+y^2}-\ell\right|$ of the DAE solution against time (log scale).

Why the constraint error grows with time. Let the constraint be written as

$$\phi(x,y) = x^2 + y^2 - \ell^2 = 0.$$

The DAE does not enforce $\phi = 0$ directly. It only enforces the twice-differentiated form,

$$\ddot{\phi} = 2\left(x\ddot{x} + y\ddot{y} + \dot{x}^2 + \dot{y}^2\right) = 0 \quad \Longleftrightarrow \quad x\ddot{x} + y\ddot{y} = -\dot{x}^2 - \dot{y}^2,$$

which is the last row of the matrix system. Integrating $\ddot{\phi} = 0$ exactly over a step starting at t_k gives

$$\phi(t) = \phi(t_k) + \dot{\phi}(t_k)\,(t - t_k).$$

So if the initial conditions satisfy $\phi(0) = 0$ and $\dot{\phi}(0) = 0$ (that is, $x_0^2 + y_0^2 = \ell^2$ and $x_0\dot{x}_0 + y_0\dot{y}_0 = 0$), the exact solution keeps $\phi = 0$ forever.

The numerical solution is not exact. Every step of `ode45` makes small errors $\delta x, \delta y, \delta\dot{x}, \delta\dot{y}$ of the order of the tolerance. These perturb both ϕ and $\dot{\phi}$:

$$\delta\phi \approx 2\left(x\,\delta x + y\,\delta y\right), \qquad \delta\dot{\phi} \approx 2\left(x\,\delta\dot{x} + y\,\delta\dot{y} + \dot{x}\,\delta x + \dot{y}\,\delta y\right).$$

Because the equations only say $\ddot{\phi} = 0$, there is nothing that pulls ϕ back to zero. A non-zero $\delta\dot{\phi}$ makes ϕ grow linearly in time, and new errors are added at every step, so the drift accumulates.

In the figure it grows from about 10^{-4} at $t = 10$ s to about 10^{-2} at $t = 100$ s. Tightening the tolerance reduces the size of each δ but does not remove the growth. A stabilised form such as $\ddot{\phi} + 2\alpha\dot{\phi} + \beta^2\phi = 0$ (Baumgarte stabilisation) would add the missing restoring term.

8.2 Q25

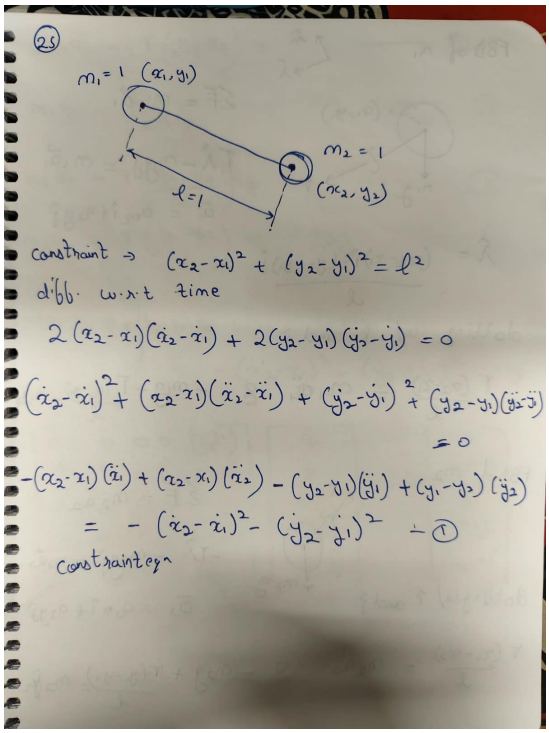
Problem Statement

25. 2D Dumbbell. Two equal masses $m = 1$ are constrained by a rod to be a distance $\ell = 1$ apart. There are no non-negligible forces, except from the constraint. At $t = 0$ they have equal and opposite velocities ($v = 1$) perpendicular to the rod. Use a set of 5 DAEs ($\vec{F} = m\vec{a}$ & the constraint equation $(x_2 - x_1)^2 + (y_2 - y_1)^2 = \ell^2$) with numerical integration to find the subsequent motion. Use plots and/or animation to help debug your code. Using what you know about systems of particles (*e.g.*, momentum, angular momentum, constraint equation, energy) quantify as many different numerical errors as you can.

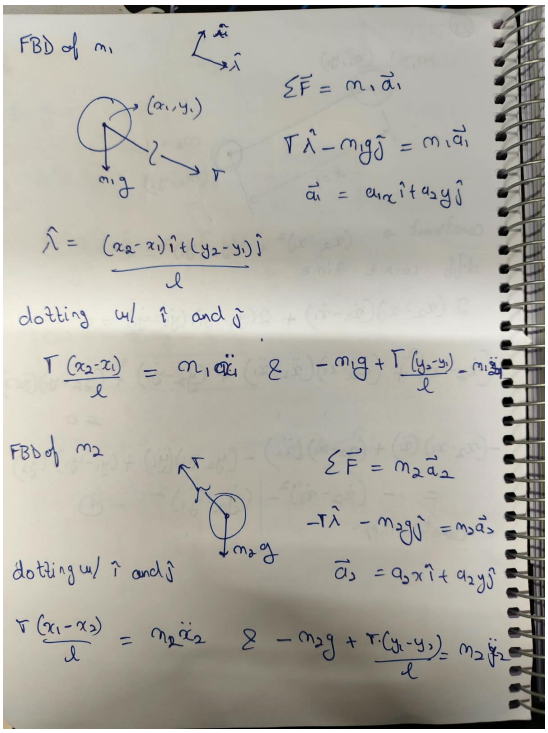
Interpretation

Each mass has two coordinates, so there are four accelerations $\ddot{x}_1, \ddot{y}_1, \ddot{x}_2, \ddot{y}_2$ plus the unknown rod tension T, which acts along the line joining the masses. The five equations are $\vec{F} = m\vec{a}$ for each mass (four scalar equations) and the constraint $(x_2 - x_1)^2 + (y_2 - y_1)^2 = \ell^2$ differentiated twice. They are linear in the five unknowns and are solved as a matrix system at every step of the integration.

Solution

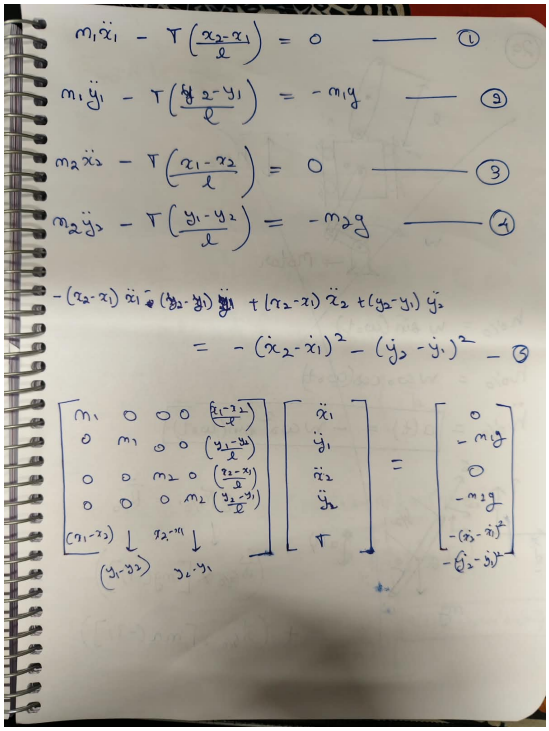


(a) The dumbbell and the constraint $(x_2 - x_1)^2 + (y_2 - y_1)^2 = \ell^2$, differentiated twice with respect to time to give constraint equation ①.

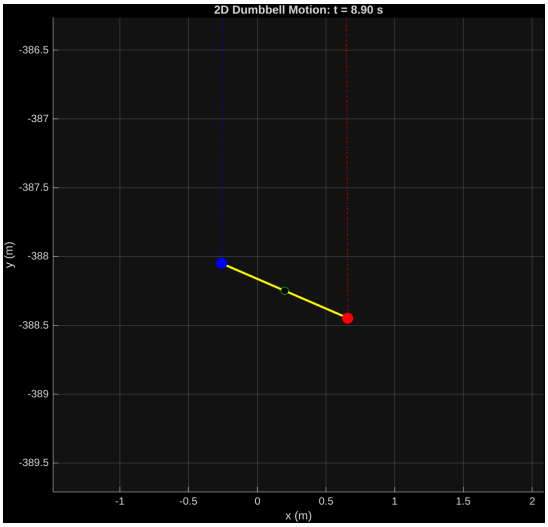


(b) Free body diagrams of m_1 and m_2 with the rod tension T along $\hat{\lambda}$, resolved into x and y components.

Figure 68: Constraint and free body diagrams for the 2D dumbbell.



(a) The four equations of motion, the acceleration-level constraint, and the matrix system in $(\ddot{x}_1, \ddot{y}_1, \ddot{x}_2, \ddot{y}_2, T)$.



(b) Screenshot of the dumbbell animation at $t = 8.90$ s. The dotted lines trace the two masses and the green marker is the centre of mass.

Figure 69: DAE set-up and animation of the dumbbell.

8.3 Q29

Problem Statement

29. Inverted pendulum with shaking base. A uniform stick with length ℓ is connected to a hinge at its lower end. That hinge is shaking up and down with a specified acceleration $a(t)$ (could be a specified position, differentiated twice).

(a) Find the equations of motion.

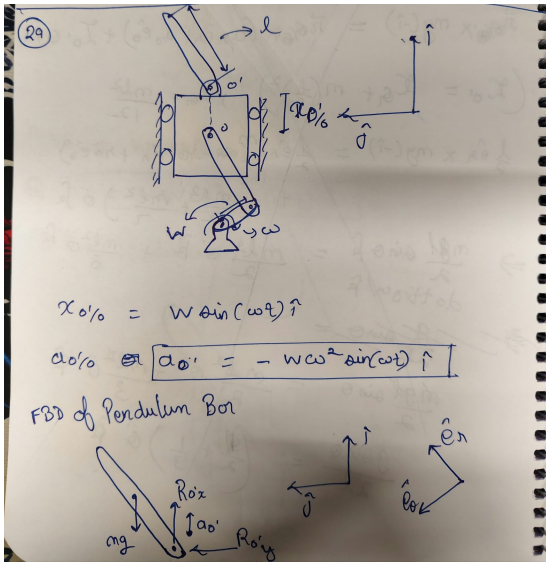
(b) Simulate the motion to show that the pendulum can be stable when upright. [Hint: Use an oscillating base with a small displacement amplitude ($\delta \ll \ell$) and a big enough frequency ($a \gg g$).]

(c) Do again with DAEs, and show that your results quantitatively agree.

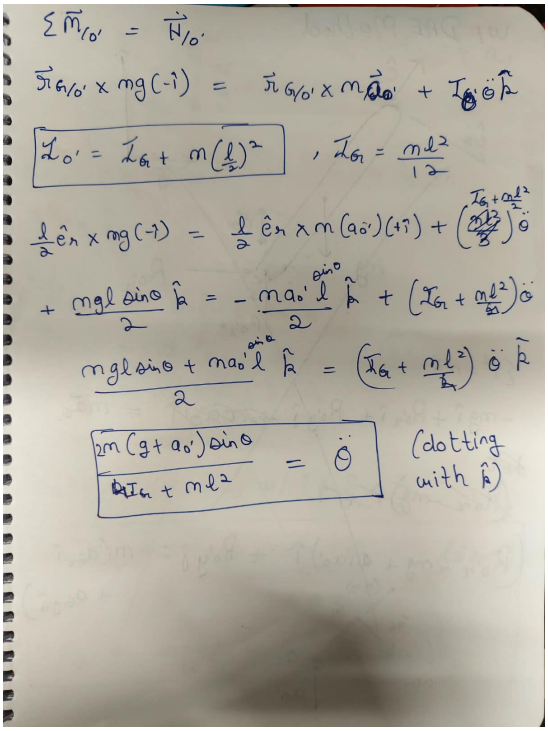
Interpretation

The hinge O' moves vertically as $x_{O'/O} = W\sin(\omega t)\,\hat{\imath}$, so $\vec{a}_{O'} = -W\omega^2\sin(\omega t)\,\hat{\imath}$. Because O' is accelerating, the angular momentum balance about O' contains the extra term $\vec{r}_{G/O'} \times m\vec{a}_{O'}$, and this is what puts $a_{O'}$ into the equation for $\ddot{\theta}$. In part (c) the stick is instead described by the tip coordinates (x, y) and the angle θ, tied together by $x = \ell\cos\theta$, $y = \ell\sin\theta$. These relations enter the DAE only through their second derivatives. The unknowns are $\ddot{x}, \ddot{y}, \ddot{\theta}$ and the reaction R along the stick.

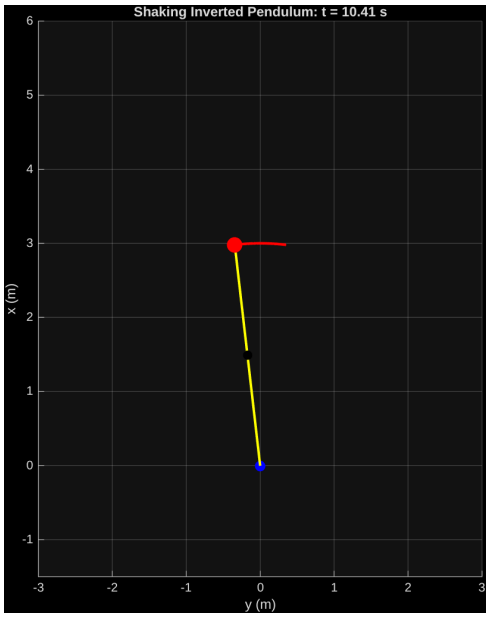
Solution



(a) The system, the base motion $x_{O'/O}$ and its acceleration $\vec{a}_{O'}$, and the free body diagram of the stick.

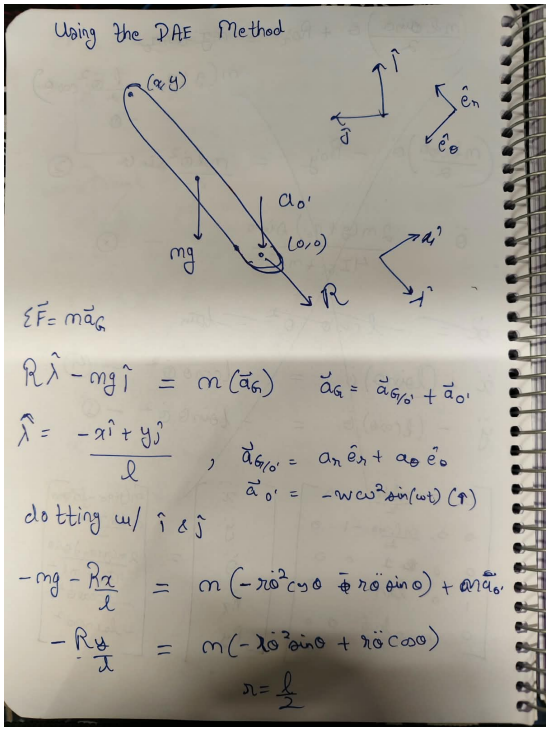


(b) Angular momentum balance about the hinge O', giving $\ddot{\theta}$ as a function of θ and $a_{O'}$.

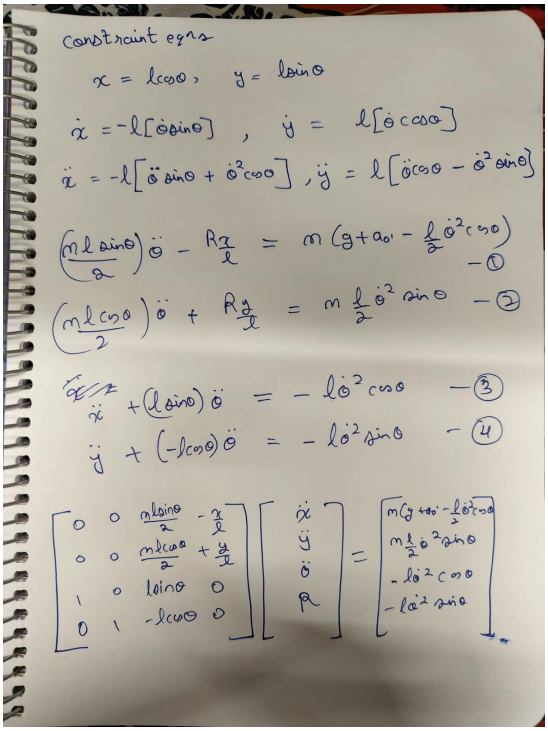


(c) Animation of the θ equation obtained from the free body diagram calculations (screenshot at $t = 10.41$ s). The stick stays close to upright while the base shakes. The red curve is the path of the tip.

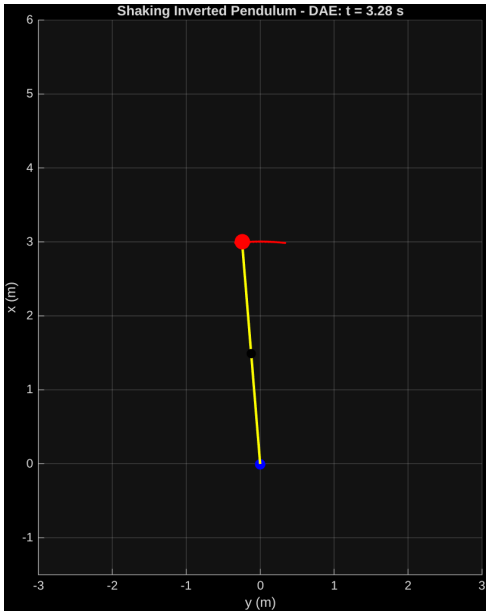
Figure 70: Equations of motion and animation using the θ coordinate.



(a) DAE method: free body diagram with the force R along the stick, and linear momentum balance resolved along $\hat{\imath}$ and $\hat{\jmath}$.

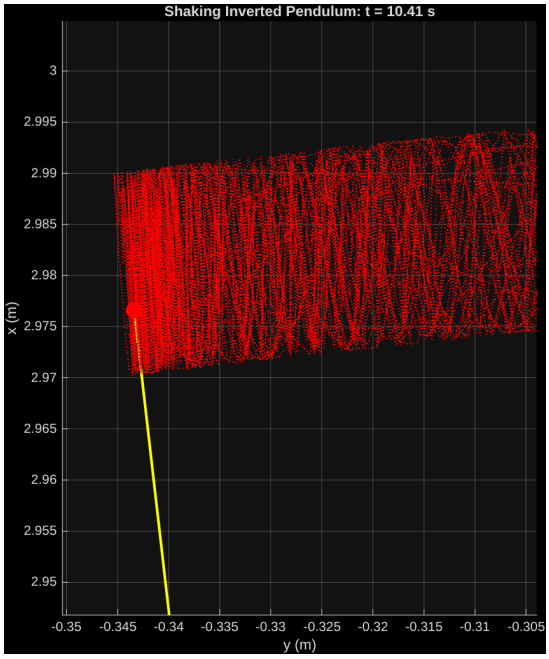


(b) Constraint equations $x = \ell\cos\theta, y = \ell\sin\theta$ differentiated twice, and the matrix system in $(\ddot{x}, \ddot{y}, \ddot{\theta}, R)$.

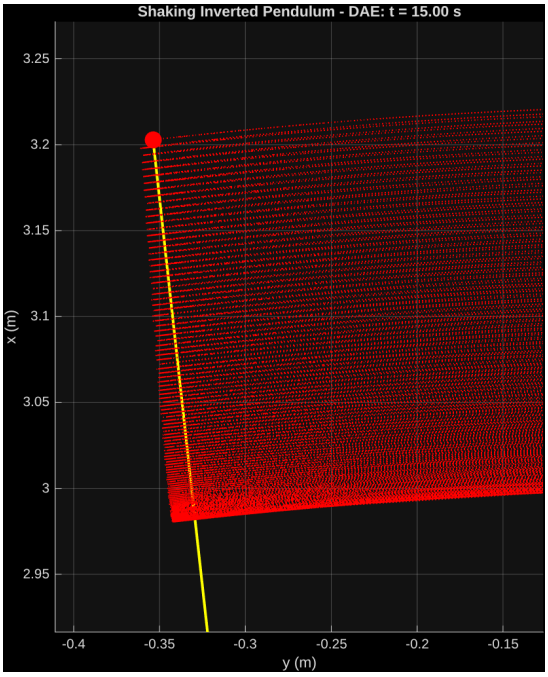


(c) Animation of the DAE method (screenshot at $t = 3.28$ s).

Figure 71: DAE formulation and animation.



(a) θ equation, zoomed in on the tip trajectory at $t = 10.41$ s. The tip stays in a narrow band around $x = 3$ m.



(b) DAE method, zoomed in on the tip trajectory at $t = 15$ s. The tip has drifted upward to $x \approx 3.2$ m.

Figure 72: Zoomed-in tip trajectories from the two methods.

Why the DAE solution drifts as time progresses. In the DAE method the quantities x, y and θ are all integrated as independent variables. They are tied together only through the two kinematic relations. Define the constraint residuals

$$e_x = x - \ell \cos\theta, \qquad e_y = y - \ell \sin\theta.$$

Differentiating twice gives

$$\ddot{e}_x = \ddot{x} + \ell \sin\theta\, \ddot{\theta} + \ell\dot{\theta}^2 \cos\theta, \qquad \ddot{e}_y = \ddot{y} - \ell \cos\theta\, \ddot{\theta} + \ell\dot{\theta}^2 \sin\theta.$$

The two constraint rows of the matrix system, $\ddot{x} + \ell \sin\theta\, \ddot{\theta} = -\ell\dot{\theta}^2 \cos\theta$ and $\ddot{y} - \ell \cos\theta\, \ddot{\theta} = -\ell\dot{\theta}^2 \sin\theta$, say exactly

$$\ddot{e}_x = 0, \qquad \ddot{e}_y = 0,$$

so on each step

$$e(t) = e(t_k) + \dot{e}(t_k)\,(t - t_k).$$

The exact solution keeps $e = 0$ if it starts with $e = \dot{e} = 0$. The numerical solution makes small errors in $x, y, \theta, \dot{x}, \dot{y}, \dot{\theta}$ at every step, which give $e \neq 0$ and $\dot{e} \neq 0$. Since only $\ddot{e} = 0$ is enforced, nothing drives e back to zero. A non-zero $\dot{e}$ makes e grow linearly, and new errors are added at every step, so the error accumulates with time. The effective stick length $\sqrt{x^2 + y^2}$ therefore drifts away from ℓ. This is why, in the zoomed-in figure, the DAE tip rises to about 3.2 m by $t = 15$ s while the stick length is $\ell \approx 3$ m.

In the θ formulation there is only one coordinate. The tip position is computed as $x = \ell \cos\theta$, $y = \ell \sin\theta$, so the constraint is satisfied exactly by construction and the tip stays at distance ℓ from the hinge.

9 Homework- Week 11

9.1 Q26

Problem Statement

Rotation with zero angular momentum. Consider a massless rigid plate with four equal masses attached to it. Two masses are fixed to the plate on its y axis at $y = \pm 2$. The other two masses move in circles of radius $r = 1$, centred at $x = \pm 2$, and their connecting line always passes through the origin. The right mass has polar angle

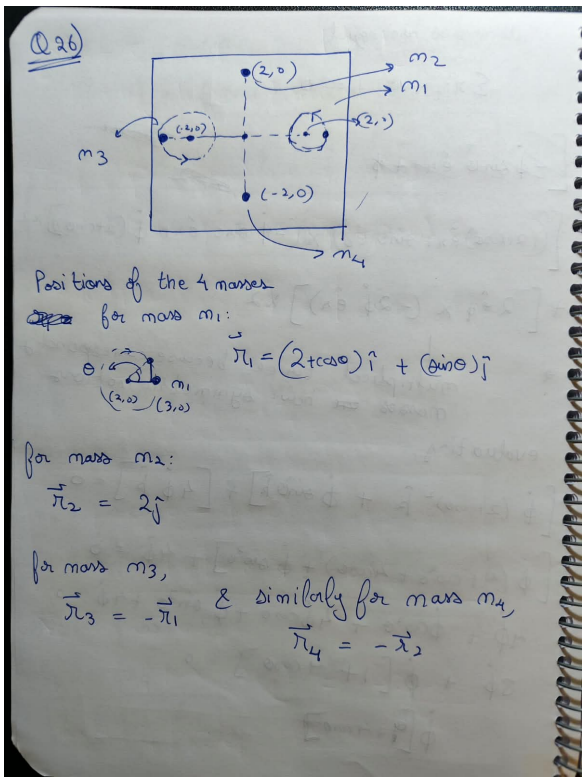
$$\theta(t) = \pi(1 - \cos t), \qquad 0 \le t \le \pi.$$

Using conservation of angular momentum, determine the net rotation ϕ of the plate (and hence of the two masses fixed to the y axis) after the moving masses complete one rotation.

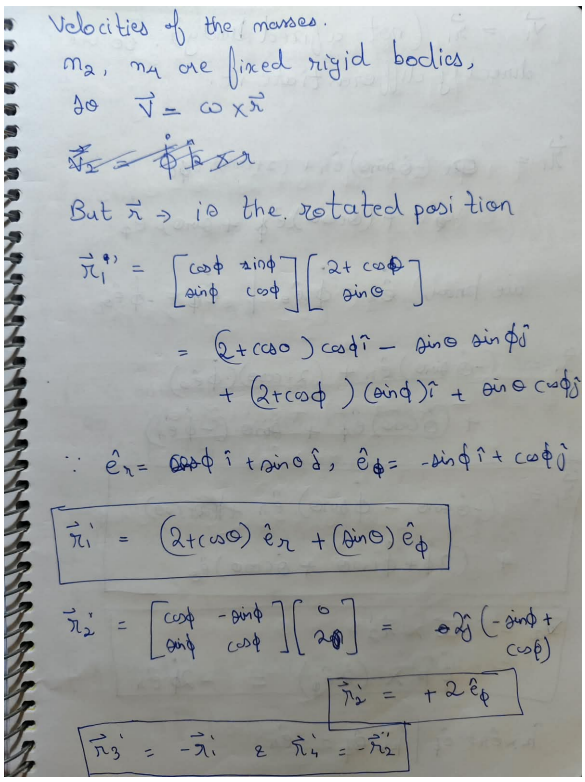
Interpretation

The system has no external torque, so its total angular momentum about the origin remains zero. The angular momentum has contributions from the two masses fixed to the plate and from the two masses moving relative to it. Setting the total angular momentum to zero gives an expression for the plate's angular velocity $\dot{\phi}$ in terms of θ and $\dot{\theta}$. Integrating this expression over the interval $0 \le t \le \pi$ gives the net plate rotation.

Solution

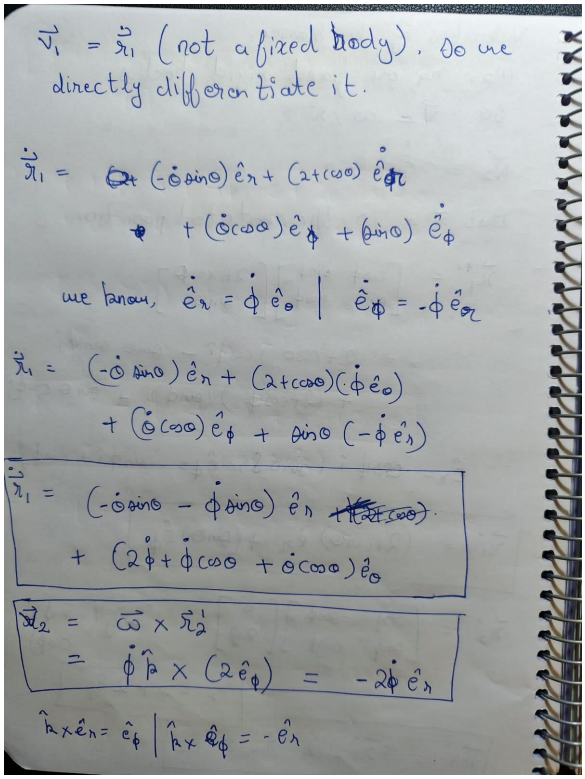


(a) Mass arrangement on the plate and the position vectors used to calculate angular momentum.

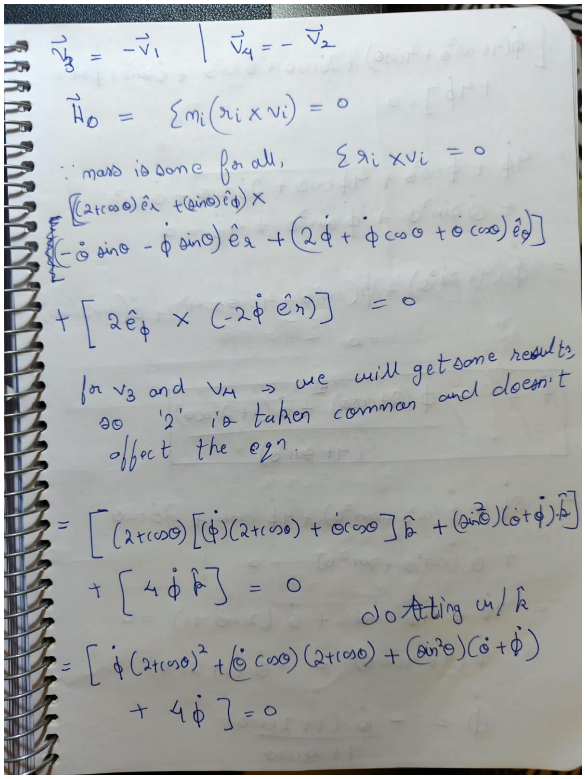


(b) Angular momentum contributions of the fixed masses and the moving masses.

Figure 73: System geometry and the initial angular momentum calculation.

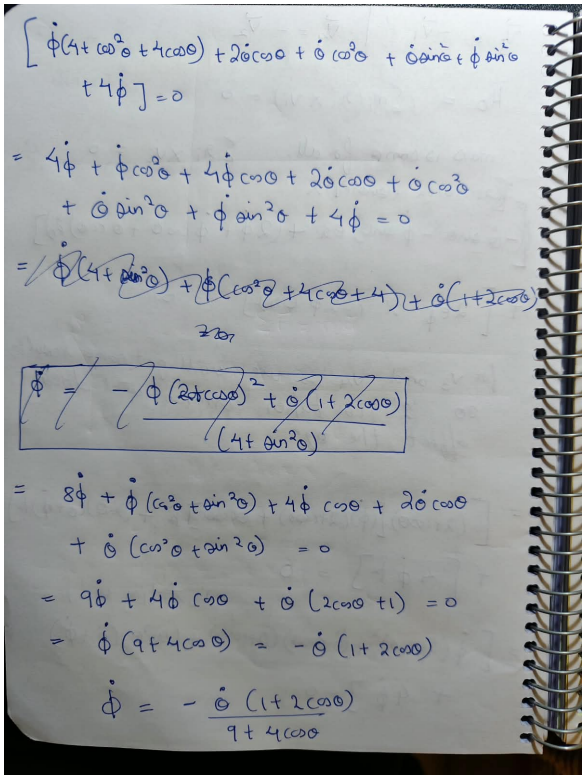


(a) Expansion and simplification of the angular momentum expression.

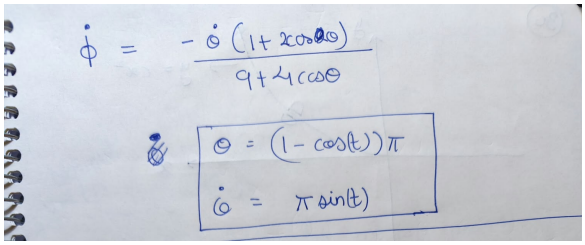


(b) Setting the total angular momentum equal to zero and isolating $\dot{\phi}$.

Figure 74: Using conservation of angular momentum to obtain the plate's angular velocity.

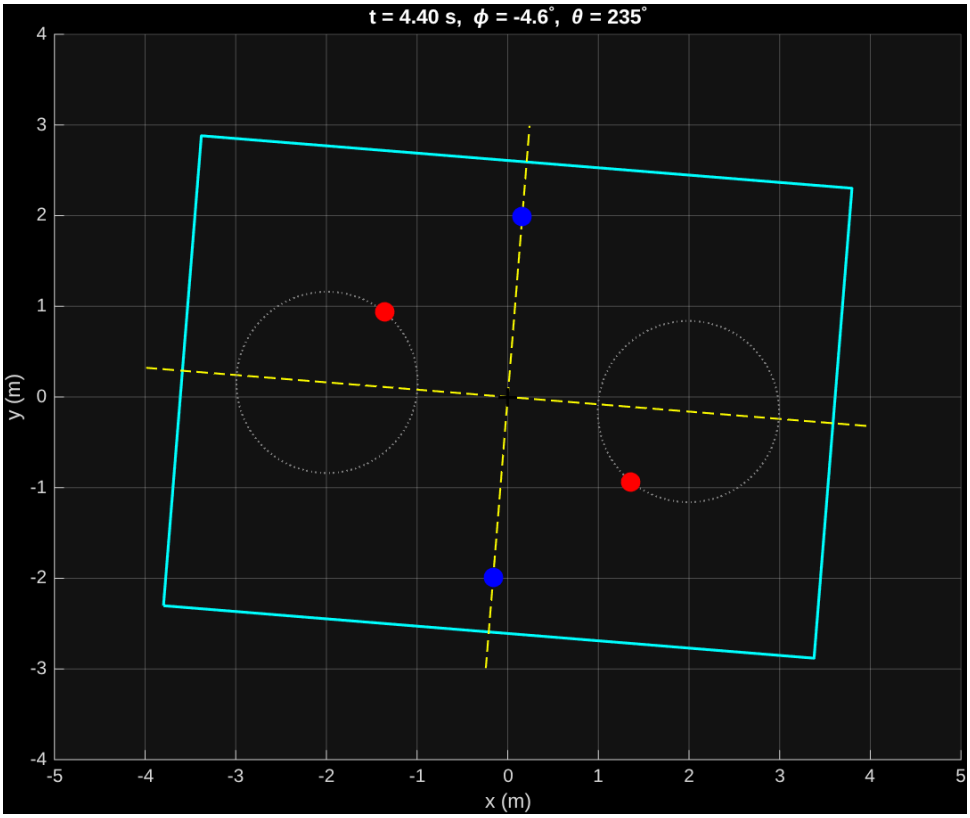


(a) Integration of $\dot{\phi}$ over the prescribed motion of the moving masses.

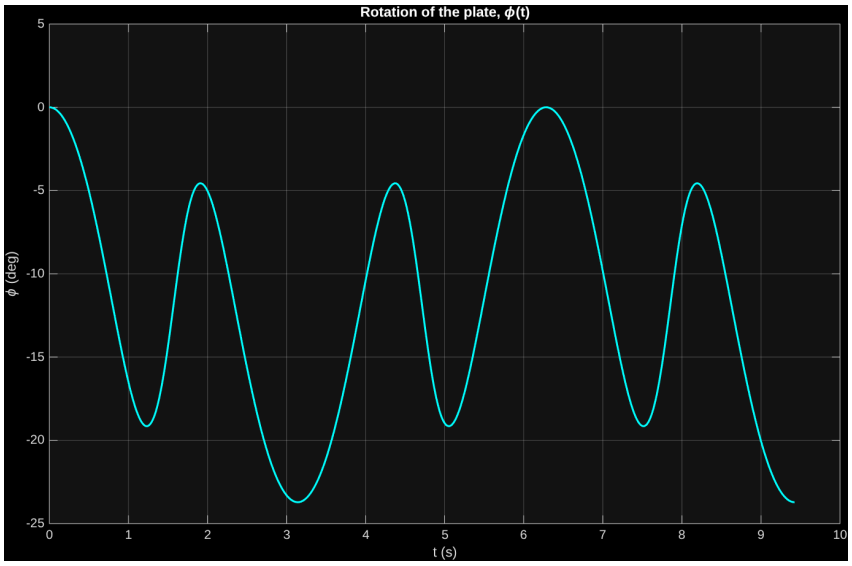


(b) Substitution of $\theta(t) = \pi(1 - \cos t)$ and $\dot{\theta}(t) = \pi \sin t$.

Figure 75: Evaluating the total change in plate orientation.



(a) Numerical animation snapshot showing the plate orientation and the positions of all four masses.



(b) Plate orientation $\phi(t)$ over the interval of the moving masses' rotation.

Figure 76: Numerical visualization of the motion and the resulting plate rotation.

9.2 Q30

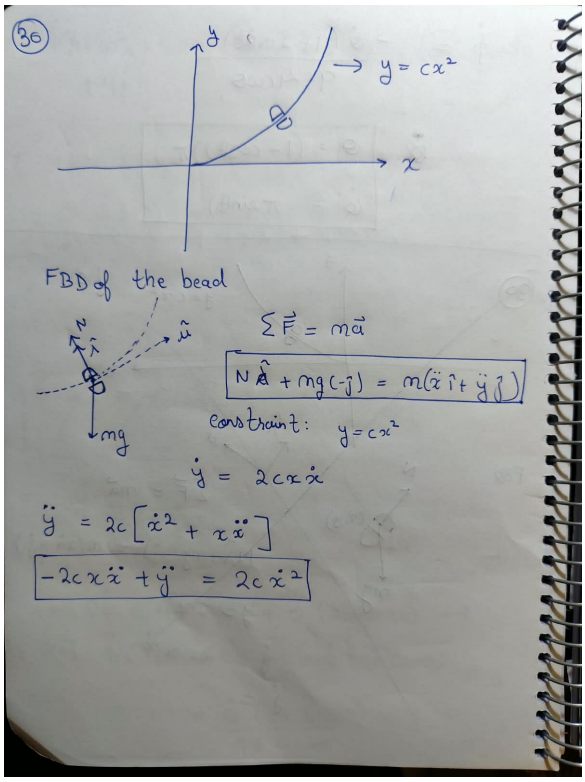
Problem Statement

Bead on parabolic wire. A frictionless point-mass bead slides on a rigid wire whose curve is $y = cx^2$, under gravity in the $-y$ direction. Derive the equation of motion using the x axis as the minimal coordinate and Newton's laws.

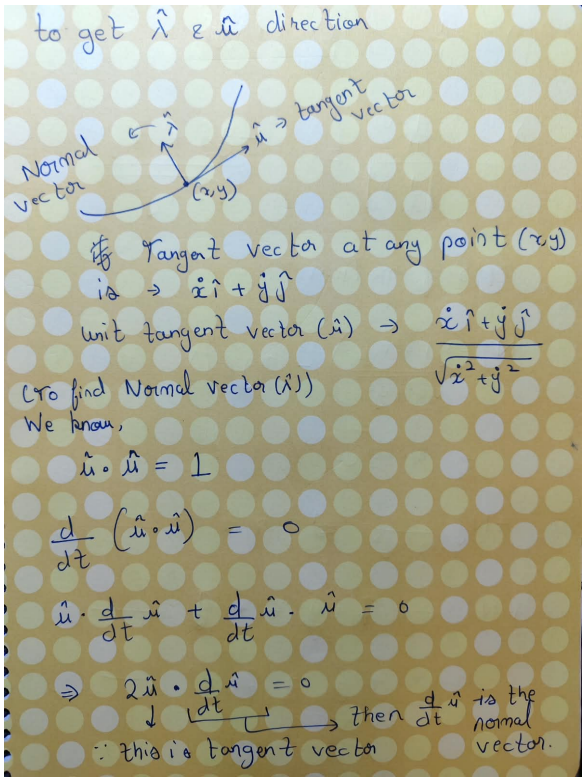
Interpretation

The bead's position is constrained by $y = cx^2$, so its motion can be described using only $x(t)$. The velocity and acceleration constraints are obtained by differentiating this geometric relation. Newton's second law is written in Cartesian components, with the wire's normal reaction resolved along a unit normal constructed from the velocity components. The acceleration equations and the differentiated constraint are then assembled into a linear system for $\ddot{x}$, $\ddot{y}$, and the normal reaction.

Solution

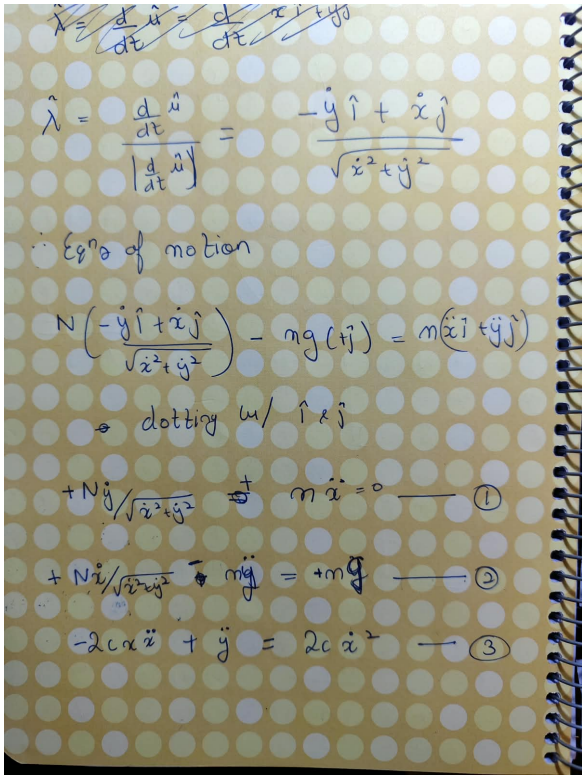


(a) Parabolic wire geometry, bead free-body diagram, and the chosen tangent/normal directions.

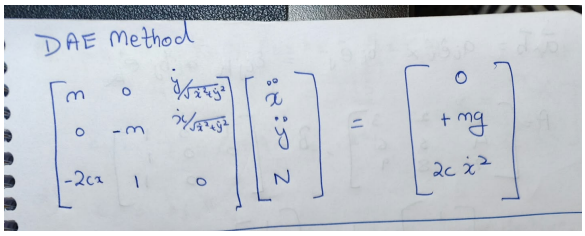


(b) Newton's second law resolved into the x and y directions, including the normal reaction.

Figure 77: Force balance and coordinate description for the bead.



(a) Differentiating $y = cx^2$ to obtain the velocity and acceleration constraints.



(b) Combining the equations into the DAE matrix system for the accelerations and normal force.

Figure 78: Constraint equations and the resulting DAE formulation.

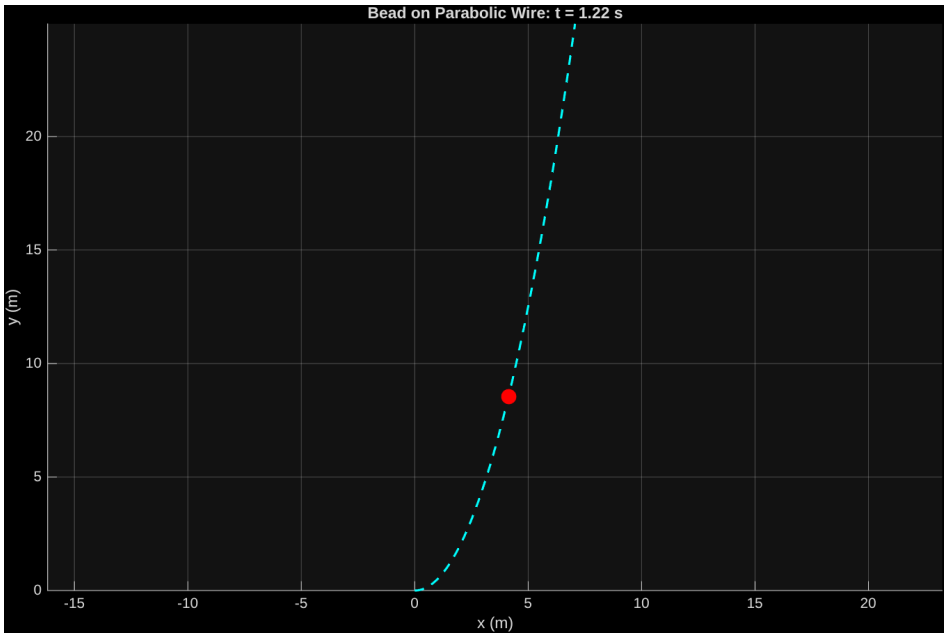


Figure 79: Numerical trajectory of the bead along the parabolic wire, with the bead position marked.

9.3 Q41

Problem Statement

Index notation and Einstein summation convention. Given

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix},$$

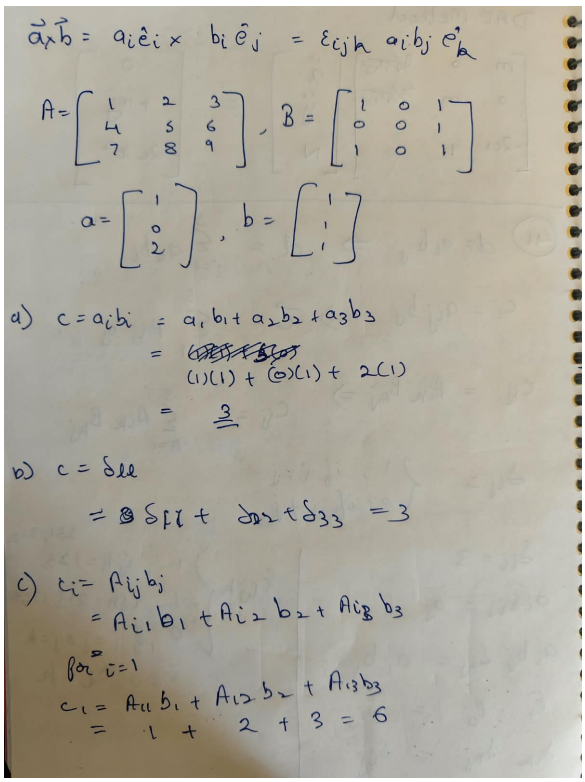
$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

evaluate the expressions in the question using the summation convention, or state when an expression is nonsensical. The expressions exercise dot products, matrix-vector and matrix-matrix products, the Kronecker delta, free and dummy indices, and the Levi-Civita symbol.

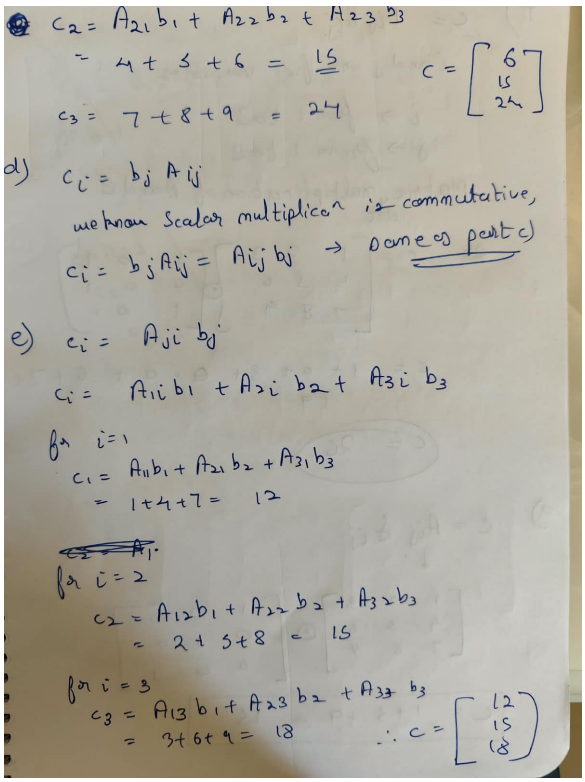
Interpretation

A repeated index in a multiplicative expression is a dummy index and is summed from 1 to 3. An index that appears only once in a term is a free index and must appear consistently on both sides of an equation. The expressions are therefore checked both for their numerical value and for whether their index structure is valid.

Solution

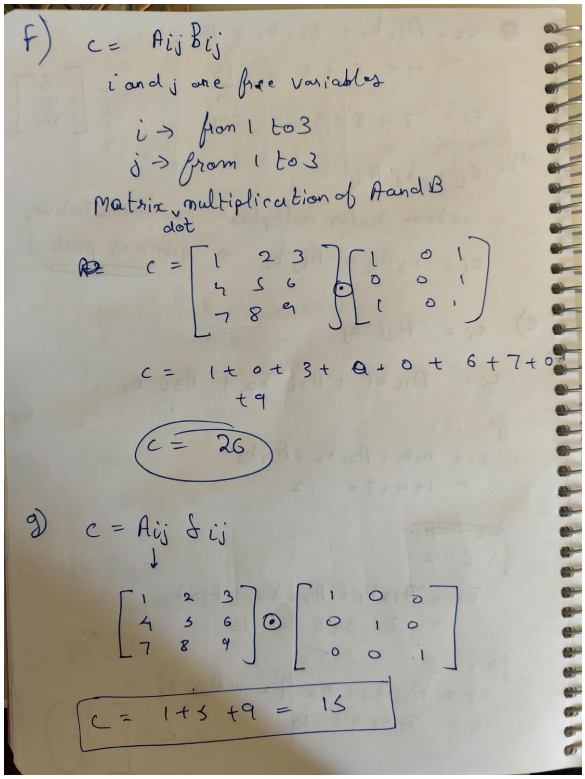


(a) Evaluating the vector dot product and the contraction of the Kronecker delta.

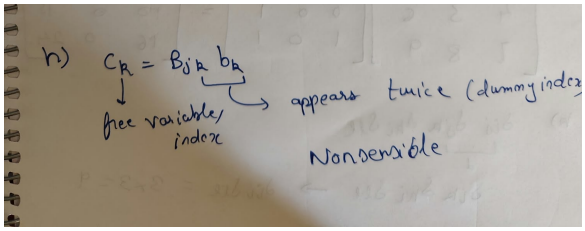


(b) Evaluating matrix-vector products and distinguishing row and column contractions.

Figure 80: Scalar and vector expressions using Einstein summation.

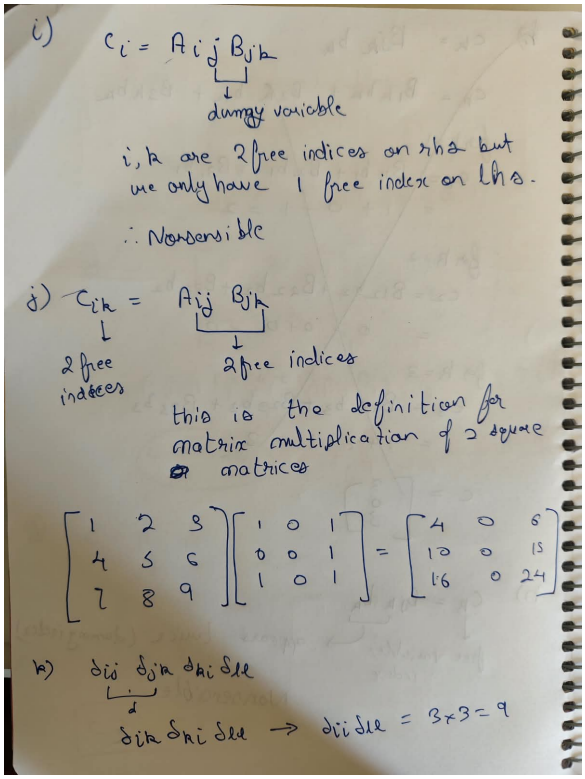


(a) Evaluating matrix contractions, including the Frobenius product and trace.

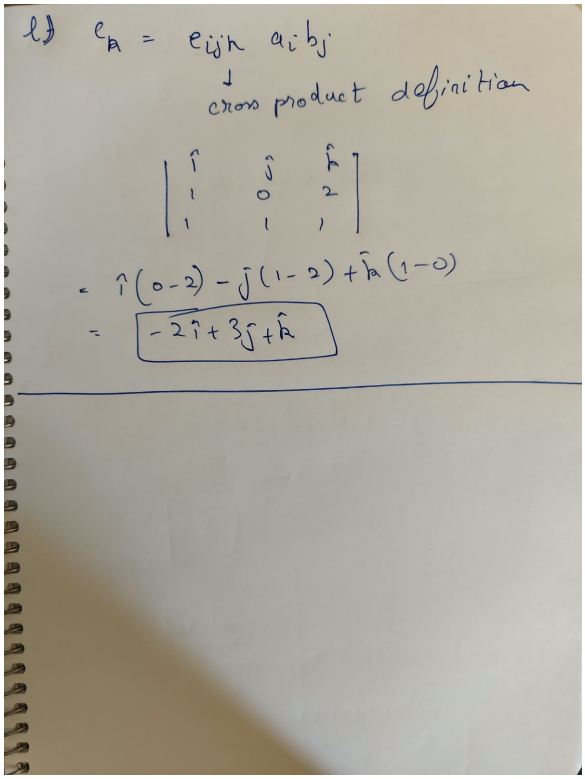


(b) Checking free-index consistency; the expression shown is nonsensical because the free indices do not match.

Figure 81: Scalar contractions and index-consistency checks.



(a) Evaluating the matrix-matrix product and simplifying the Kronecker-delta expression.



(b) Using the Levi-Civita symbol to evaluate the cross product $\mathbf{a} \times \mathbf{b}$.

Figure 82: Matrix multiplication, delta identities, and the Levi-Civita symbol.